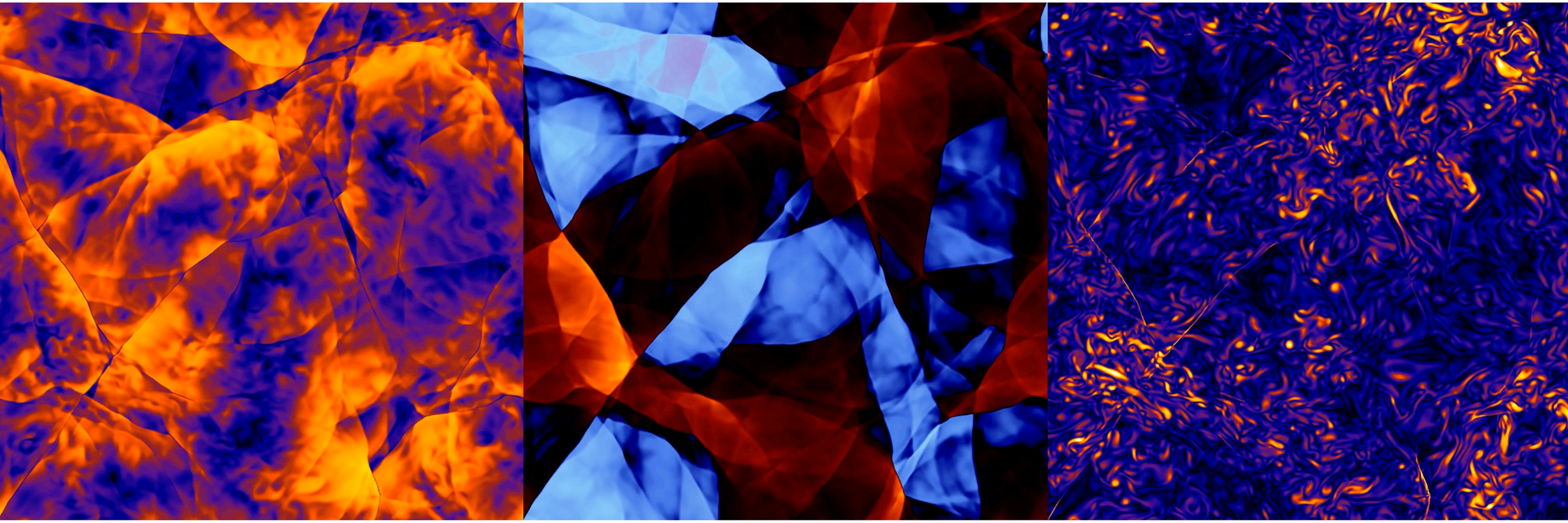
Learned Closure for the Dynamics of Cosmological Phase Transitions with Convolutional Neural Networks

A KAAI Workshop · Carnegie Mellon University, Nov. 4, 2026



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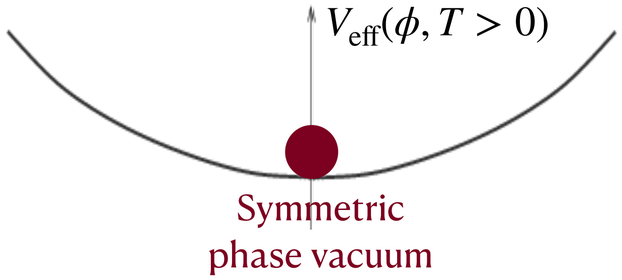
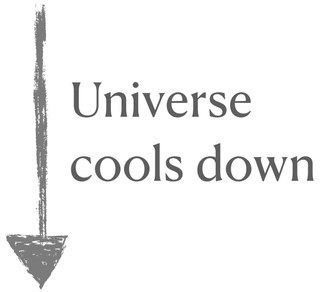
VNIVERSITAT DE VALÈNCIA

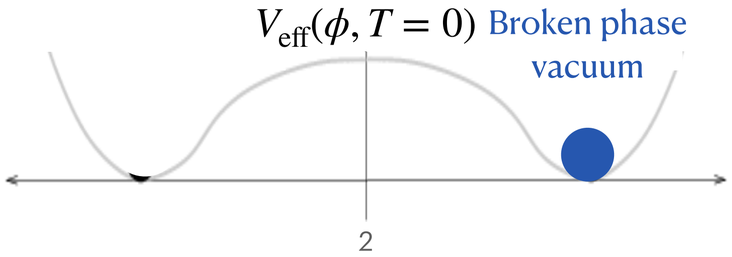
Cosmological phase transitions $V_{\text{eff}}^{T}(\phi, T)$

High temperature effective potential





Low temperature effective potential



Cosmological phase transitions $V_{\text{eff}}^{T}(\phi, T)$

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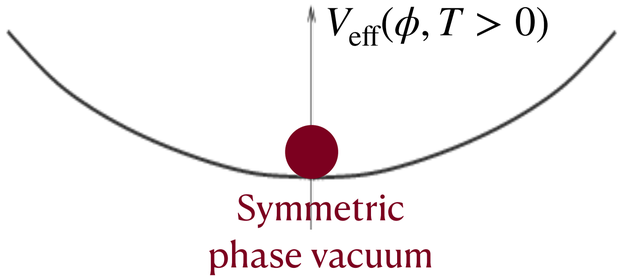
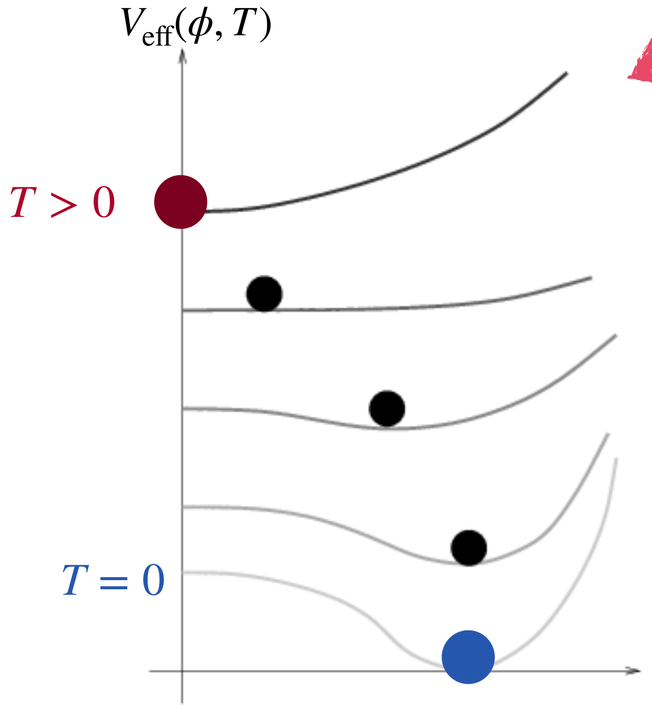


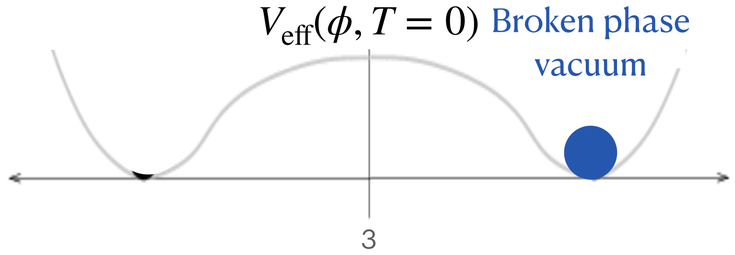
2-order PT



Smooth cross-over

Universe
cools down

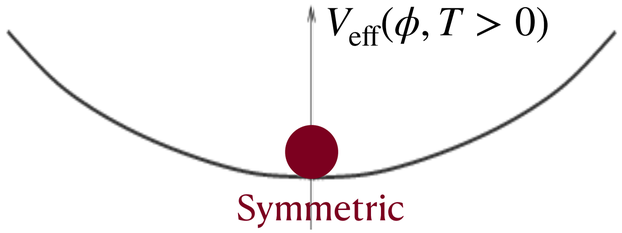
Low temperature effective potential



Cosmological phase transitions

$V_{\rm eff}^{T}(\phi, T)$

High temperature effective potential



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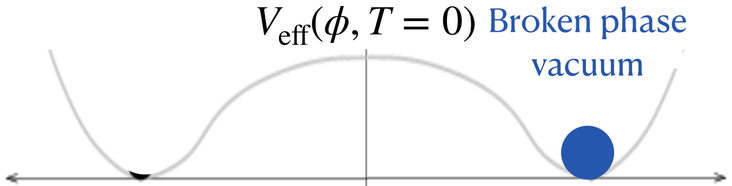
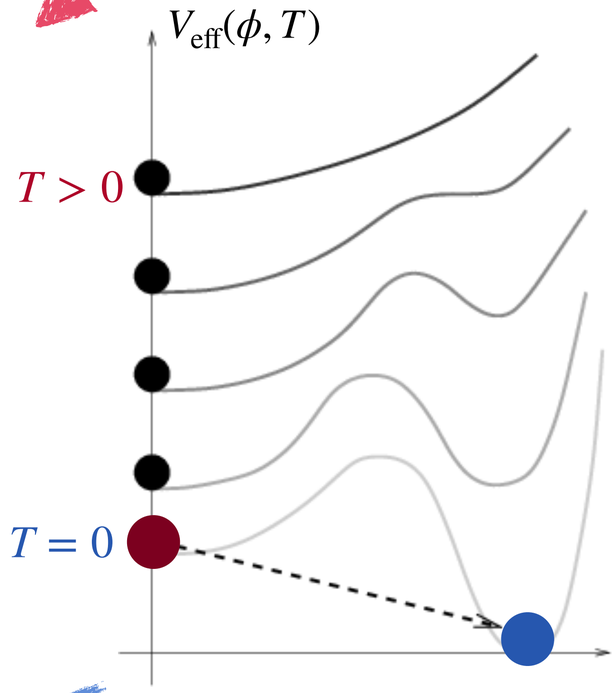
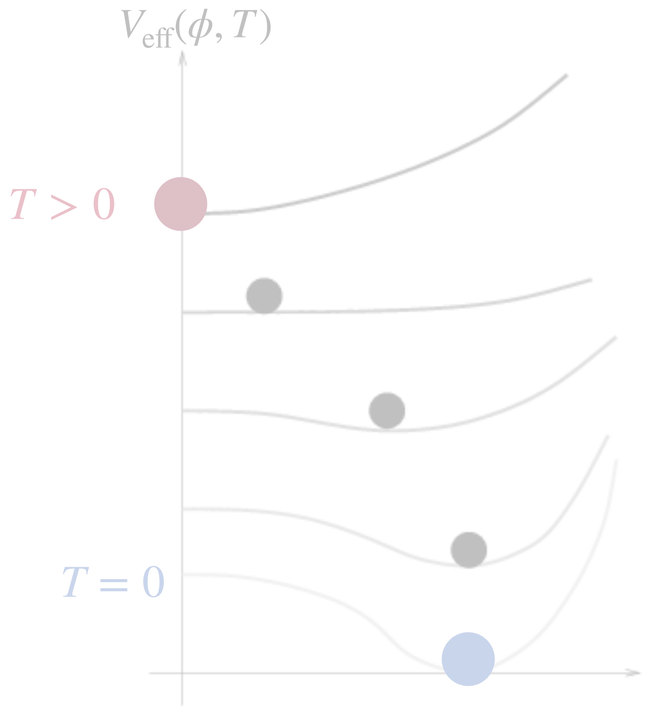


1-order PT



Quantum/Thermal Tunneling

2-order PT



Smooth cross-over

Cosmological phase transitions

$V_{\text{eff}}^{T}(\phi, T)$

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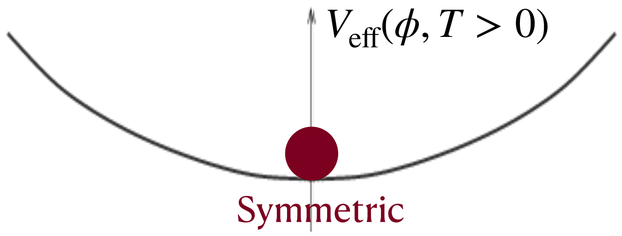

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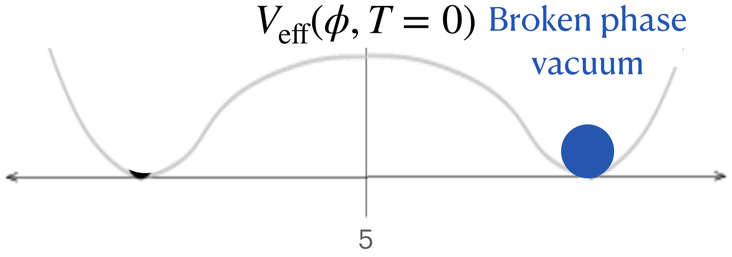

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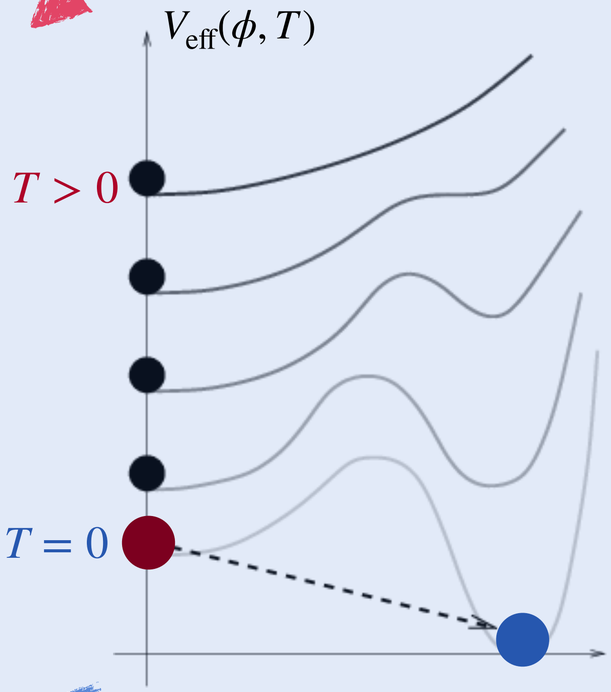

Quantum/Thermal Tunneling

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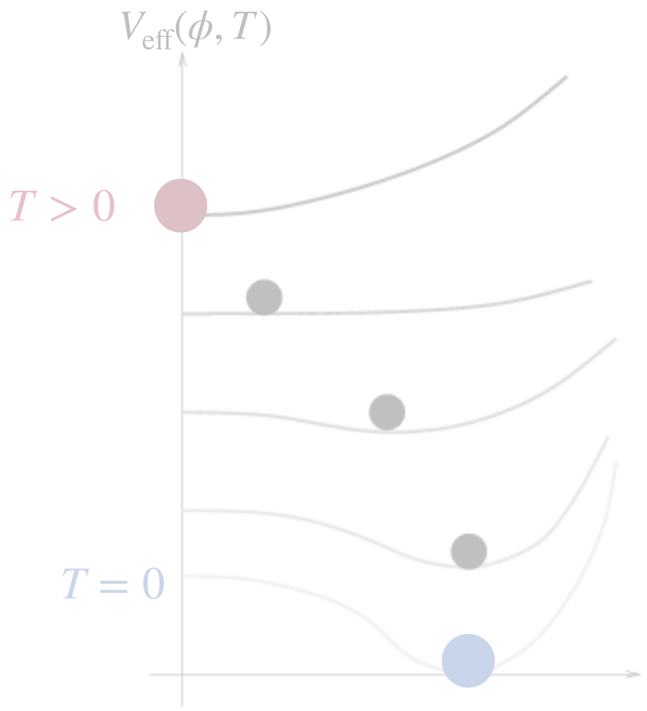
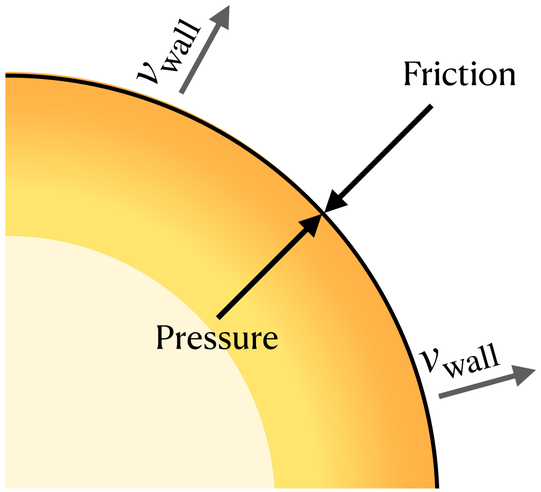


Smooth cross-over

Single bubble evolution

- The **evolution** of a **single bubble** depends upon a **balance** between **pressure** and **friction** between **scalar** field and the **plasma**.



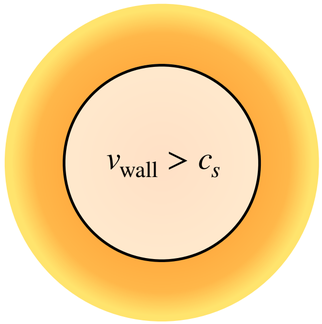
Self-similar profiles

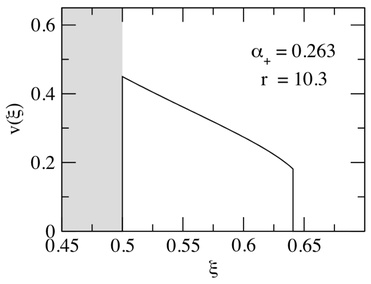
Self-similar solutions and runaway walls

Wall velocity

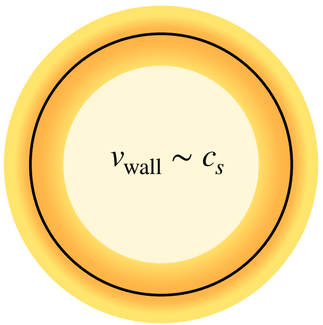
$v_{\text{wall}} < c_s$ | $v_{\text{wall}} \sim c_s$ | $v_{\text{wall}} > c_s$ | v_{wall} runaway

Deflagrations

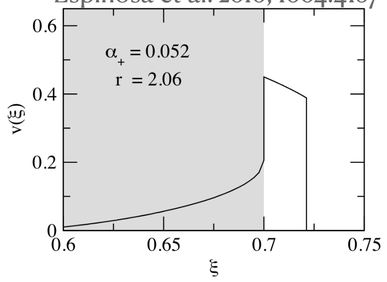




Hybrids



Espinosa et al. 2010, 1004.4187



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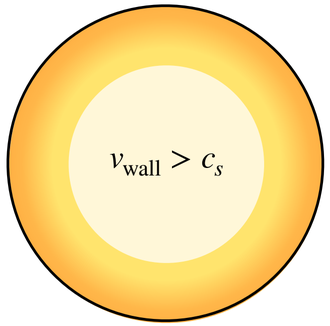
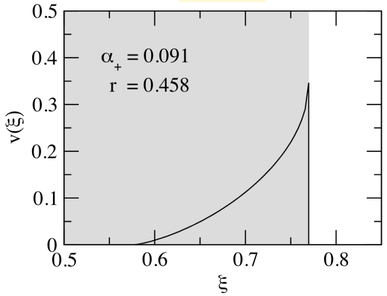
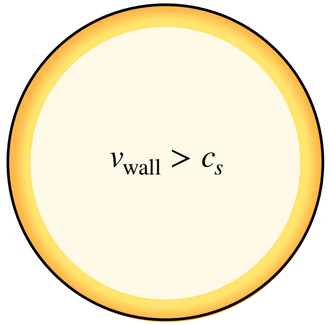




Strong detonations



Runaway

- Friction fails to balance pressure
- Walls runaway: $v_{\text{wall}} \to c$
- Energy carried by walls

Strength

$\alpha \ll 1$ | $\alpha \sim 1$ | $\alpha \gg 1$

Self-similar profiles

Self-similar solutions and runaway walls

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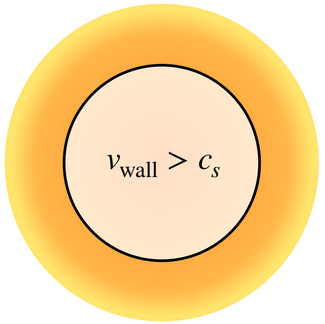

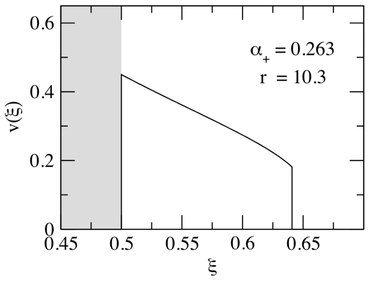

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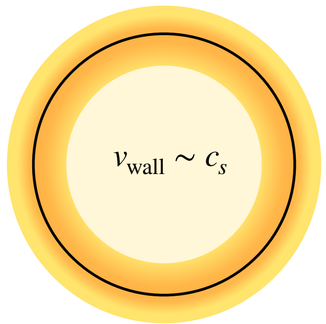

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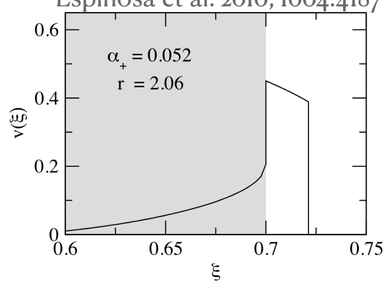

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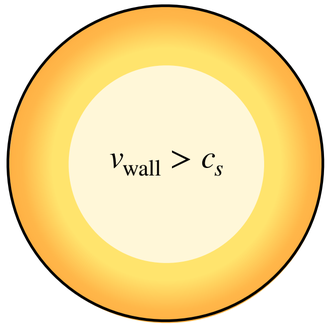

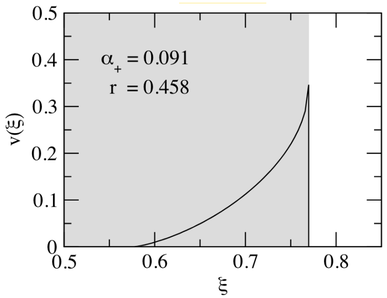

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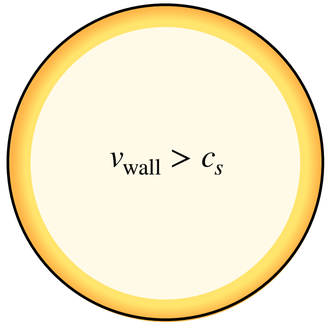


Runaway

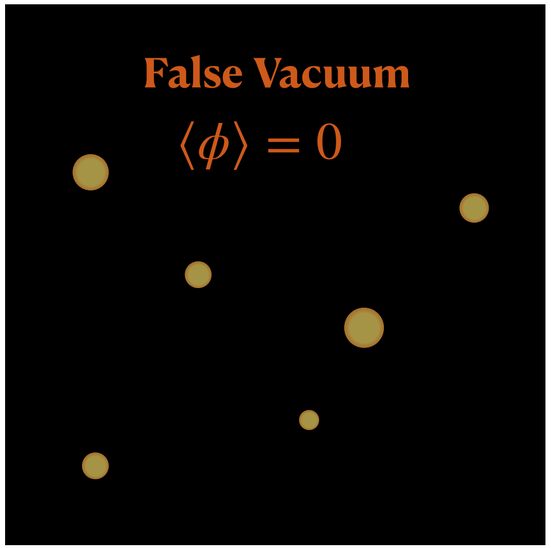
- Friction balances pressure
- Walls reach terminal velocity
- Sound-shells approach self-similar solutions
- Energy carried by plasma
- Focus of my work

Strength

$\alpha \ll 1$ | $\alpha \sim 1$ | $\alpha \gg 1$

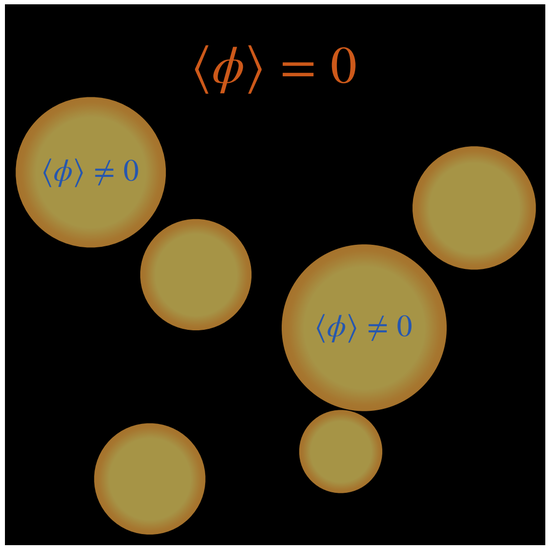
Schematics of a first-order PT

Bubble nucleations



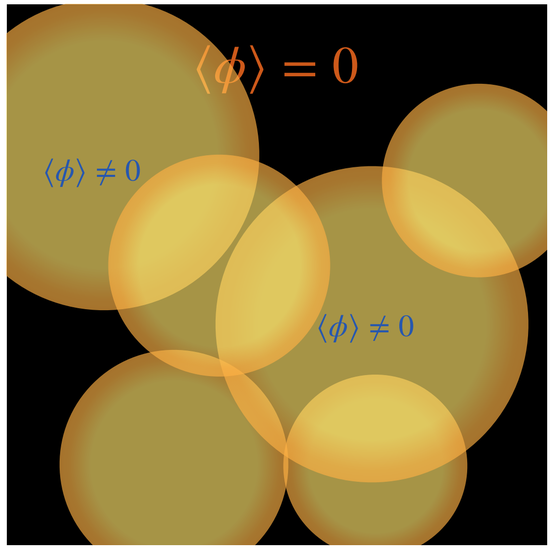
- Universe **initially homogeneous**
- T drops below T_* and ϕ locally **tunnels** to TV
- Bubbles **nucleate** and begin **expanding**

Bubbles expand



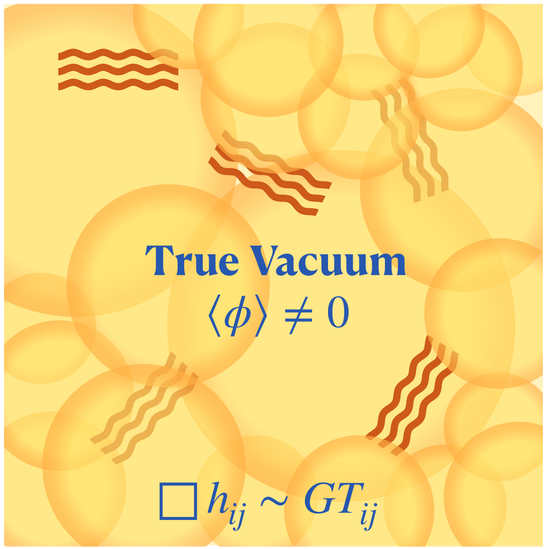
- **Expansion** proceeds
- **Vacuum energy deposits** in fluid
- Fluid **responds**
- **Self-similar** profiles develop

Bubbles collide



- Bubbles **collide**
- **GW production** from **anisotropic** stress T_{ij}^{plasma}

Sound waves



- **PT completion**
- Long-lasting **dynamics (sound waves)** source GWs
- **Non-linear** evolution may produce **shocks** and **turbulence**

PT parameters

Particle physics model

- Lagrangian $\mathscr{L}$
- Thermal field theory
- Effective potential

Theory

Simulation or theory

PT parameters

- **Temperature** $T_* :\Longrightarrow H_* \to$ Horizon size $\to f_{\text{typcial}}$
- Inverse **duration** β : as in nucleation rate $\Gamma_{\text{nuc}} \propto e^{\beta t} \to R_*(\beta)$,
  $\beta/H \sim \mathcal{O}(100) \Rightarrow$ neglect expansion, $\omega_{\text{peak}} \simeq \frac{\beta}{100H}\frac{T}{100\text{GeV}}\text{mHz}$
- **Strength** $\alpha := \rho_{\text{vac}}/\rho_{\text{rad}}$
- **Wall velocity** $v_w :\longrightarrow$ energy budget and self-similar profiles

Hydrodynamics

- Character of **fluid perturbation**
- **Linear** sound-waves or **non-linear** evolution
- Detailed **evolution** of T_{ij}

GW spectrum

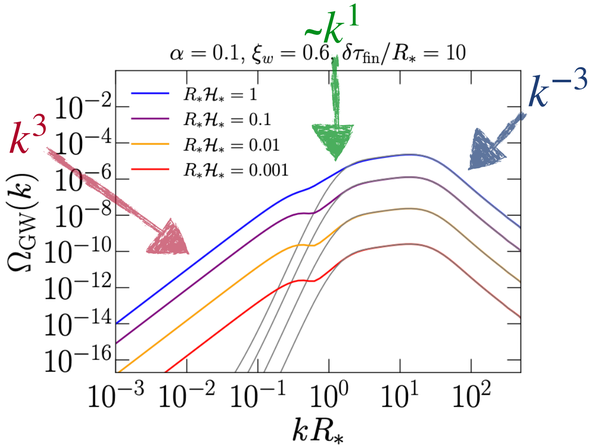
- $\Omega_{\text{GW}}(f \mid H_*, \beta, \alpha, v_w)$

Sound-shell model

Pros

- Semi-analytical estimation of GW spectra
- Results efficiently and easily obtainable for any (hydrodynamically valid) choice of model parameters

Hindmarsh 2016, Hindmarsh et al. 2019,
Roper Pol et al. 2308.12943, Ramkishor et al. 2308.12916



Cons

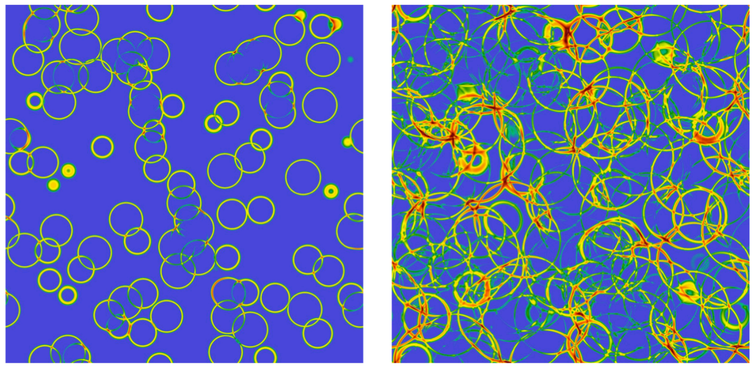
- Assumes linearly interacting freely evolving sound waves
- Not valid when non-linearities become important
- —> cannot probe strong PTs

Plasma + ϕ simulations

Pros

- Robust GW spectra predictions
- Few simplifying assumptions
- Models bubble friction
- Non-linear dynamics

Hindmarsh, Huber, Rummukainen, Weir (13,15,17)



Cons

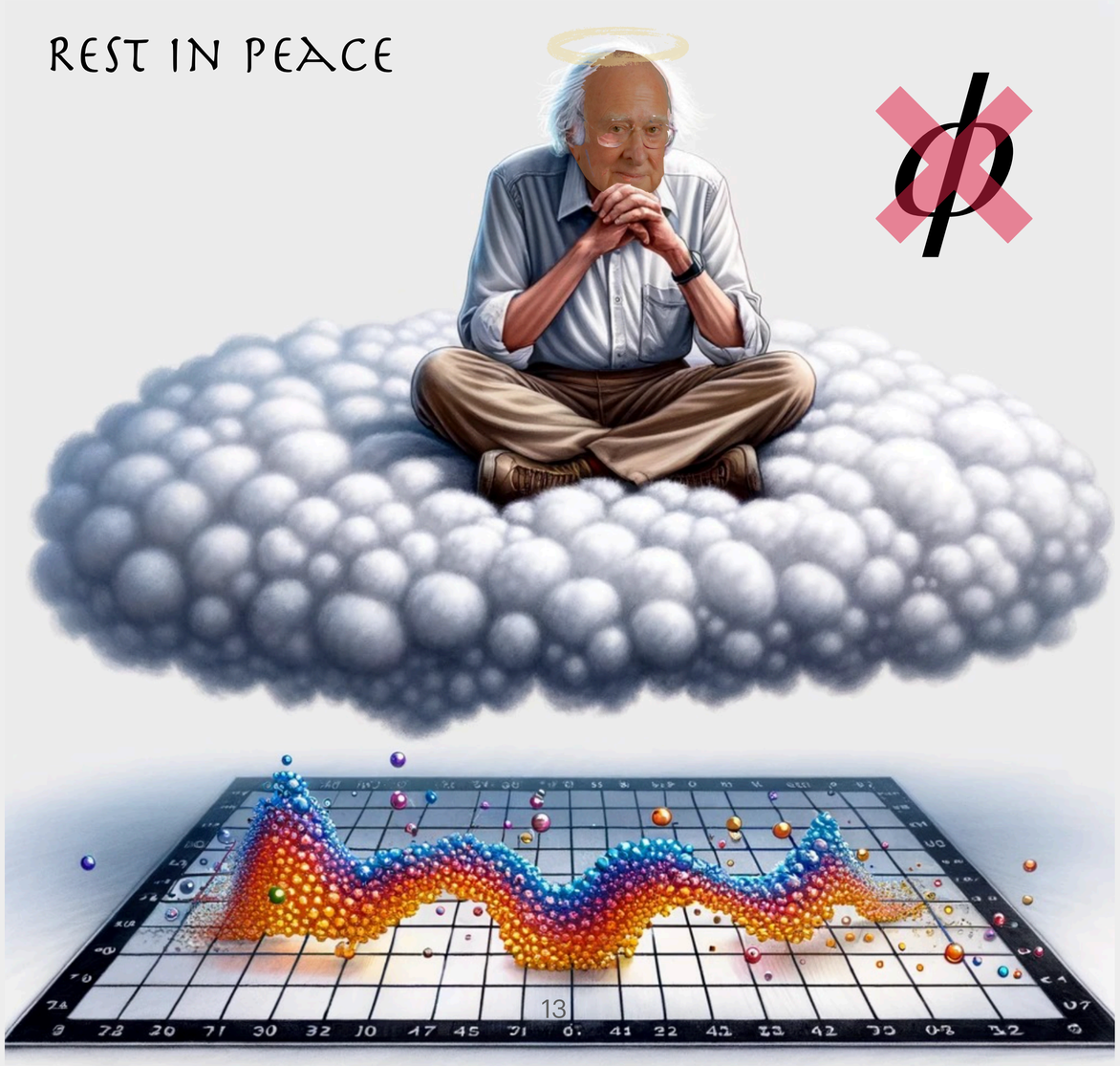
- Numerically costly to run (need high res.)
- Spectra for only few parameter points
- Hard to obtain reliable interpolation between points

Space of opportunity

There is (was) a need for a tool that can
efficiently produce GW predictions covering a vast array of parameter points,
without compromising physicality with too many simplifying assumptions,
and being non-linear to capture the realistic long-term fluid evolution.

Remove the Higgs field...

... and run exclusively hydrodynamical simulations

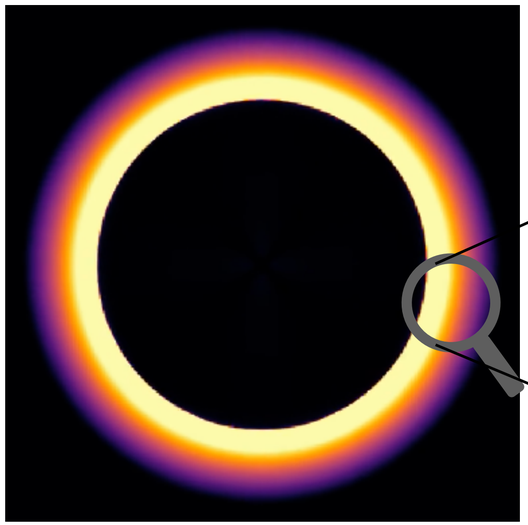


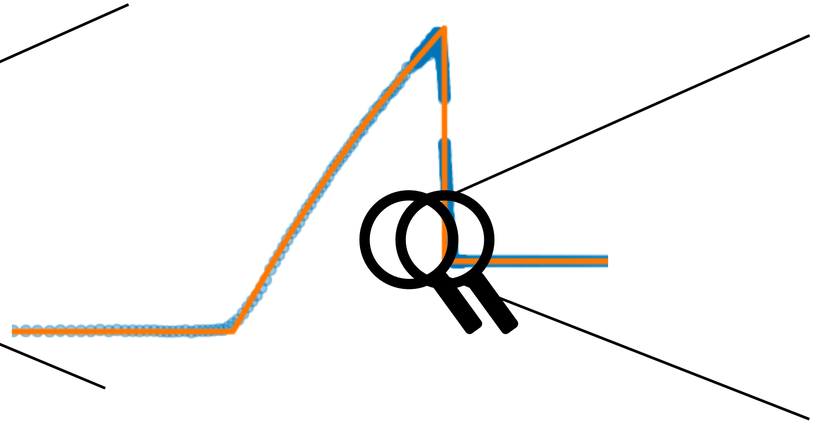
Why Higgsless?

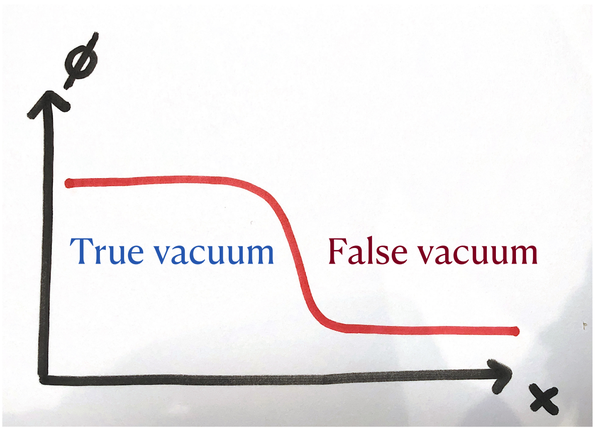
Reason 1: Separation of scales

Few % of Hubble horizon | **% of Hubble horizon** | 1/100 **GeV (EW)**

Fluid perturbations > Sound-shell width ⋙ Scalar field wall profile







Difficult to resolve all three scales in a numerical simulation.

Why Higgsless?

Reason 2: GWs from ϕ subdominant

- If wall velocity v_w constant:
  - $E_\phi \propto r_{\text{bubble}}^2$
  - But $E_{\text{released}} \propto r_{\text{bubble}}^3$
- $\Longrightarrow$Most **energy** is deposited **into plasma**

GW contribution from ϕ is negligible

Why Higgsless?

Conclusion

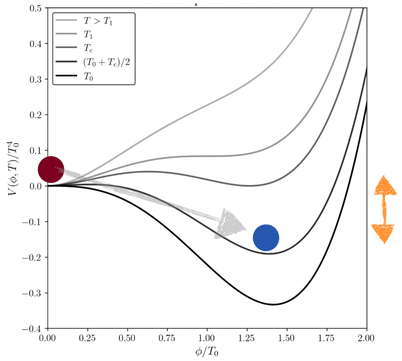
Removing the Higgs-field **circumvents** the **difficulty** of **resolving** enormously **separated scales** in a single numerical simulation while capturing well the **hydrodynamics** and **anisotropic stresses,** and by extension the resulting **GW background.**

How achieve this?

Higgsless simulations

Physical setup

Relativistic hydrodynamics

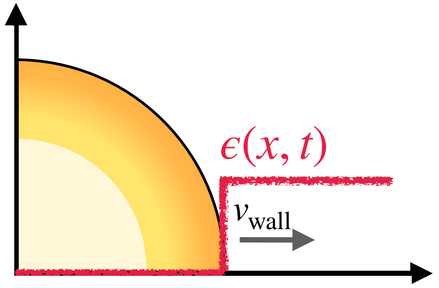


- **Perfect fluid with vacuum contribution:** $T^{\mu\nu} = (p+\rho)u^\mu u^\nu - p g^{\mu\nu}$
- Bag **E.o.S.**: $\rho = aT^4 + \epsilon$ and. $p = \frac{1}{3}aT^4 - \epsilon$ (Thermal, Vacuum)

where $\epsilon = \begin{cases} \varepsilon_+ \equiv \frac{3\alpha}{4}w, & \text{in the symmetric phase,} \\ 0, & \text{in the broken phase} \end{cases}$

is the energy **difference between** the **symmetric** and **broken** phases.

- **E.o.M.** given by **total energy-momentum conservation:** $\partial_\mu T^{\mu\nu} = 0$
- 4 equations: $\begin{aligned} \partial_t K^0 + \nabla_i K^i &= 0 \\ \partial_t K^j + \nabla_i T^{ij}\left[K^\mu\right] &= 0 \end{aligned}$ where $T^{ij}\left[K^\mu\right] = \frac{K^i K^j}{K^0 + p} + p[K^\mu]\,\delta^{ij}$
- 4 dynamical variables: $K^\mu \equiv T^{0\mu}$ $\Rightarrow$ $\partial_t U + \nabla_i F^i(U) = 0, \quad U = \begin{pmatrix} K^0 \\ K^j \end{pmatrix}, \quad F^i = \begin{pmatrix} K^i \\ T^{ij} \end{pmatrix}$



Physical setup

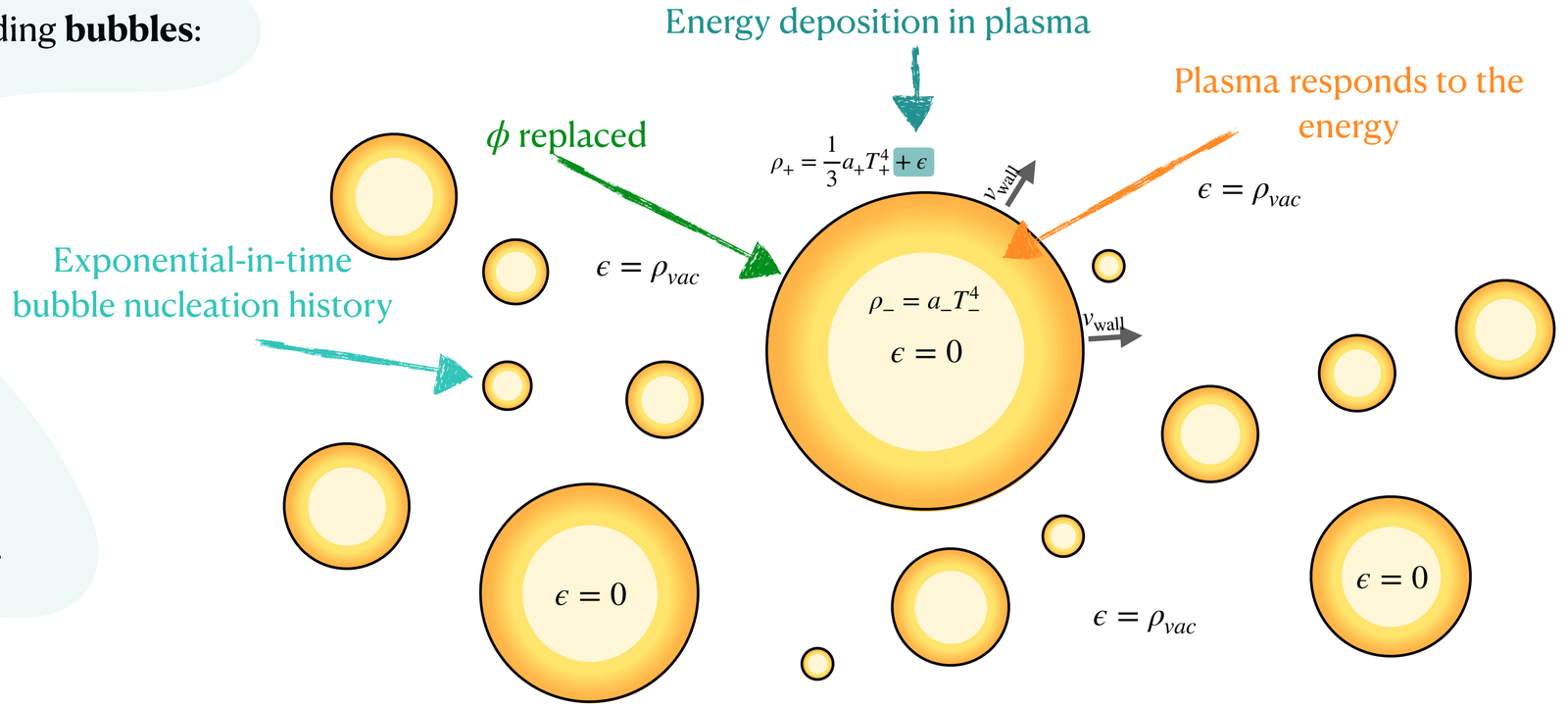
Bubbles and the Equation of State

- **Bag E.o.S.** suggests natural way to **embed** the expanding **bubbles**:
  - Prescribe that $\epsilon = \epsilon(t, \vec{x})$ is a **function** of **space** and **time**
  - **Bubbles** are **radially expanding** regions where $\epsilon = 0$

so that $\epsilon(t, \vec{x} \,|\, \alpha) = \begin{cases} 0, & \textbf{inside bubbles}, \\ \frac{3\alpha}{4} w, & \textbf{outside bubbles} \end{cases}$

- To determine $\epsilon(t, \vec{x})$, we **need** to specify: v_w, α, and an exponential **bubble nucleation history.**

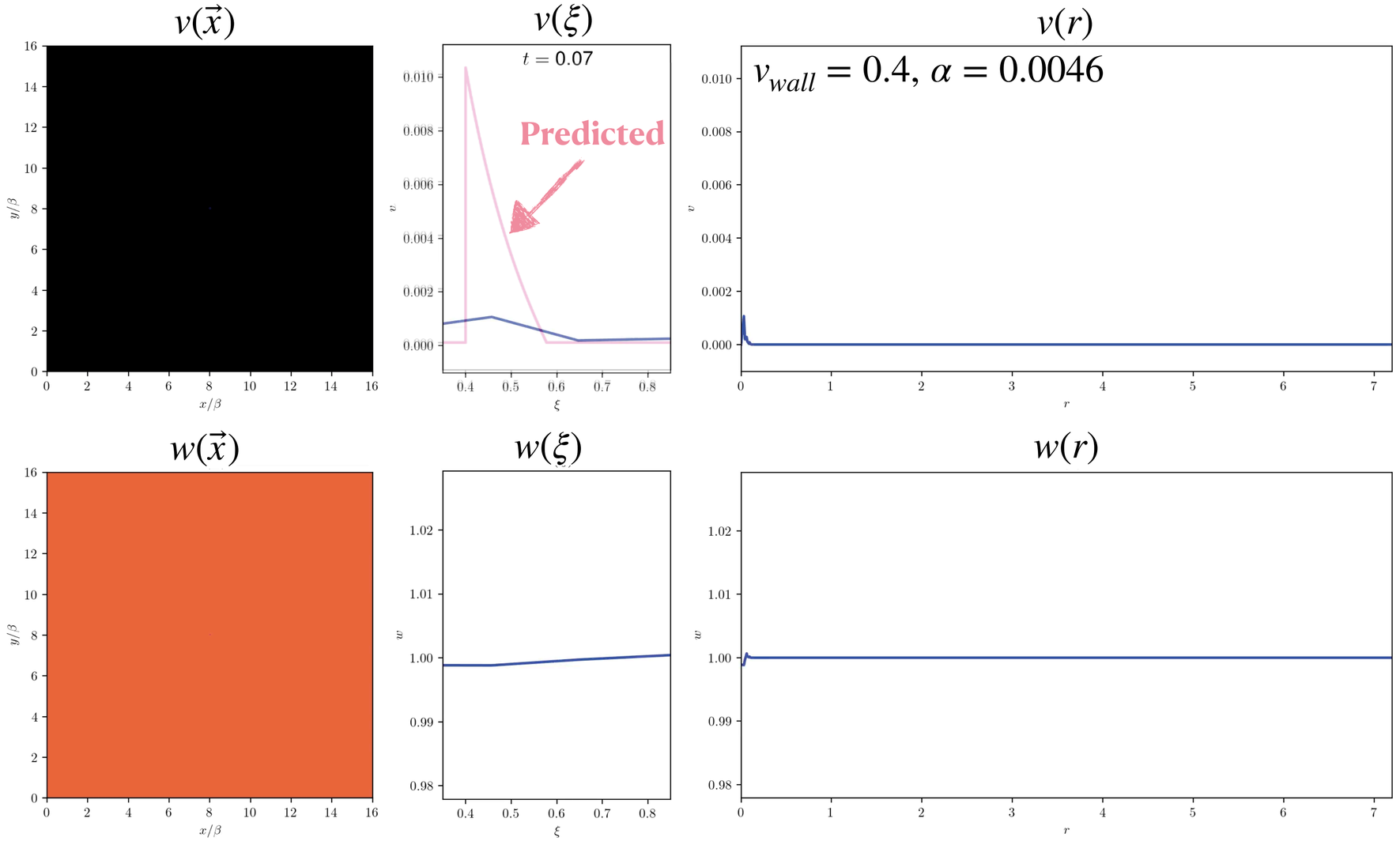
- **Nucleation rate** $\Gamma(t) \simeq \Gamma_* e^{\beta(t - t_*)}$



- Thus, with these simple **3 ingredients,** $\epsilon(t, \vec{x} \,|\, \alpha)$ is **predetermined input** to the simulation, effectively **replacing** $\phi \to \epsilon(t, \vec{x} \,|\, \alpha)$
- **Couples to fluid** through $p = \frac{1}{6}\left[-2K^0 - 4\epsilon(t, \vec{x} \,|\, \alpha) + \sqrt{(4K^0 - 4\epsilon(t, \vec{x} \,|\, \alpha))^2 - 12K^i K^i}\right]$

as in $T^{ij}\left[K^\mu\right] = \frac{K^i K^j}{K^0 + p} + p$

Convergence of self-similar solutions

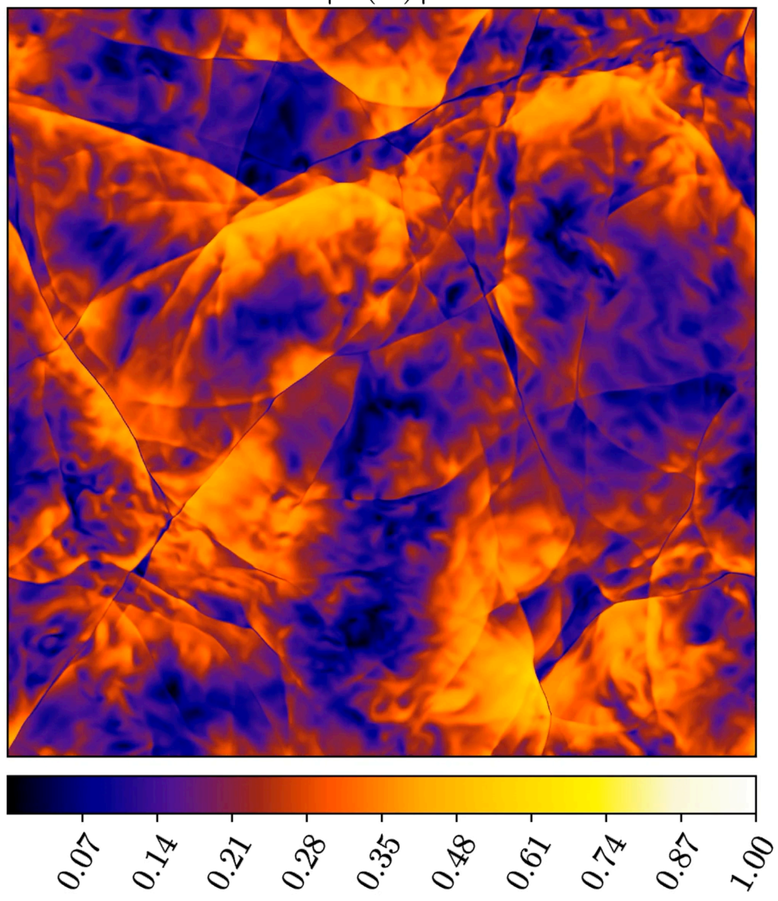


Example

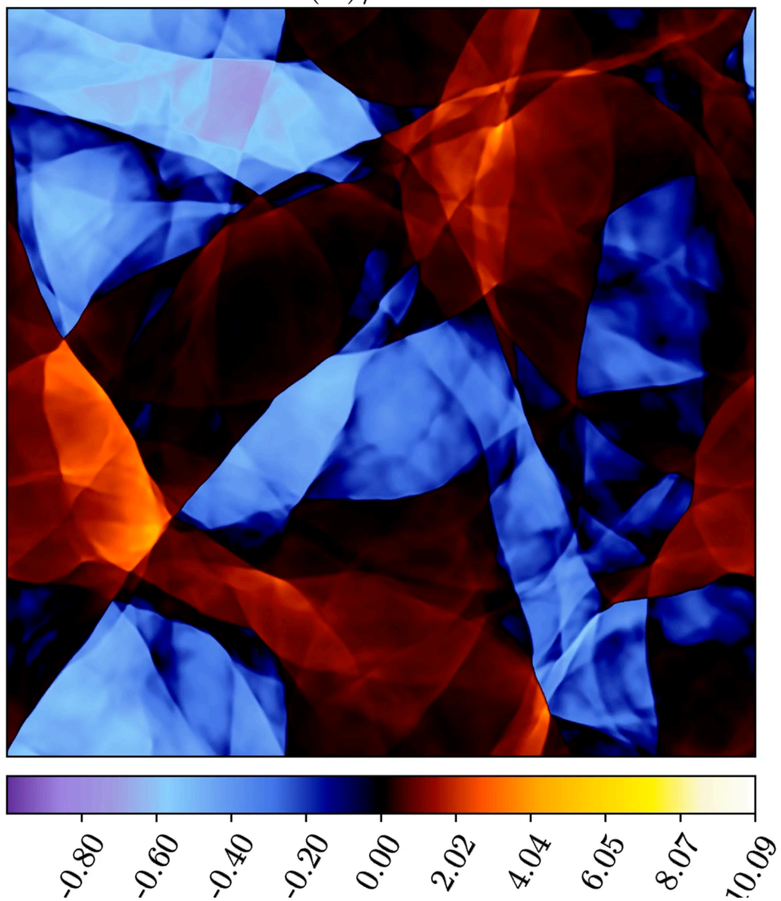
$\alpha = 0.5,\ v_w = 0.36,\ J = 1024,\ L/v_w = 20.0$

$\tilde{t} = 20.75$

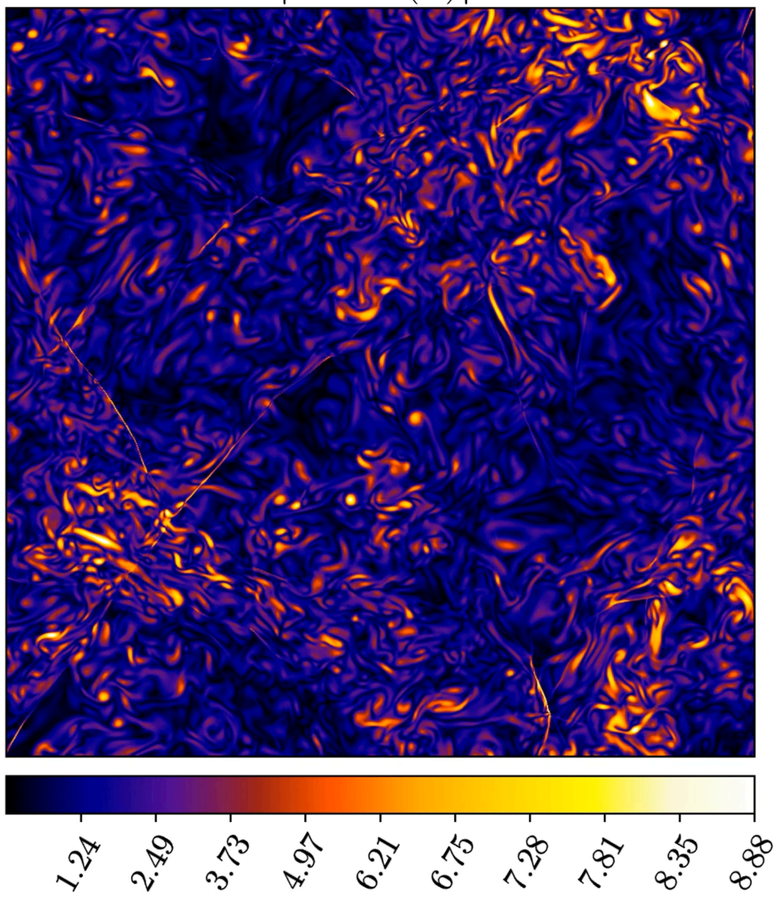
$|\mathbf{v}(\tilde{\mathbf{x}})|$



$w(\tilde{\mathbf{x}})/\bar{w} - 1$



$|\tilde{\nabla} \times \mathbf{v}(\tilde{\mathbf{x}})|$

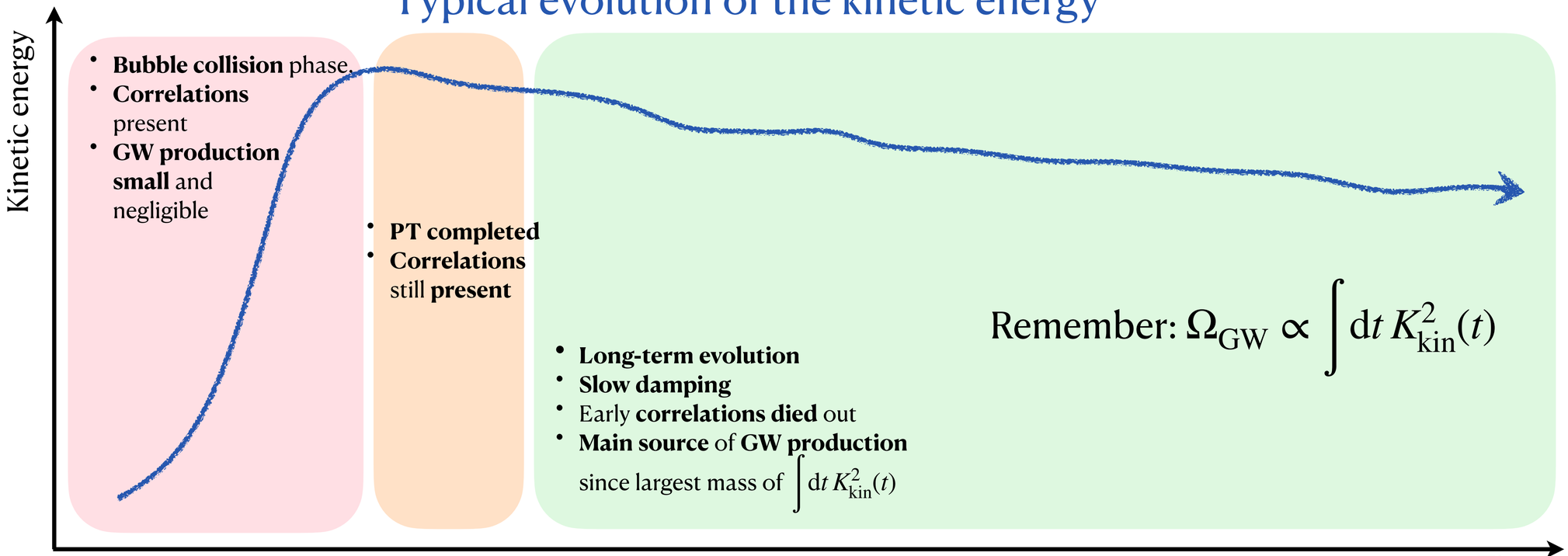


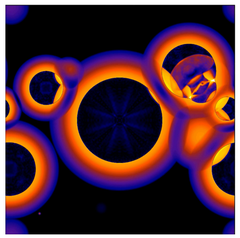
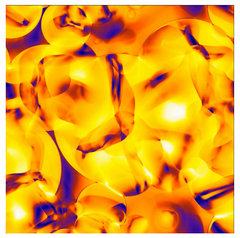
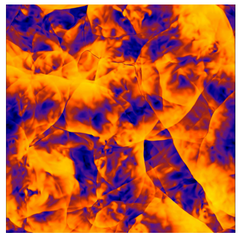
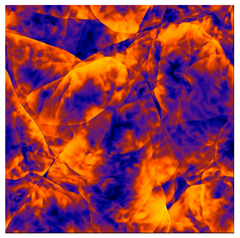
GW production

Strategy 1:

Assume hydrodynamics is a long-lived GW source

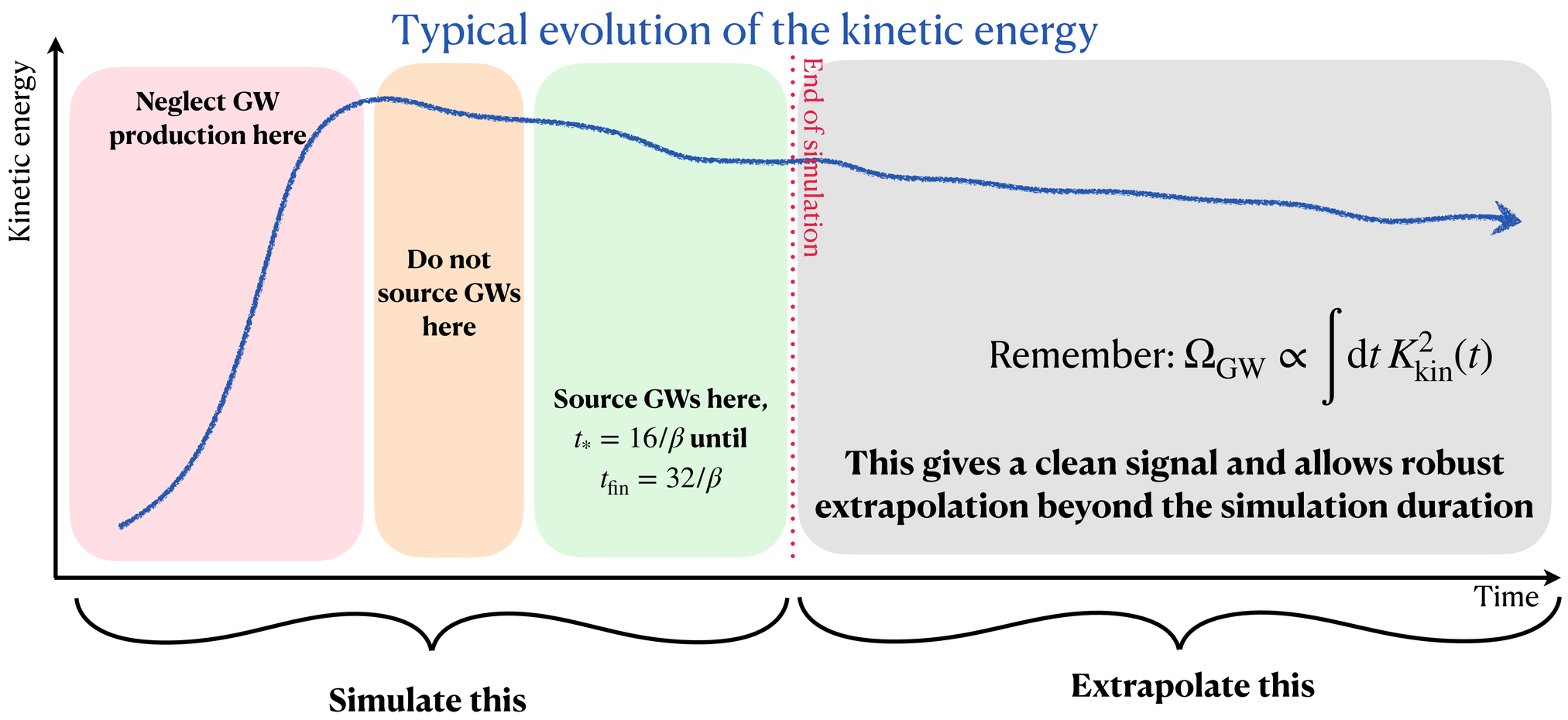
Typical evolution of the kinetic energy



Strategy 1:

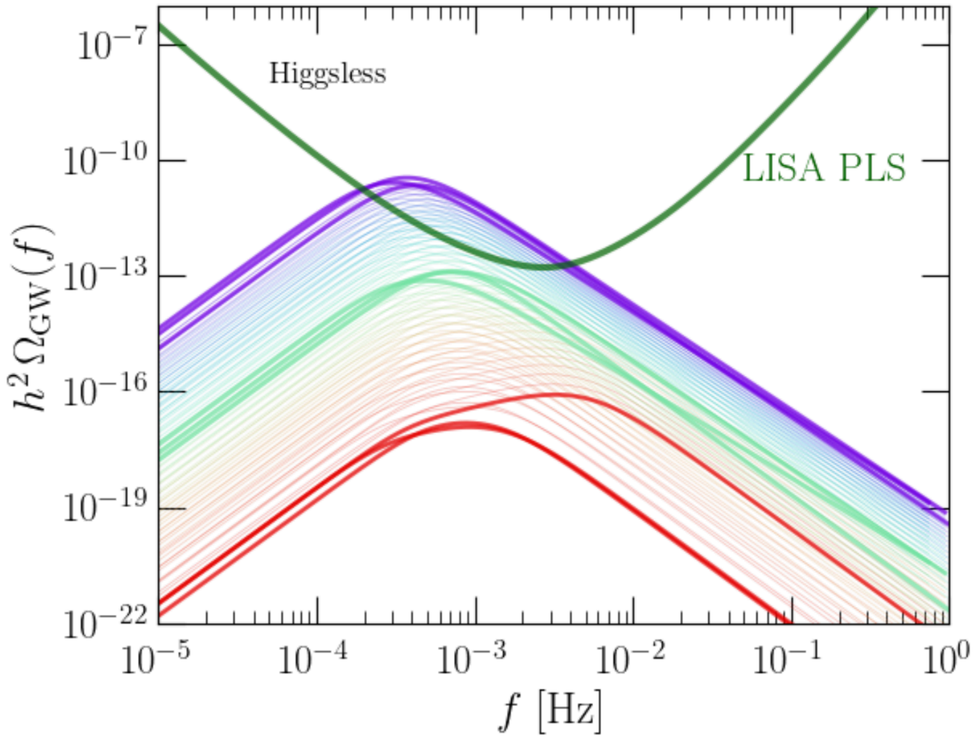
Assume hydrodynamics is a long-lived GW source



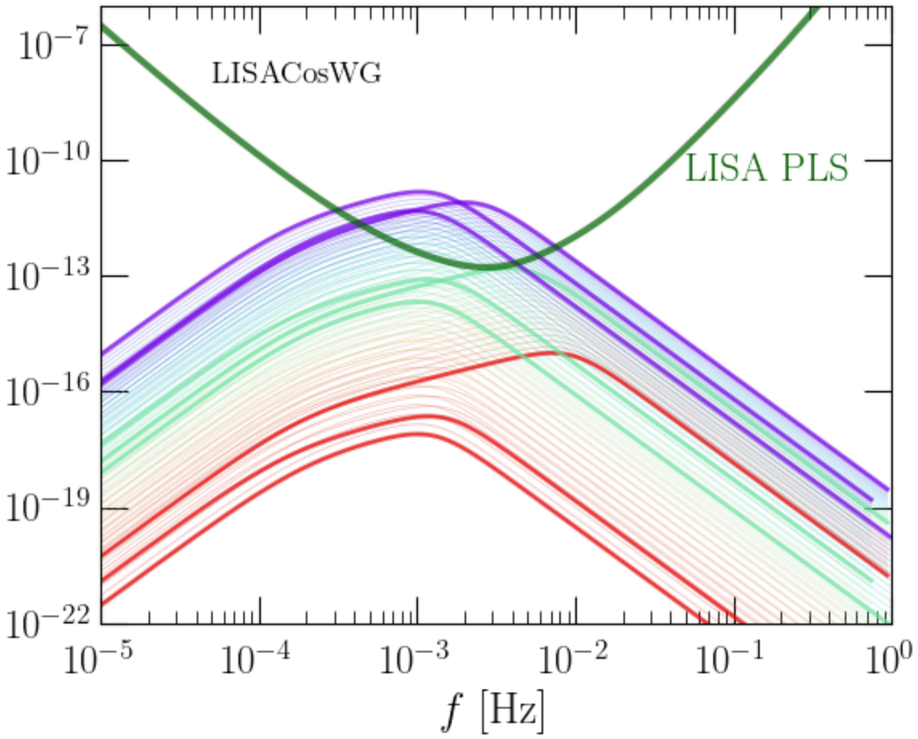
CosmoGW

Making the results readily accessible

Templates from as outlined in 2508.04263



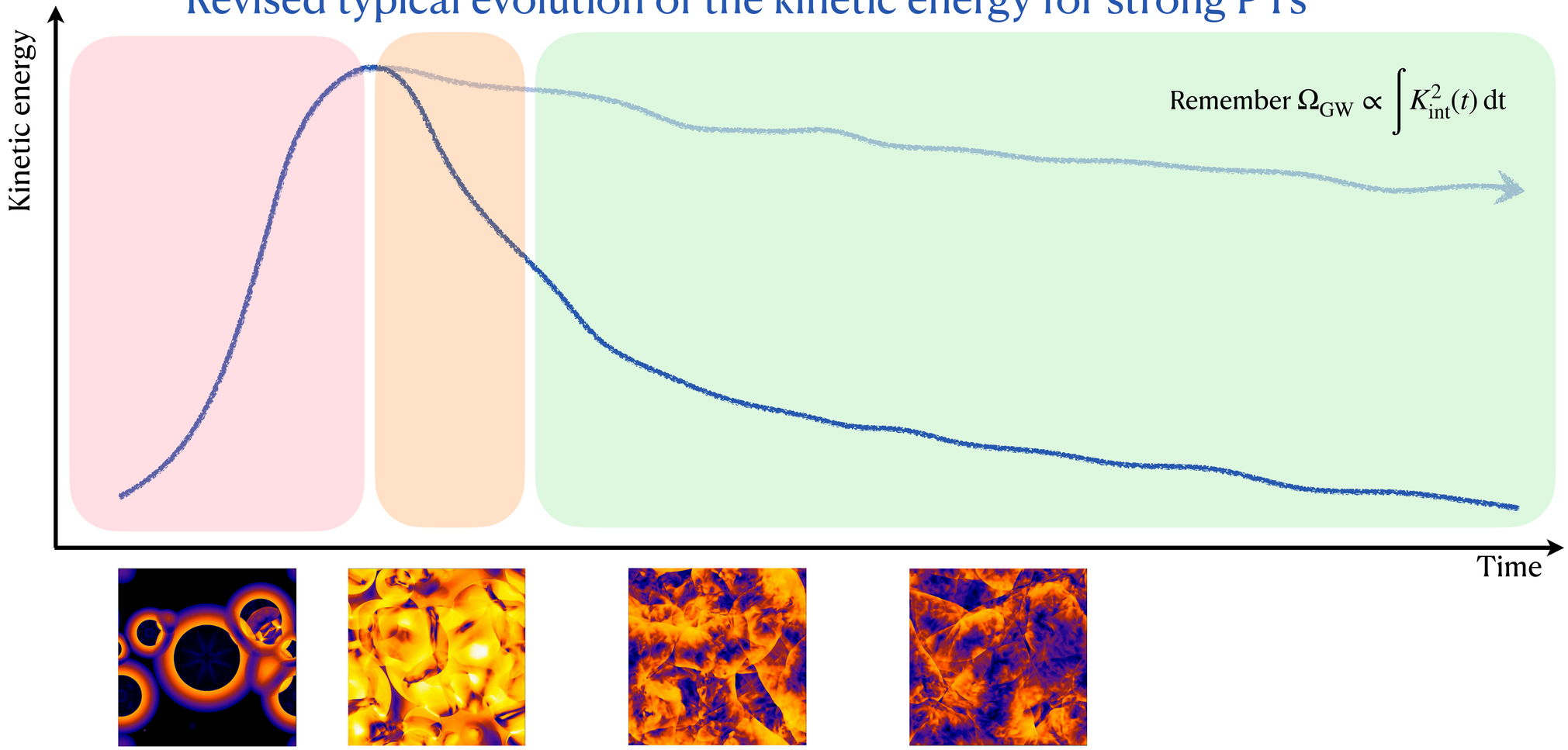
2403.03723



Strategy 2:

Source GWs through the entire simulation duration

Revised typical evolution of the kinetic energy for strong PTs



Strategy 2:

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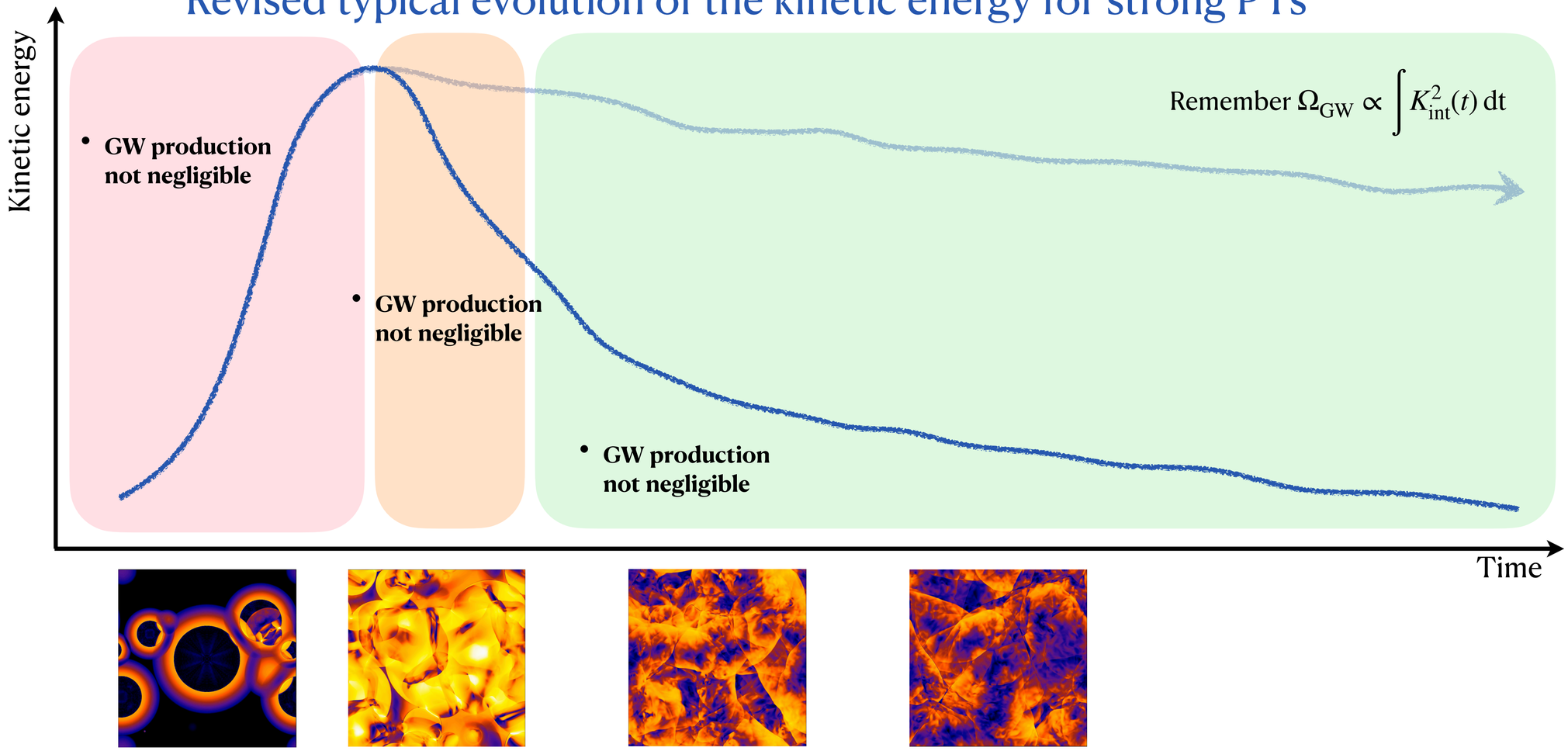

Strategy 2:

Source GWs through the entire simulation duration

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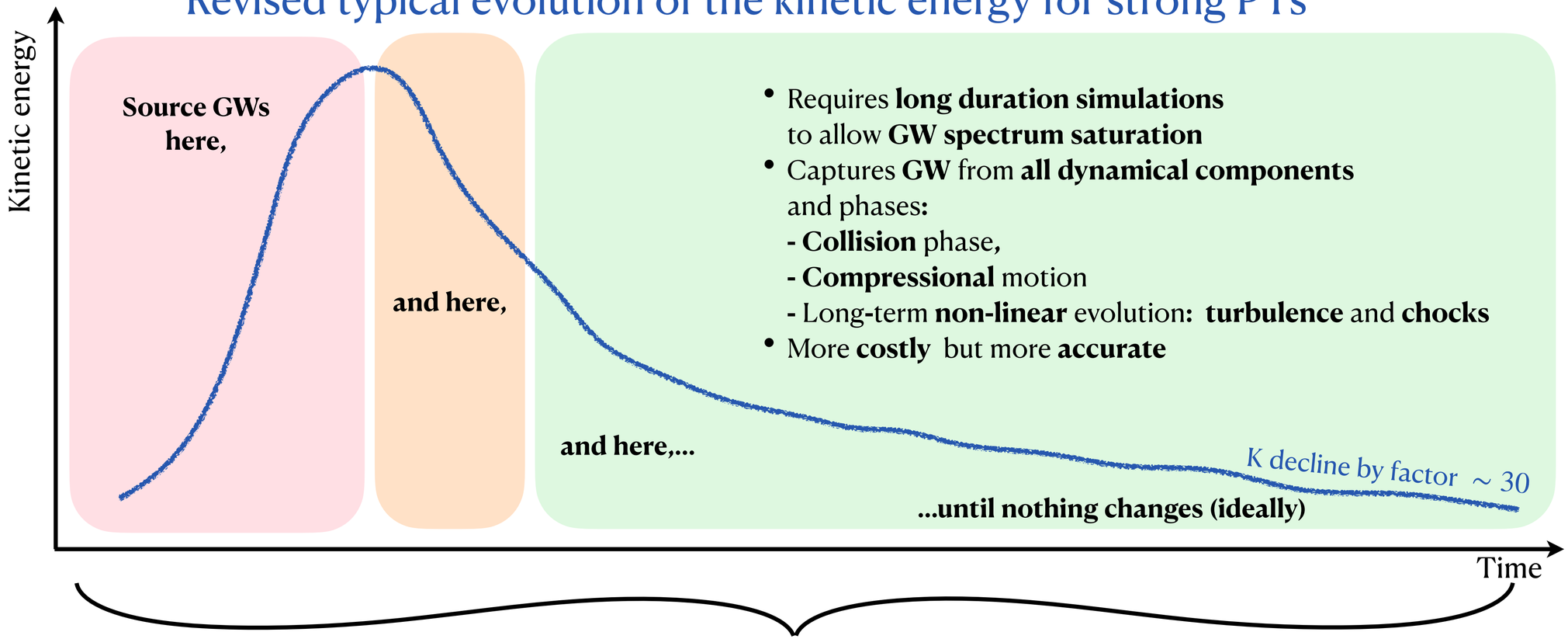


- Requires **long duration simulations** to allow **GW spectrum saturation**
- Captures **GW** from **all dynamical components** and phases:
  - **Collision** phase,
  - **Compressional** motion
  - Long-term **non-linear** evolution: **turbulence** and **chocks**
- More **costly** but more **accurate**

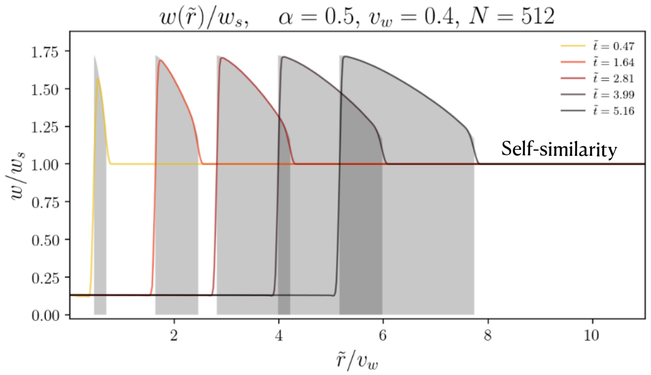
Simulate everything until the end*.

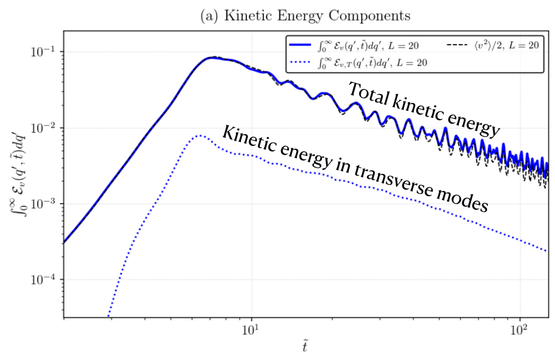
*end = when the GW spectrum saturates.

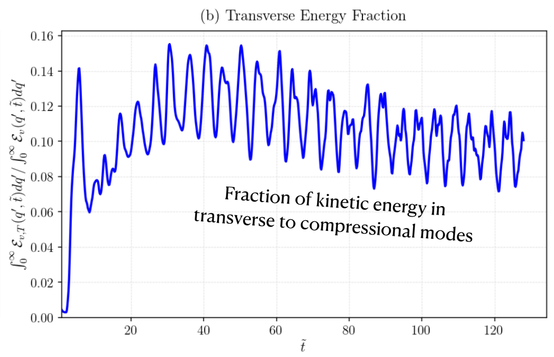
Results

Kinetic energy and velocity spectra

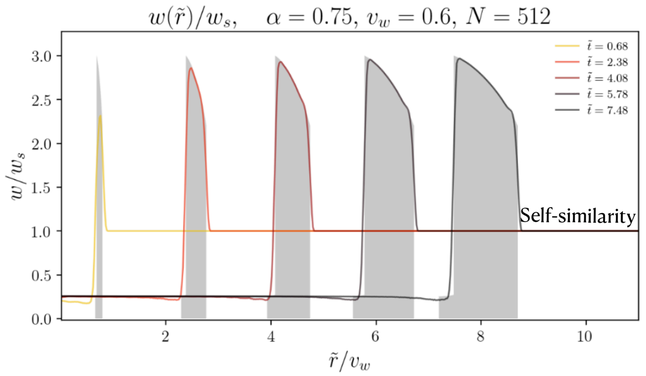
$\alpha = 0.5$
$v_w = 0.4$

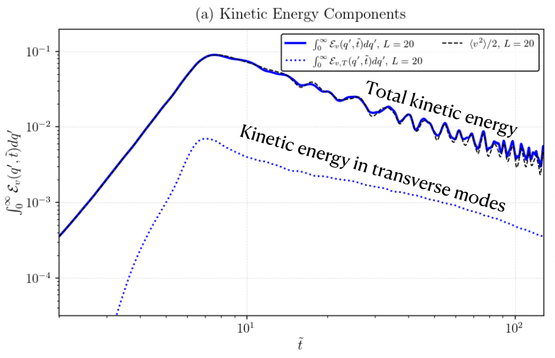


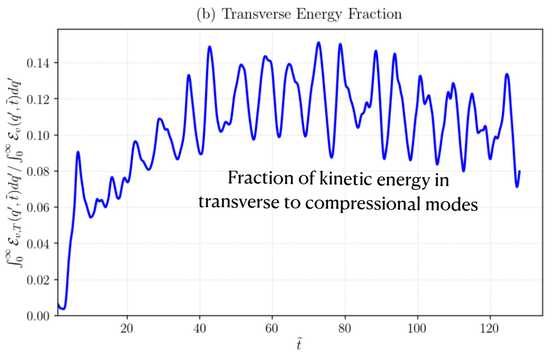




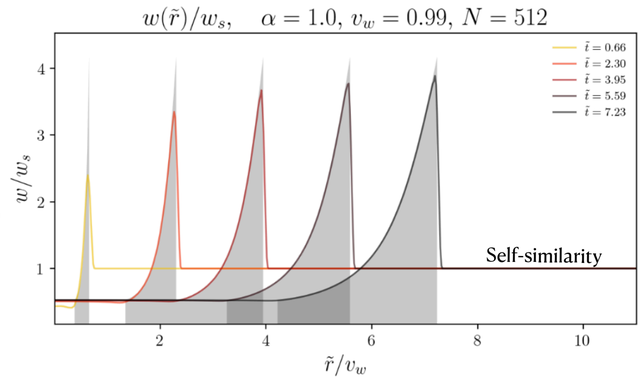
$\alpha = 0.75$
$v_w = 0.6$

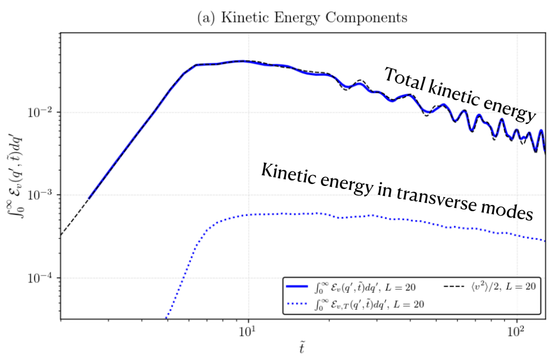


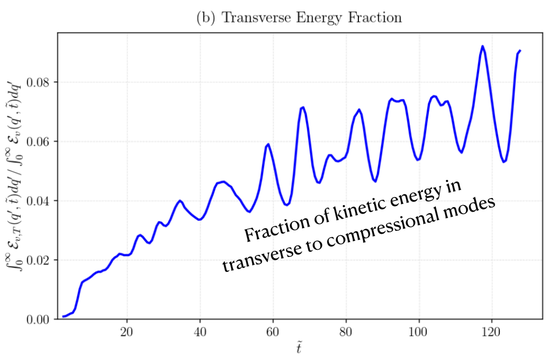




$\alpha = 1.0$
$v_w = 0.99$







Two deflagrations

- **Vorticity generated** in **initial** shell **collisions**
- **Plateaus** after som time
- Reaches a **maximum fraction** $\sim 14\,\%$

Detonation

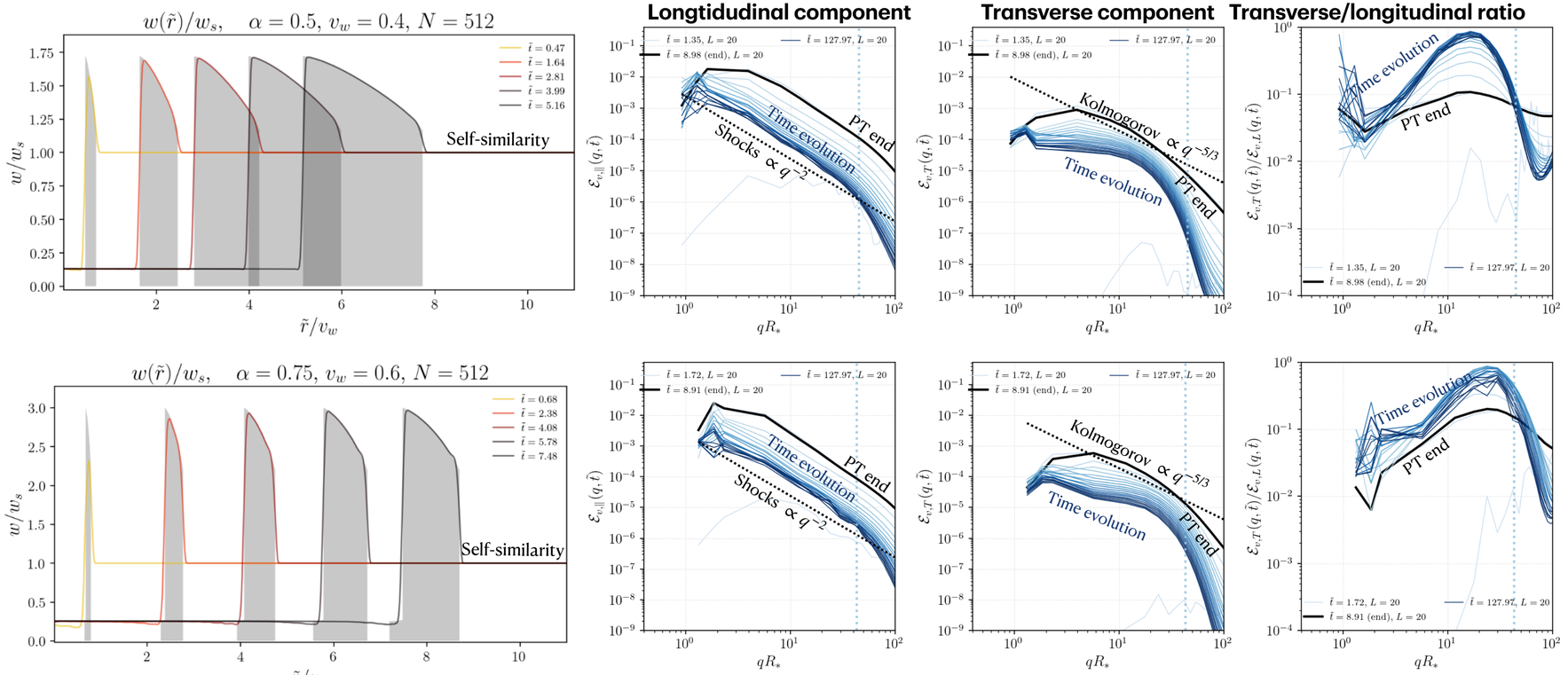
- **Vorticity** is **not produced** in **initial** shell **collisions**
- **Keeps growing**
- **Final fractions** much **lower than** in **deflagrations** of similar α

Results

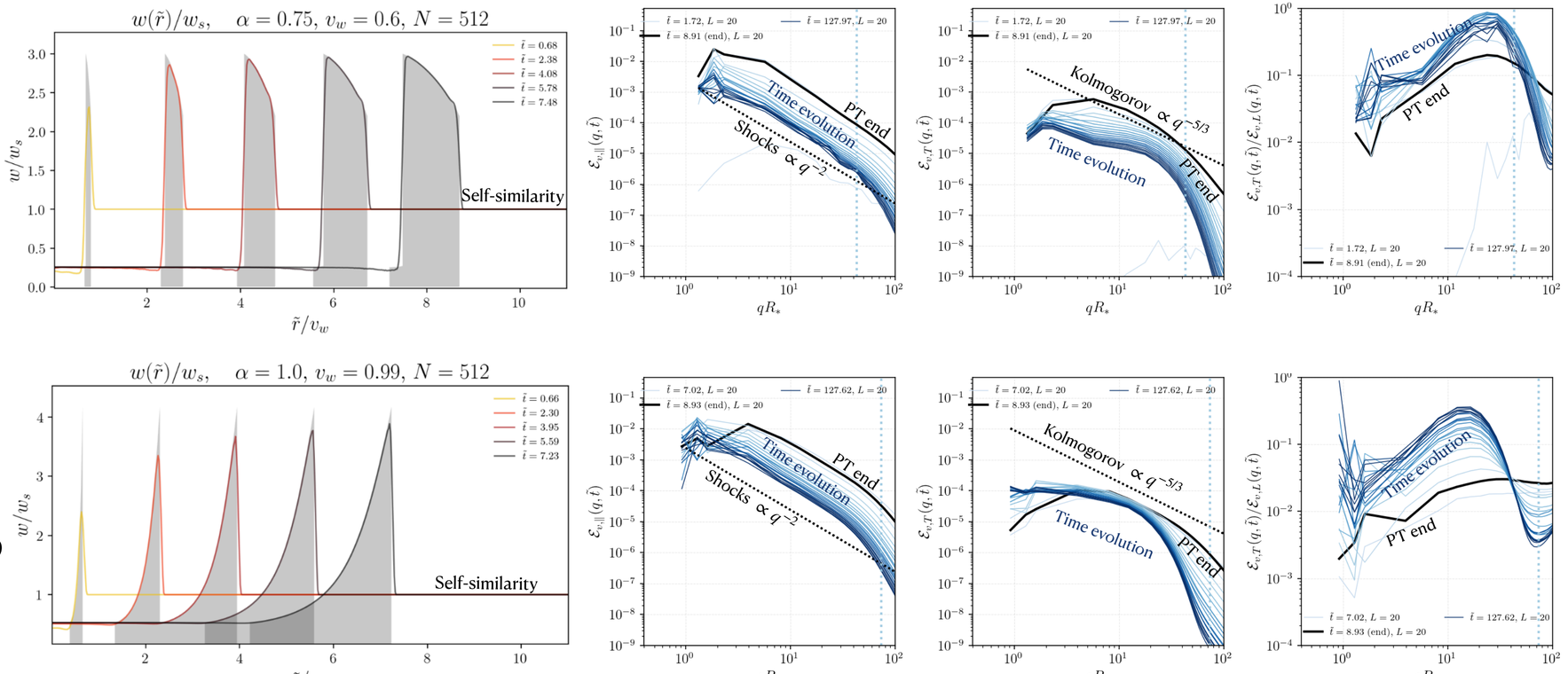
Kinetic energy and velocity spectra

Longtidudinal component | Transverse component | Transverse/longitudinal ratio

$\alpha = 0.5$
$v_w = 0.4$



$\alpha = 0.75$
$v_w = 0.6$



$\alpha = 1.0$
$v_w = 0.99$

Two deflagrations

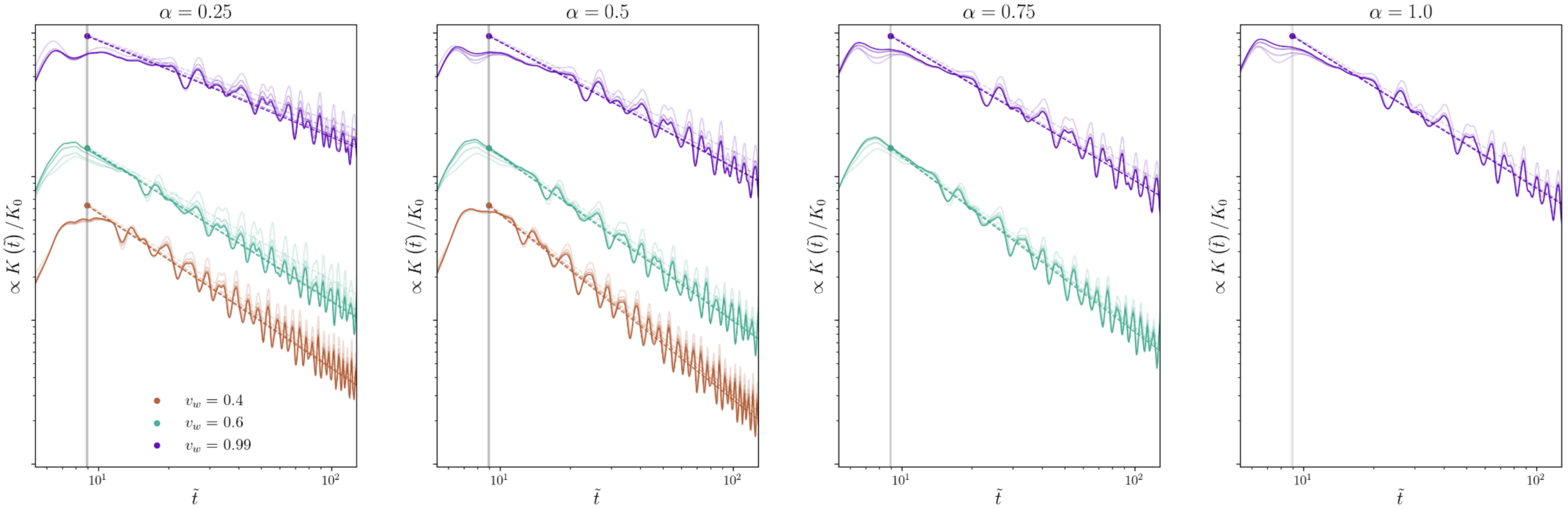
- **Power** in **transverse** and **longitudinal** modes **equal** at certain scales

Detonation

- **Power** in **transverse much smaller** than in **longitudinal** modes at all scales

Results

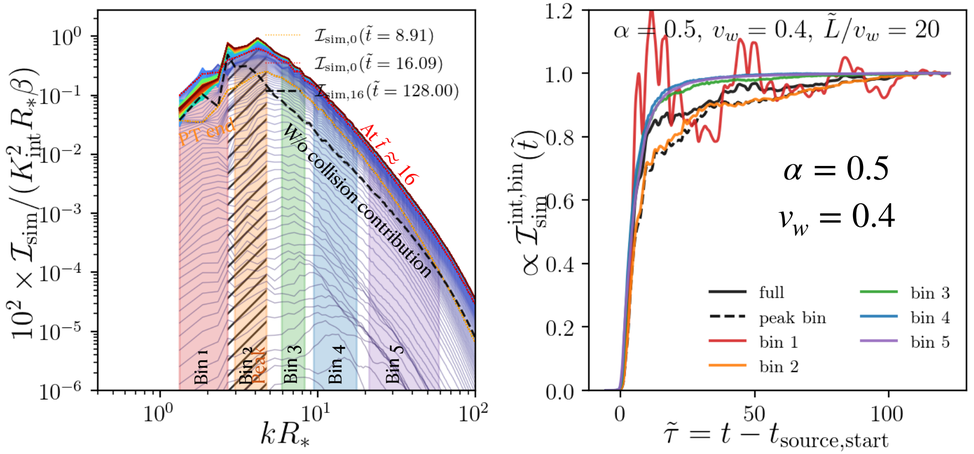
Kinetic energy and spectra

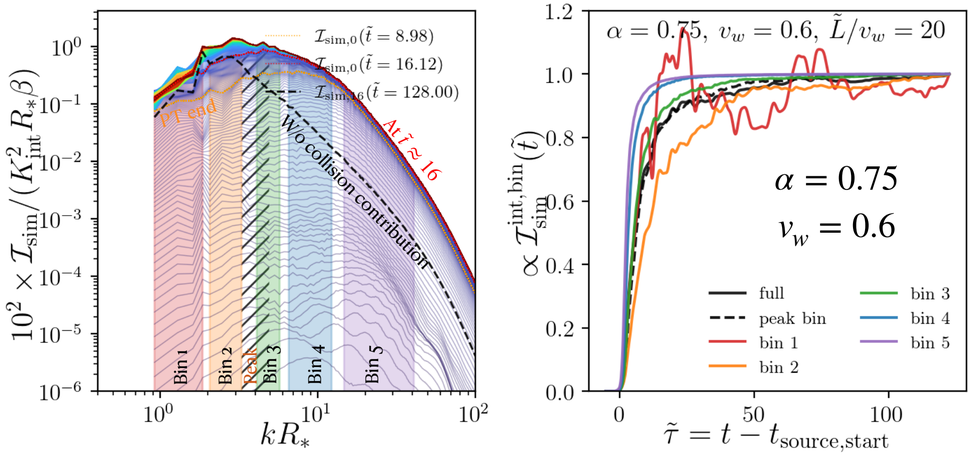


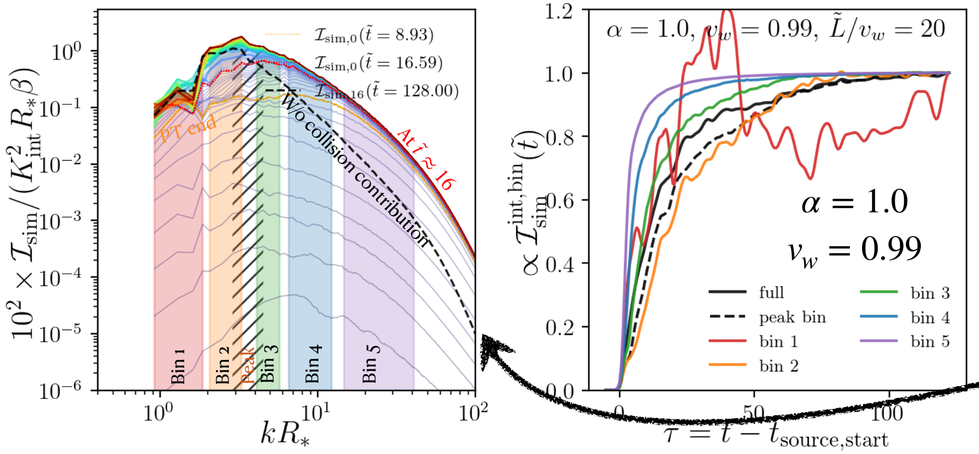
- Observe a clear **power law decay** across all parameters
- **Kinetic energy decayed** by a **factor 30** at $\tilde{t} = 128/\beta$
- **GW amplitude** is nearly fully **saturated** since since K^2 decreased by factor ~ 900

Results

GW spectrum evolution and saturation







- **Spectrum** obtained **from** the **full set** of **dynamical components** *
- GW **spectrum saturation** across all frequencies
- **Dynamically emerged peak** structure
  - Initially, growth at higher f
  - Later, growth at smaller f
  - Substantial differences from the spectrum neglecting initial contribution

Conclusion: need to derive spectra from the full dynamical process until eventual saturation for robust and accurate GW spectrum predictions.

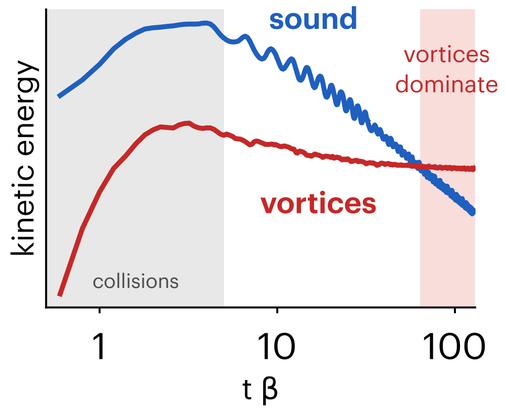
Compare the shape with black-dashed line, neglecting initial $0 < \tilde{t} < 16$

Are we done now?

The price of strategy 2

Longer, bigger and higher resolution, all at once

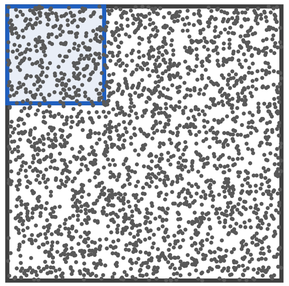
longer



Run until the **GW spectrum saturates**: the sound decays, and the **vortices take over** the source.

×

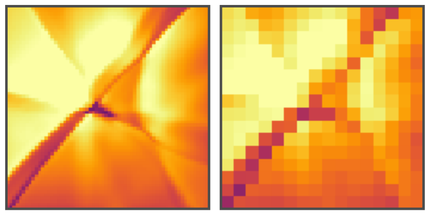
bigger



The **IR tail** and **statistics**: Part 1 used **~2500 bubbles** for the IR, ~300 for the UV.

×

higher resolution



160 cells/bubble 40 cells/bubble

Shells, **shocks** and **vortices**.

Non-linear dynamics → **Vorticity** → **Turbulence** → **GW source**

shell collisions, shocks, shear → made in collisions, then by shocks → vortices merge and cascade over the **whole run** → a **growing share** as the sound decays

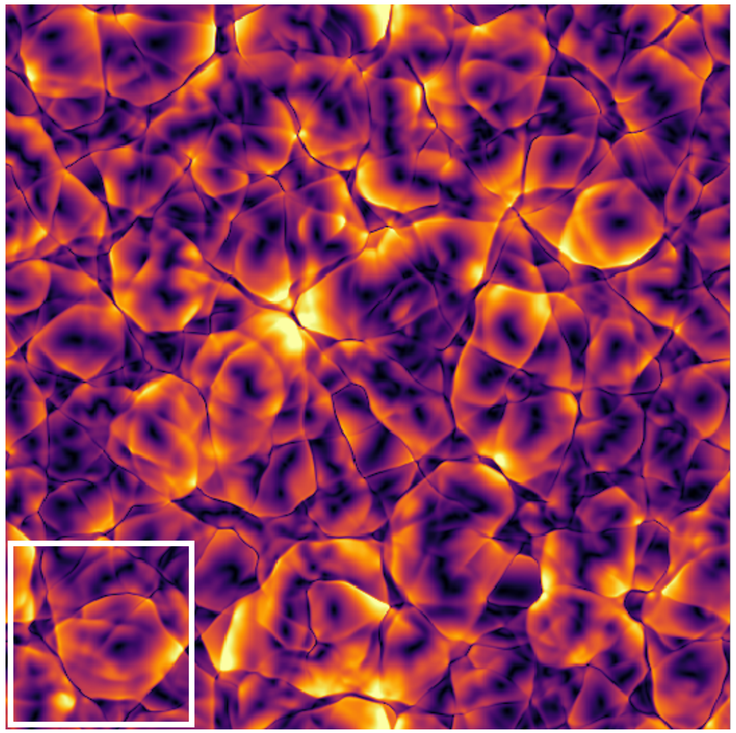
The longer the run, the more of the source the vortices carry, and they are the first thing low resolution smears out.

$$\text{cost} \propto \frac{T}{\Delta t}\left(\frac{L}{\Delta x}\right)^3 \propto \frac{TL^3}{\Delta x^4} \propto \text{resolution}^4 \qquad (\Delta t \propto \Delta x)$$

Strategy 2 could afford **N = 512**, for **selected points only**.

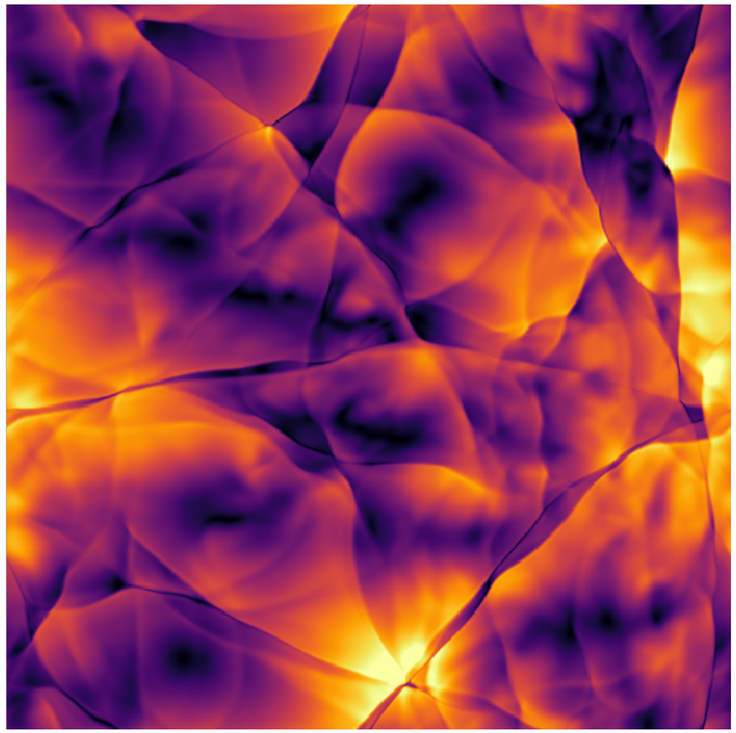
The gap

At the cost strategy 2 can afford: many bubbles or high resolution, not both



either

163 bubbles, 40 cells per bubble

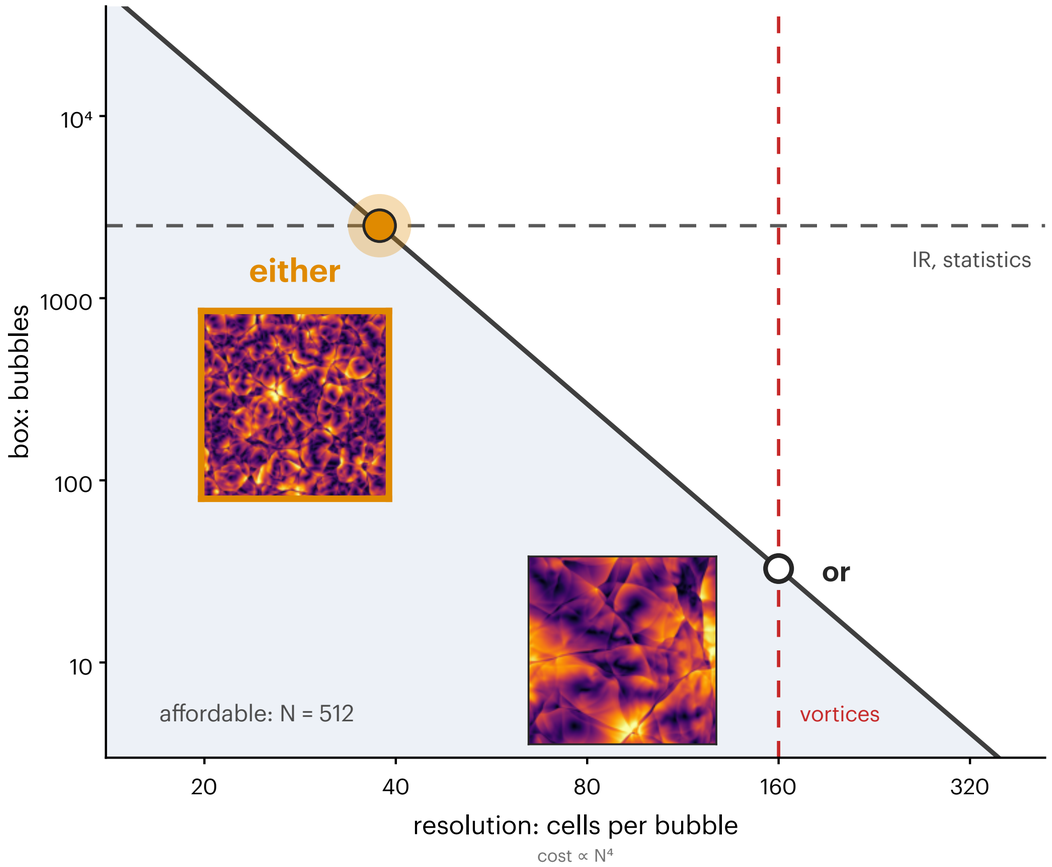


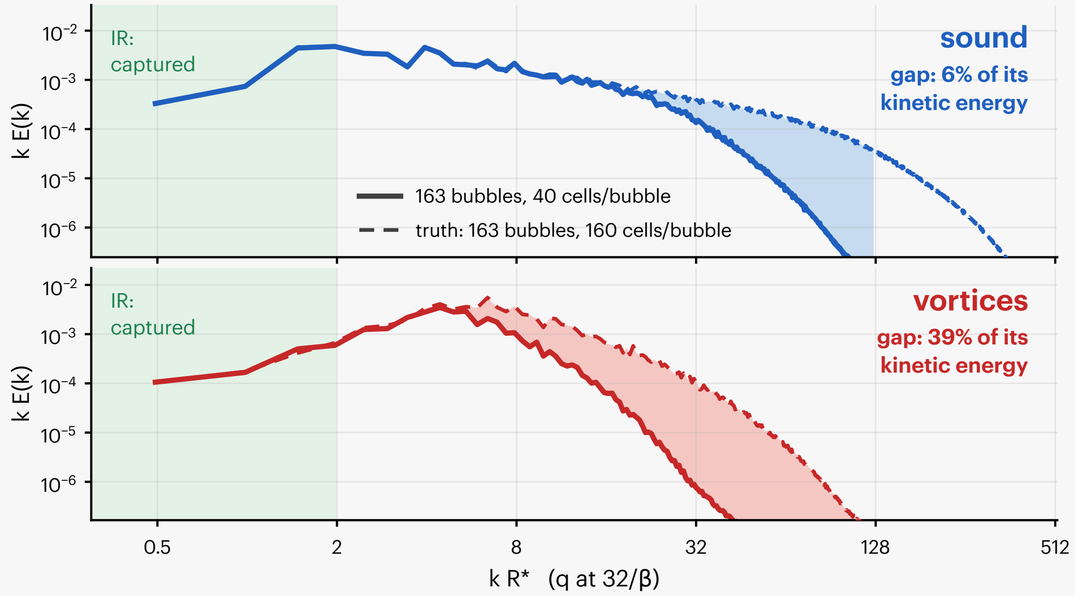
or

10 bubbles, 160 cells per bubble

The gap

At the cost strategy 2 can afford: many bubbles or high resolution, not both





Either: many bubbles

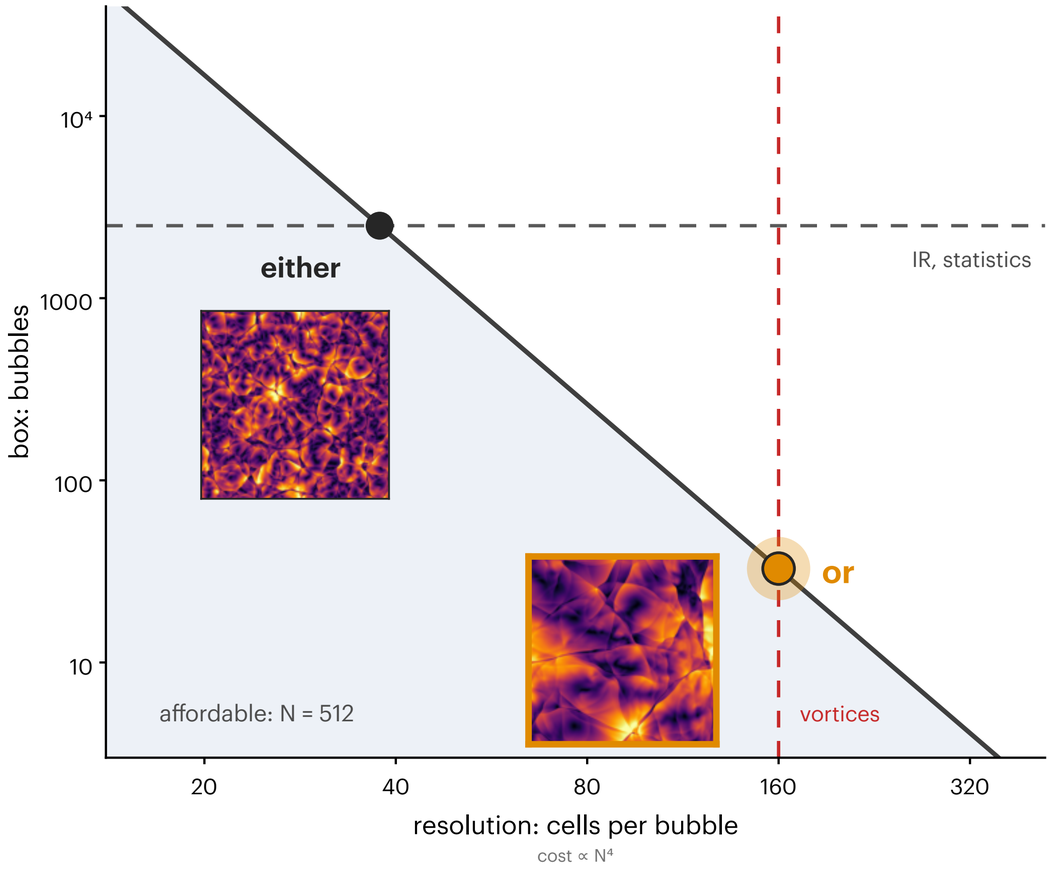
163 bubbles at 40 cells per bubble

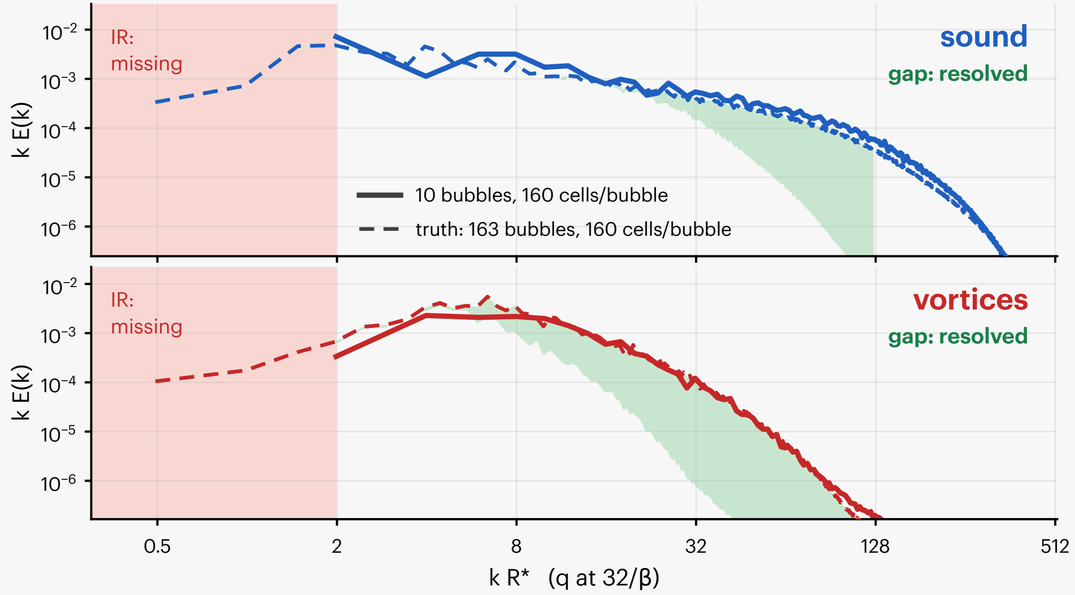
Keeps the IR and the statistics.

Loses the small scales, most of all the vortices: made at 61%, down to 35% of the truth by 128/β, when they carry the source.

The gap

At the cost strategy 2 can afford: many bubbles or high resolution, not both





Or: high resolution

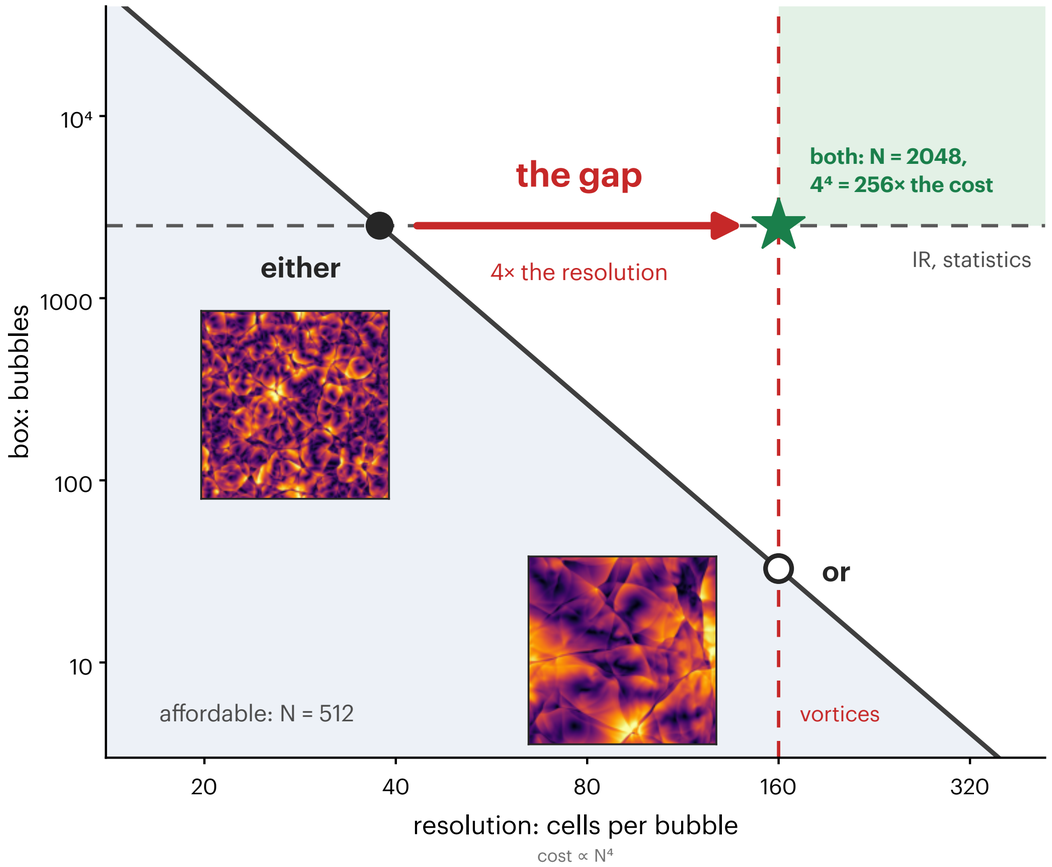
10 bubbles at 160 cells per bubble

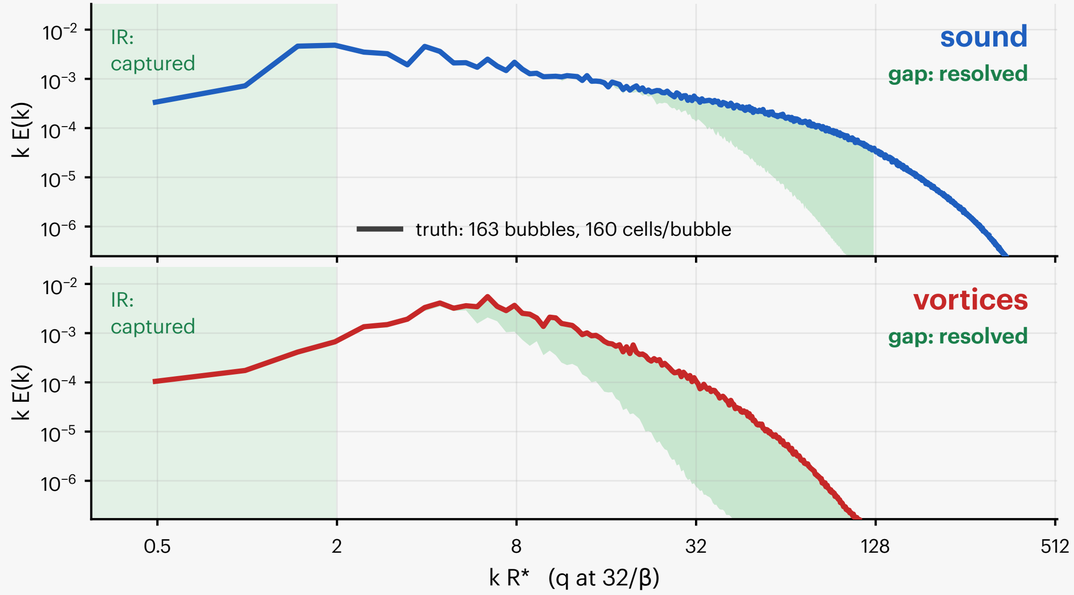
Keeps the small scales and the vortices.

Loses the IR below the bubble scale, and the statistics of many bubbles.

The gap

At the cost strategy 2 can afford: many bubbles or high resolution, not both





Where we want to go: both

163 bubbles at 160 cells per bubble

The IR, the statistics and the vortices, for the whole run.

N = 2048, 4^4 = 256× what strategy 2 can afford.

The gap: from 40 to 160 cells per bubble in a box big enough for the IR, for the whole run

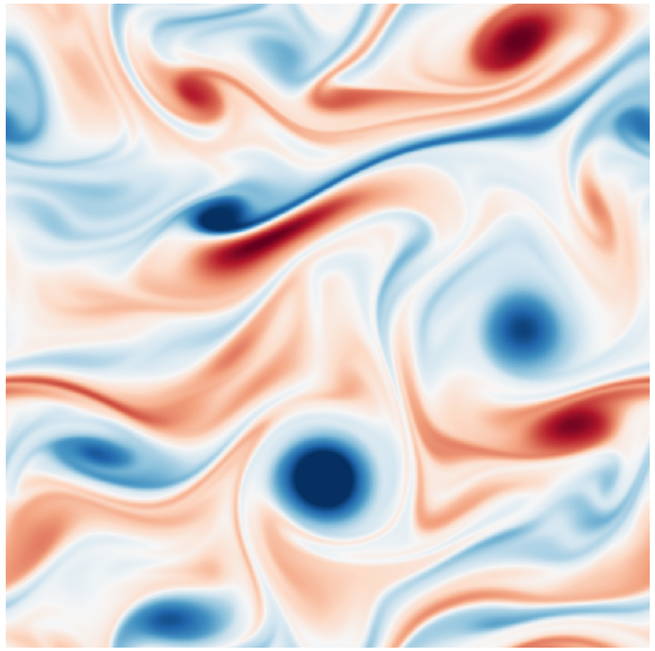
Bridging the gap with learned closure

Based on work in progress

Learning to simulate fluids

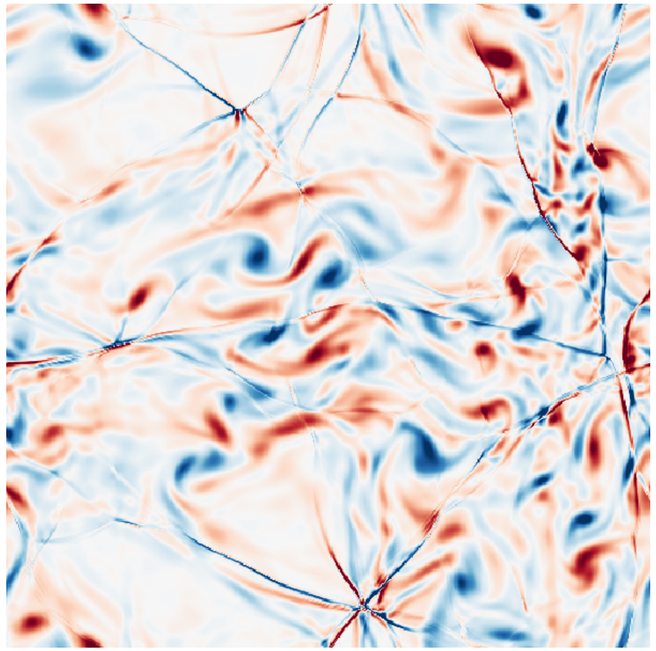
From incompressible turbulence to relativistic fluids

Forced turbulence



smooth vortices, resolved

Phase transition, 5/β



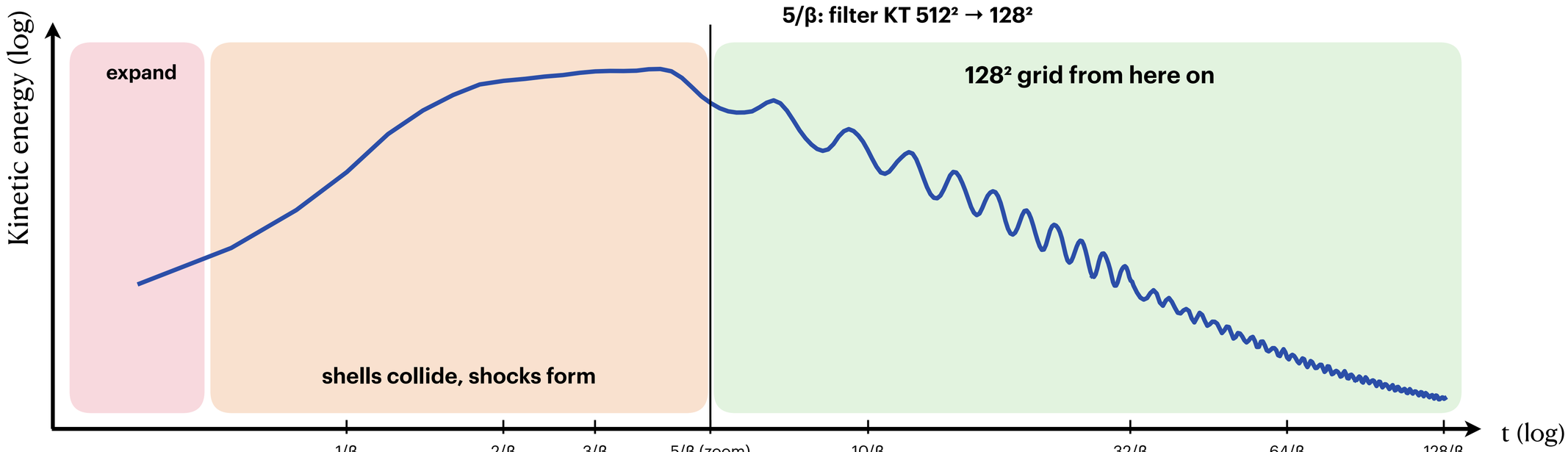
shocks, filaments, 1-cell cores

Kochkov et al. (2021): a **learned correction**, as accurate as **8–10× finer grids**.

D. Kochkov, J. A. Smith, A. Alieva, Q. Wang, M. P. Brenner, S. Hoyer, PNAS 118, e2101784118 (2021)

Which problem to solve?

2D, α = 0.5, vw = 0.4, 10 bubbles: from nucleation to the 128² initial condition



high resolution: KT 512², filtered

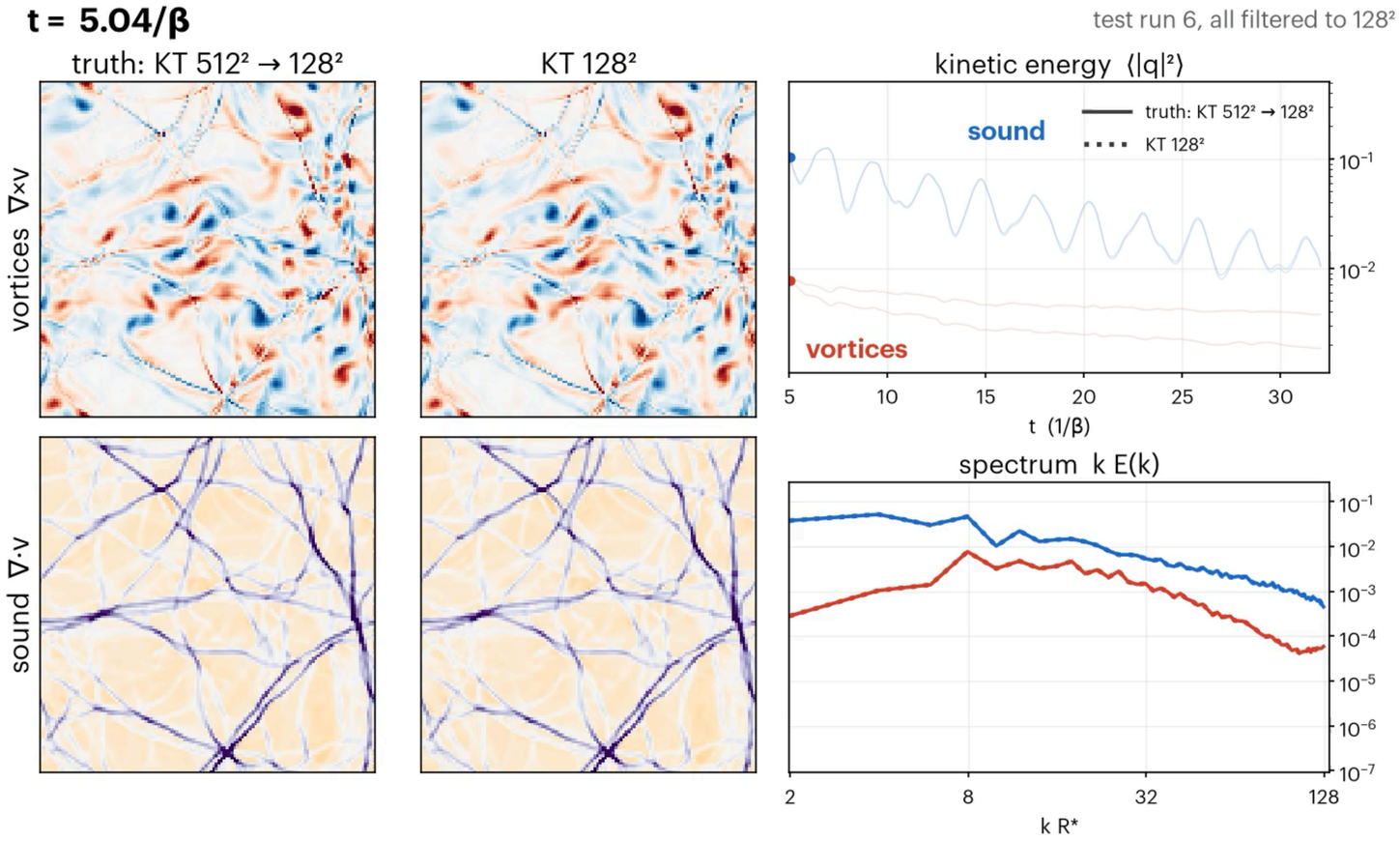
4×4 filter ↓

low resolution: bare KT 128² from 5/β

Vorticity, test run 6, one colour scale per column. The low-resolution run loses the vortices that carry the late source.

What low resolution loses

Truth against bare KT 128²



$q = \sqrt{w}\,\gamma\,\mathbf{v}, \quad |q|^2 = w\gamma^2 v^2$ (kinetic energy density); $\quad q = q^{\mathrm{c}} + q^{\mathrm{s}}, \quad \nabla \times q^{\mathrm{c}} = 0$ (sound), $\quad \nabla \cdot q^{\mathrm{s}} = 0$ (vortices)

What is at stake for GW predictions

Intermission

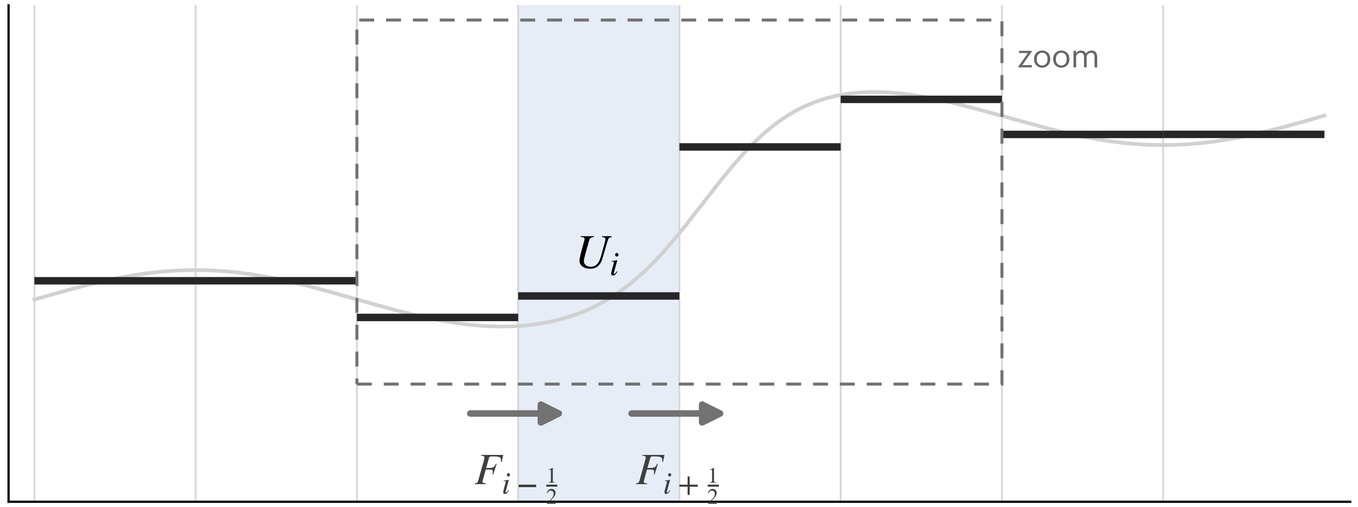
Lost at low resolution (bare KT 128^2)	Effect on GW production	What a closure must deliver
Vortices smeared out 40–45% of the vortical energy gone by $20–32/\beta$	**Turbulent GW source too weak** a growing share of the source as the sound decays	**Keep the vortices** the vortical energy and spectrum of high resolution, at low resolution
Source decays too fast the GW source Π_{TT} at the peak scales 55% low by $32/\beta$	**Amplitude biased low** the signal integrates the source in time: errors accumulate until saturation	**Keep the source alive** dissipate as high resolution does; conserve energy and momentum exactly, stably
Small scales lost ~95% of the GW source Π_{TT} at $kR^* > 32$ gone	**UV tail too steep** its slope set by the resolution, not by the physics	**Restore the small scales** the UV statistics of high resolution: spectra, not exact phases
Smaller box instead? More cells per bubble, but few bubbles, no IR	**IR tail and peak uncertain** they need big boxes and many bubbles; small boxes scatter	**Work in big boxes** local and general: any box size, nucleation history and wall speed

GW spectra need the statistics of the source, not the phases of small vortices: that is what a closure must target.

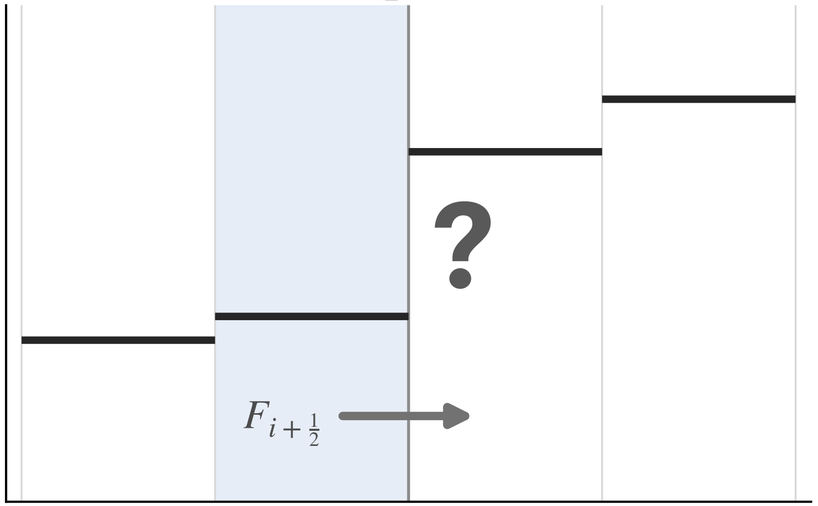
Where a closure acts

A cell changes only by what flows through its faces

cell averages and the fluxes through their faces



face $i+\frac{1}{2}$, zoomed



$$\partial_t U + \partial_x F(U) = 0$$

continuous

average over cell i

$$\frac{dU_i}{dt} = -\frac{F_{i+\frac{1}{2}} - F_{i-\frac{1}{2}}}{\Delta x}$$

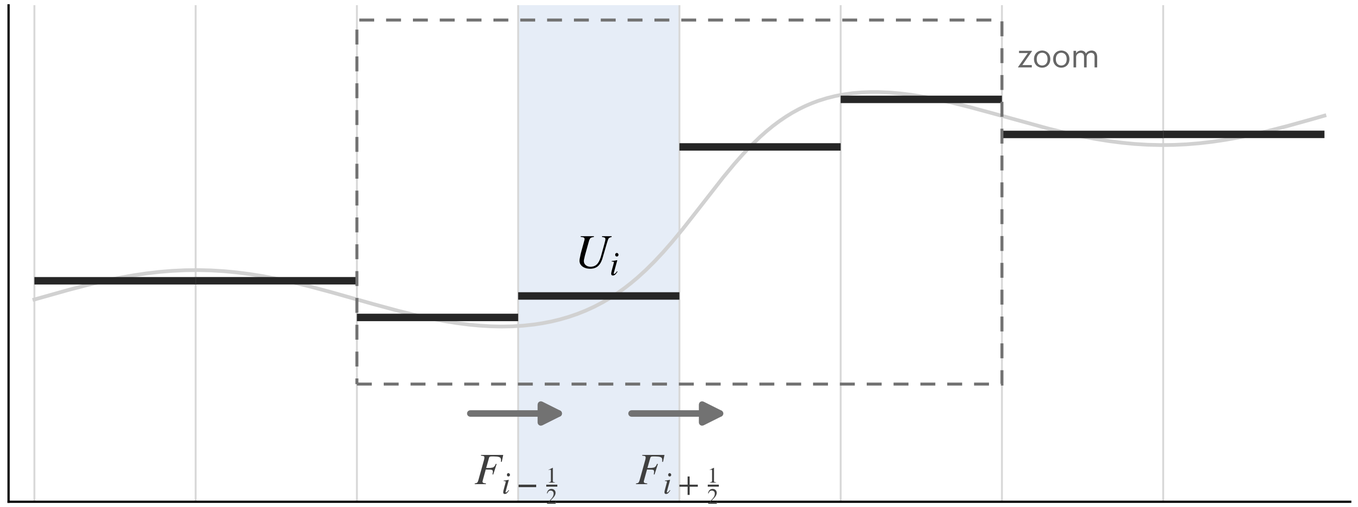
semi-discrete

The **face flux** is unknown, and a **scheme** must reconstruct it.

Where a closure acts

A cell changes only by what flows through its faces

cell averages and the fluxes through their faces



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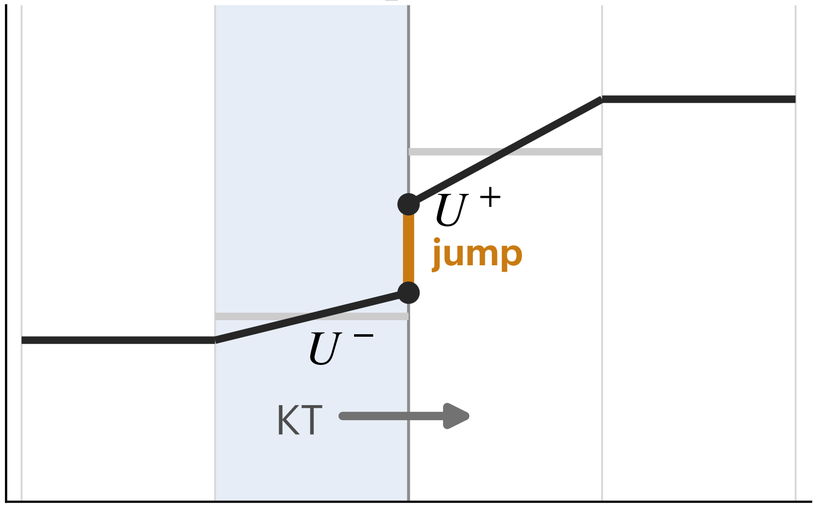


Kurganov & Tadmor 1999

$$\partial_t U + \partial_x F(U) = 0$$

continuous

average over cell i →

$$\frac{dU_i}{dt} = -\frac{F_{i+\frac{1}{2}} - F_{i-\frac{1}{2}}}{\Delta x}$$

semi-discrete

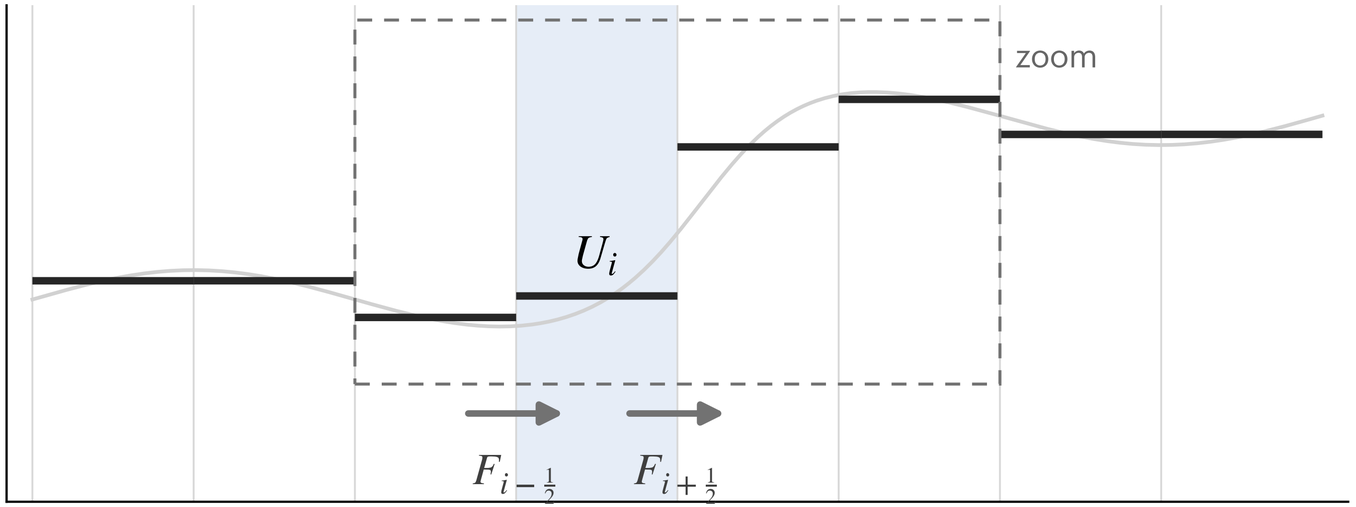
$$F_{i+\frac{1}{2}} = \frac{F(U^-) + F(U^+)}{2} - \frac{a}{2}\left(U^+ - U^-\right)$$

KT damps **sound and shear jumps alike**, at the **fastest signal speed a**: **vortices** are damped like sound.

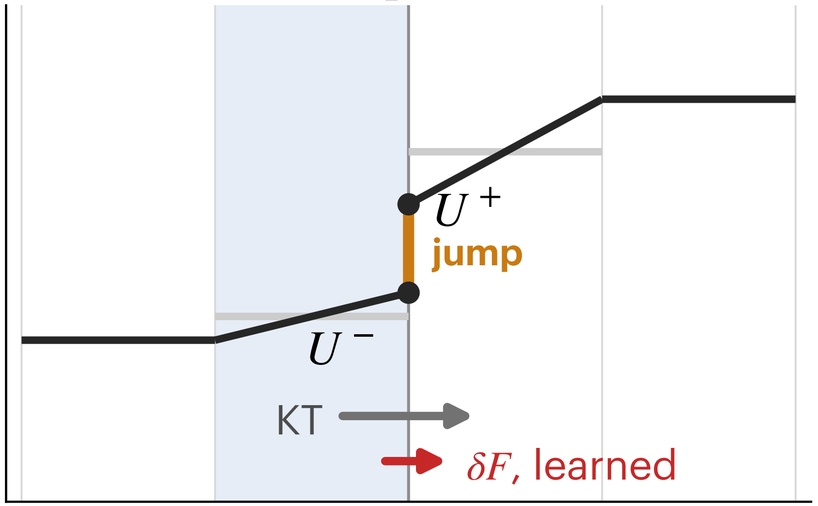
Where a closure acts

Finite volumes: a cell changes only by what flows through its faces

cell averages and the fluxes through their faces



face $i+\frac{1}{2}$, zoomed



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semi-discrete

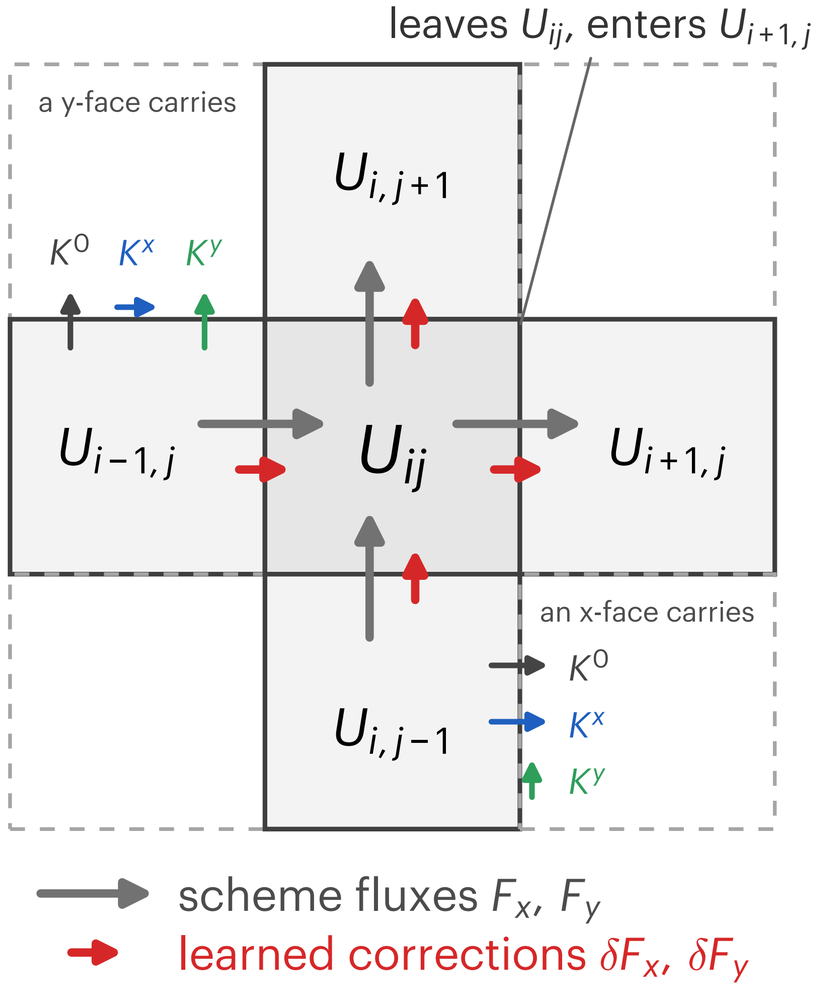
$$F_{i+\frac{1}{2}} = \frac{F(U^-) + F(U^+)}{2} - \frac{a}{2}\left(U^+ - U^-\right)$$

$$F_{i+\frac{1}{2}} \;\to\; F_{i+\frac{1}{2}} + \delta F_{i+\frac{1}{2}}(U;\theta)$$

From the cells around it, a **learned closure** corrects the **face flux** towards the **high-resolution truth**.

A learned flux correction

The same in 2D: a learned δF on every x- and y-face



$$\frac{dU_{ij}}{dt} = -\frac{F_{x,i+\frac{1}{2},j} - F_{x,i-\frac{1}{2},j}}{\Delta x} - \frac{F_{y,i,j+\frac{1}{2}} - F_{y,i,j-\frac{1}{2}}}{\Delta y}$$

$$F_x \to F_x + \delta F_x(U;\theta), \qquad F_y \to F_y + \delta F_y(U;\theta)$$

U and the fluxes, as on slide 23, now in 2D

$$U = \left(K^0,\, K^x,\, K^y\right)^{\mathsf{T}}$$

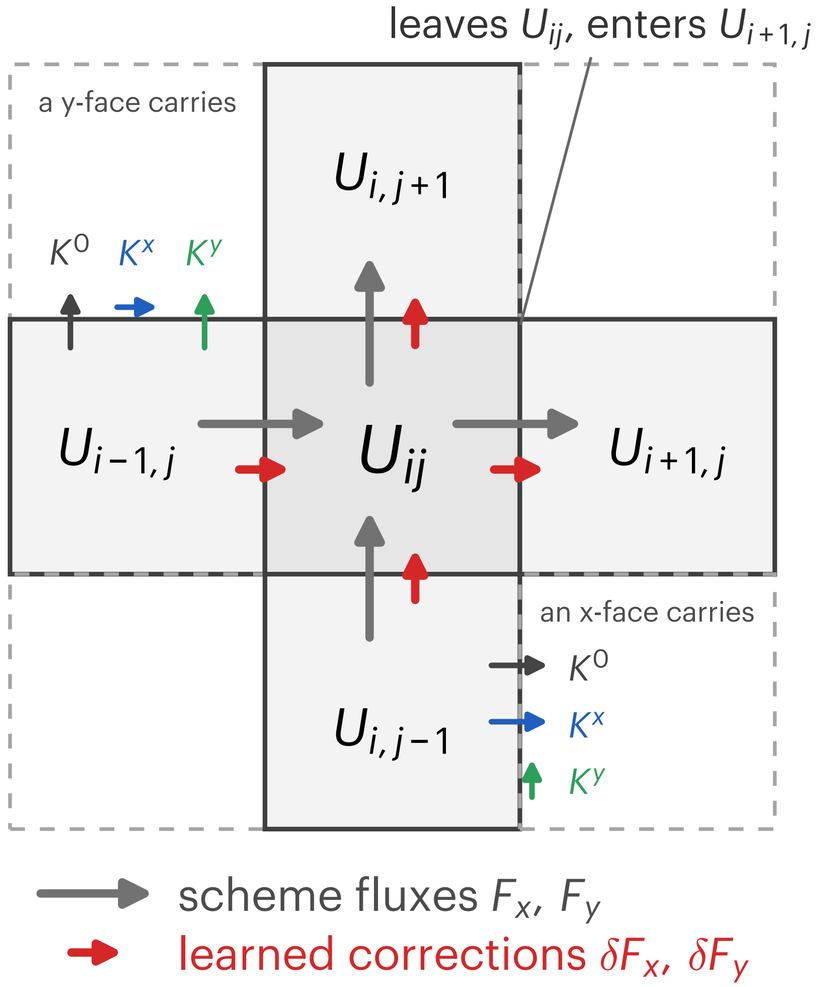
$$F_x = \left(K^x,\, T^{xx},\, T^{xy}\right)^{\mathsf{T}}$$

$$F_y = \left(K^y,\, T^{yx},\, T^{yy}\right)^{\mathsf{T}}$$

$$T^{ij} = \frac{K^i K^j}{K^0 + p} + p\,\delta^{ij}$$

A learned flux correction

The same in 2D: a learned δF on every x- and y-face



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$$F_x \to F_x + \delta F_x(U;\theta), \qquad F_y \to F_y + \delta F_y(U;\theta)$$

Scheme-agnostic
any scheme supplies the face fluxes (here KT), any time integrator

A rate, not an increment

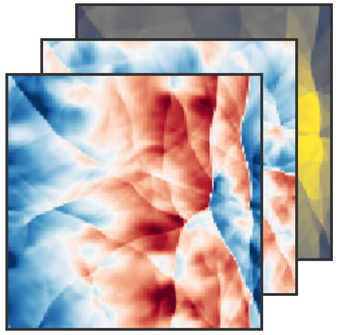
independent of Δt, added wherever the scheme evaluates its fluxes

Conservative by construction
each face flux leaves one cell and enters the next: energy and momentum exact

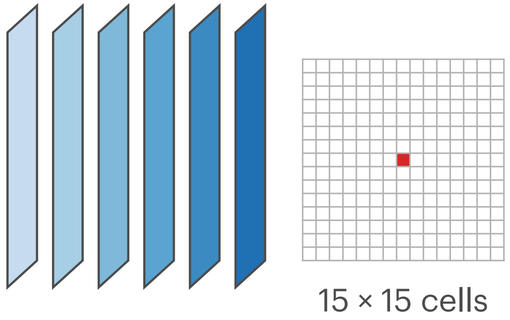
The network, input and output

From the low-resolution state to a safe face correction

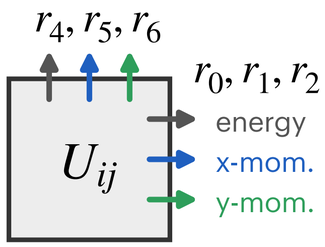
10 inputs per cell



A small CNN



6 dimensionless outputs per cell



Local flux scales

energy
$\times\, w(|v| + c_s)$
momentum
$\times\, w(|v| + c_s)^2$

Flux correction

$\delta F_x,\ \delta F_y$

γv, $\ln(w/\langle w \rangle)$, $\gamma - 1$ and one-cell **slopes**: **dimensionless** and bounded

6 layers of 3×3 convolutions, 48 channels, 90k parameters; each output **sees 15×15 cells**

Dimensionless order-one corrections on its two upper faces, for **energy** and both **momentum fluxes**

outputs become **fluxes**, in quiet regions and shocks alike

x and **y flux corrections**

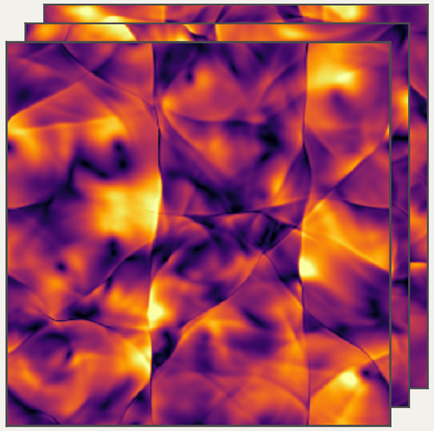
$$\delta F_x = \left(w(|v| + c_s)\, r_0,\ w(|v| + c_s)^2\, r_1,\ w(|v| + c_s)^2\, r_2 \right)$$

$$\delta F_y = \left(w(|v| + c_s)\, r_3,\ w(|v| + c_s)^2\, r_4,\ w(|v| + c_s)^2\, r_5 \right)$$

The training data

From high-resolution runs to the coarse snapshots the closure learns from

1 **28 runs, KT 512²**



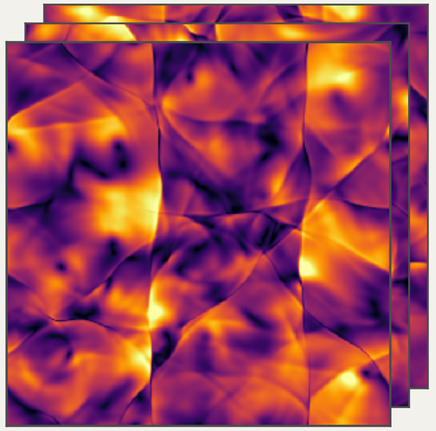
each from nucleation, with
~10 bubbles at random places

28 runs at **high resolution**, KT 512², each from nucleation with its own bubbles.

The training data

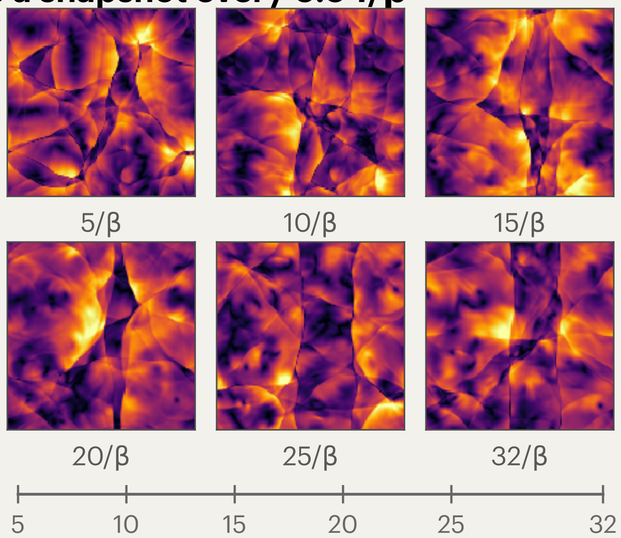
From high-resolution runs to the coarse snapshots the closure learns from

1 28 runs, KT 512²



each from nucleation, with
~10 bubbles at random places

2 Keep a snapshot every 0.04/β



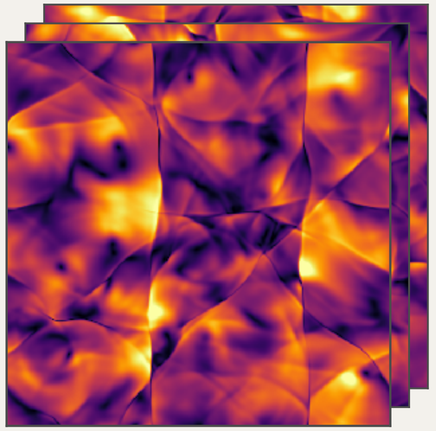
684 snapshots per run, 32 fine or 8 coarse steps apart

From each run we keep **a snapshot every 0.04/β**: 32 fine or 8 coarse steps apart.

The training data

From high-resolution runs to the coarse snapshots the closure learns from

❶ 28 runs, KT 512²



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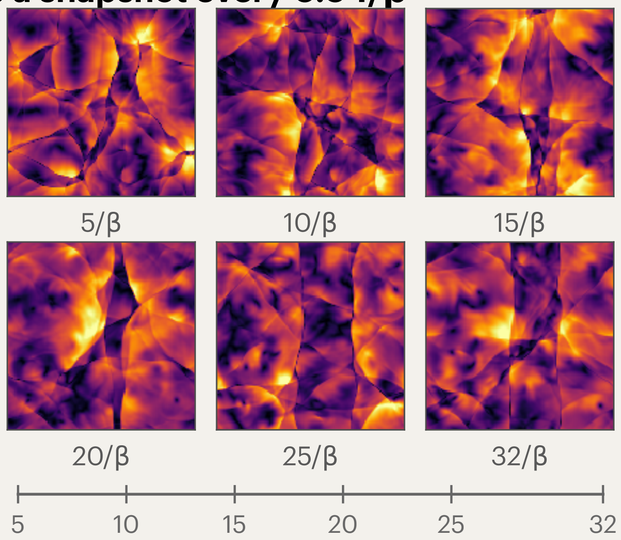
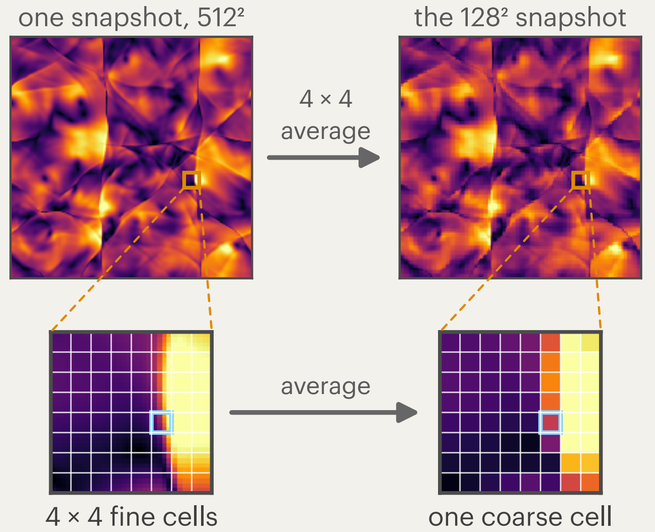


684 snapshots per run, 32 fine or 8 coarse steps apart

❸ Average each onto the 128² grid



the cell averages: the finite-volume state of the coarse grid

Each snapshot is averaged over **4 × 4 fine cells**: the **128² cell averages**, the coarse state.

The training data

From high-resolution runs to the coarse snapshots the closure learns from

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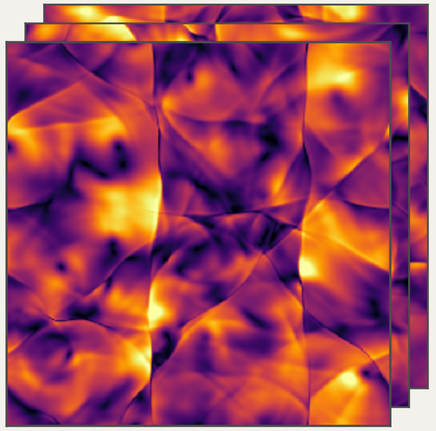

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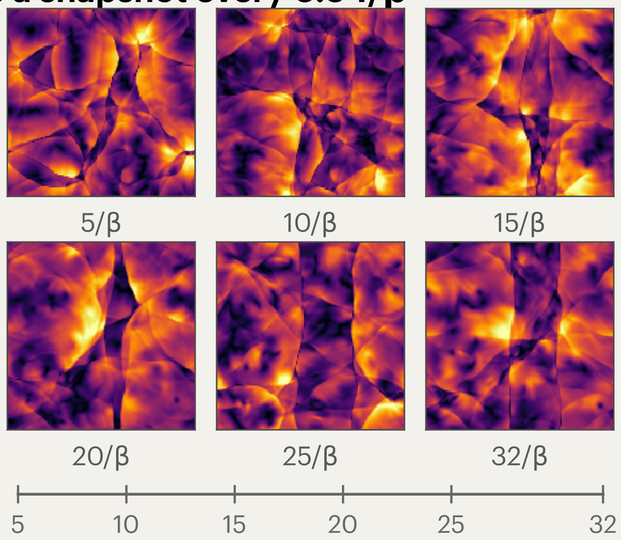

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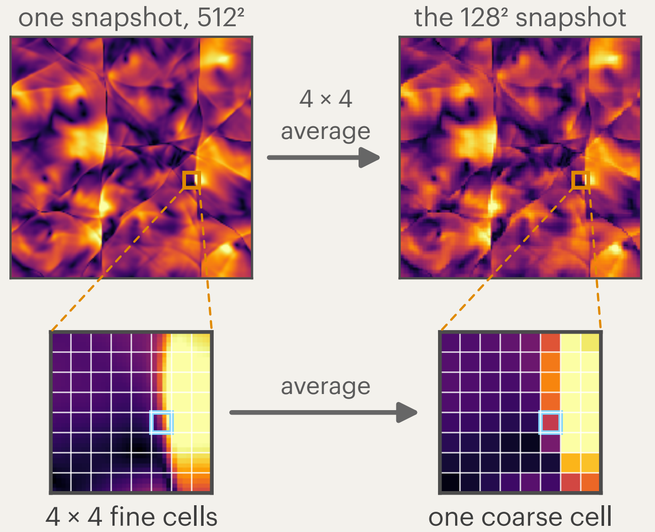
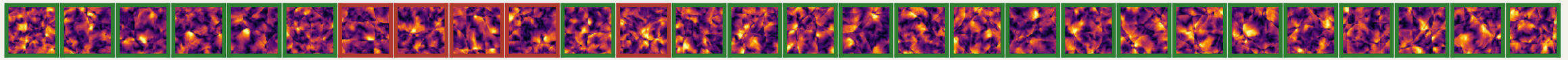


the cell averages: the finite-volume state of the coarse grid

The 28 runs, split once: 23 for training 5 for validation, never trained on



The runs are split once: **23 for training**, **5 for validation**, **never trained on**.

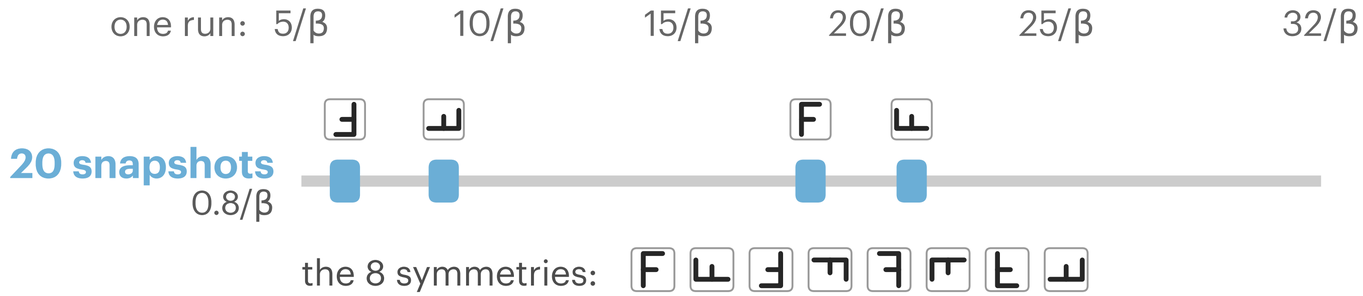
The curriculum

Short windows first, longer ones as training progresses

1) Learn short-window corrections (20 snapshots)

repeat

- pick a random **training run**, **start** in 5–32/β, and **symmetry**
- start the coarse solver **on that truth snapshot**; **correct at every time step**
- **compare with the truth** at every snapshot: $(U - U^{\text{true}})/\sigma$
- the **gradient of the loss averaged over the window** → **one weight update**
- teaches **pointwise accuracy** and **short horizons**



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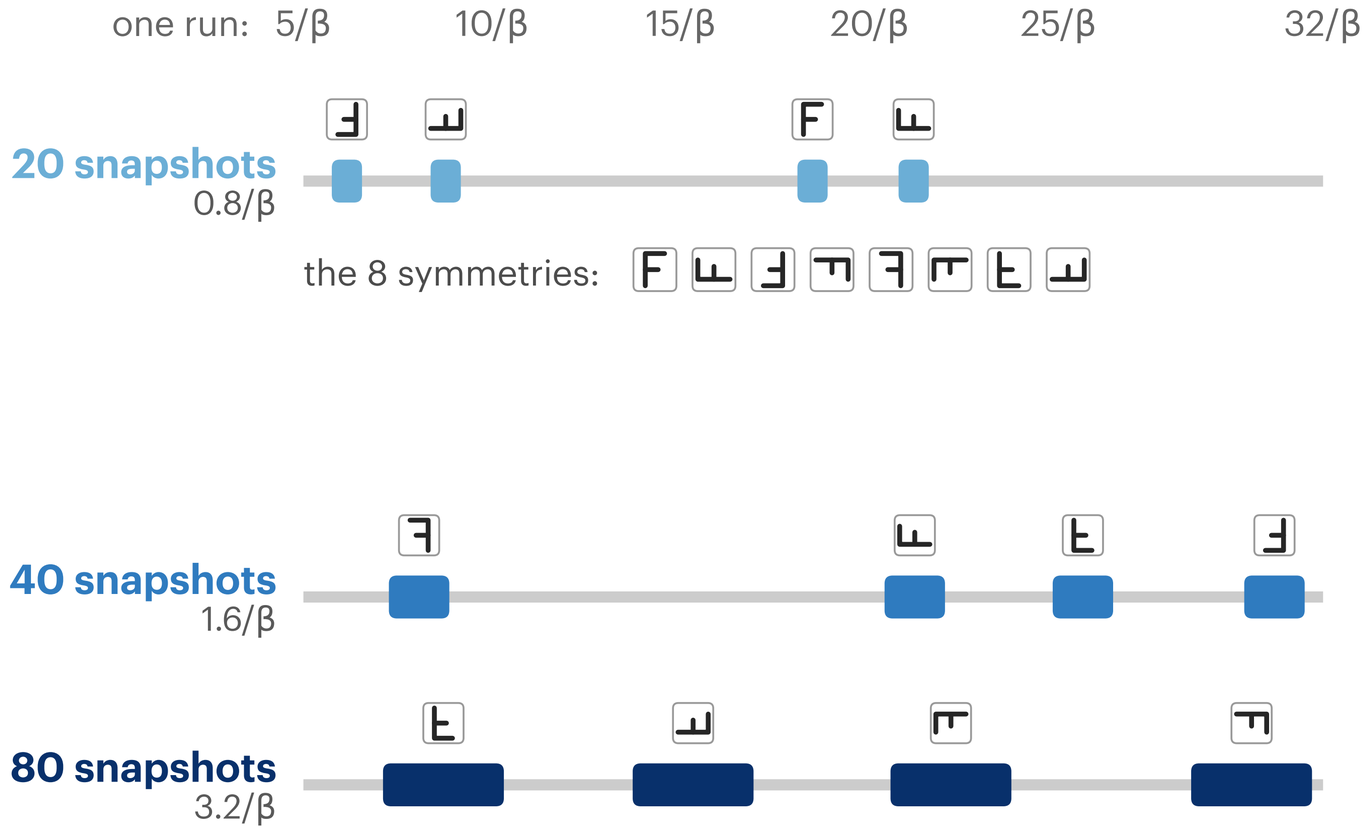
2) Same, with longer windows: 40, then 80

repeat

- each from the **best network so far** (lowest error on the held-out run)
- teaches **stability** and the **slow drift**
- until the **horizons stop growing**



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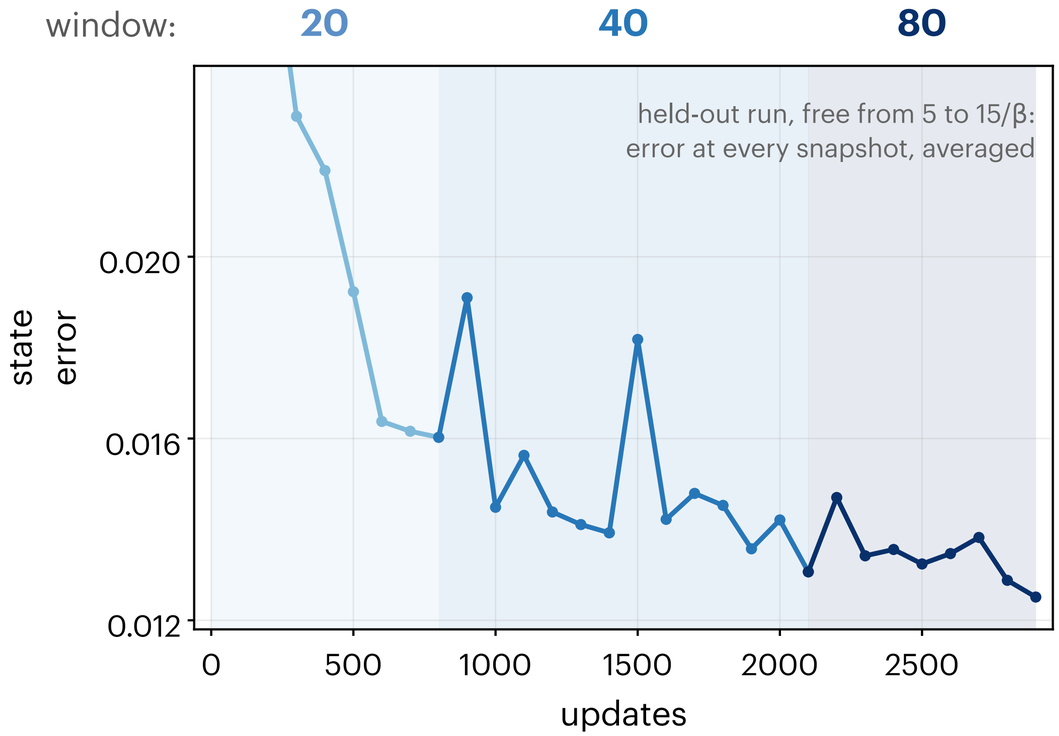
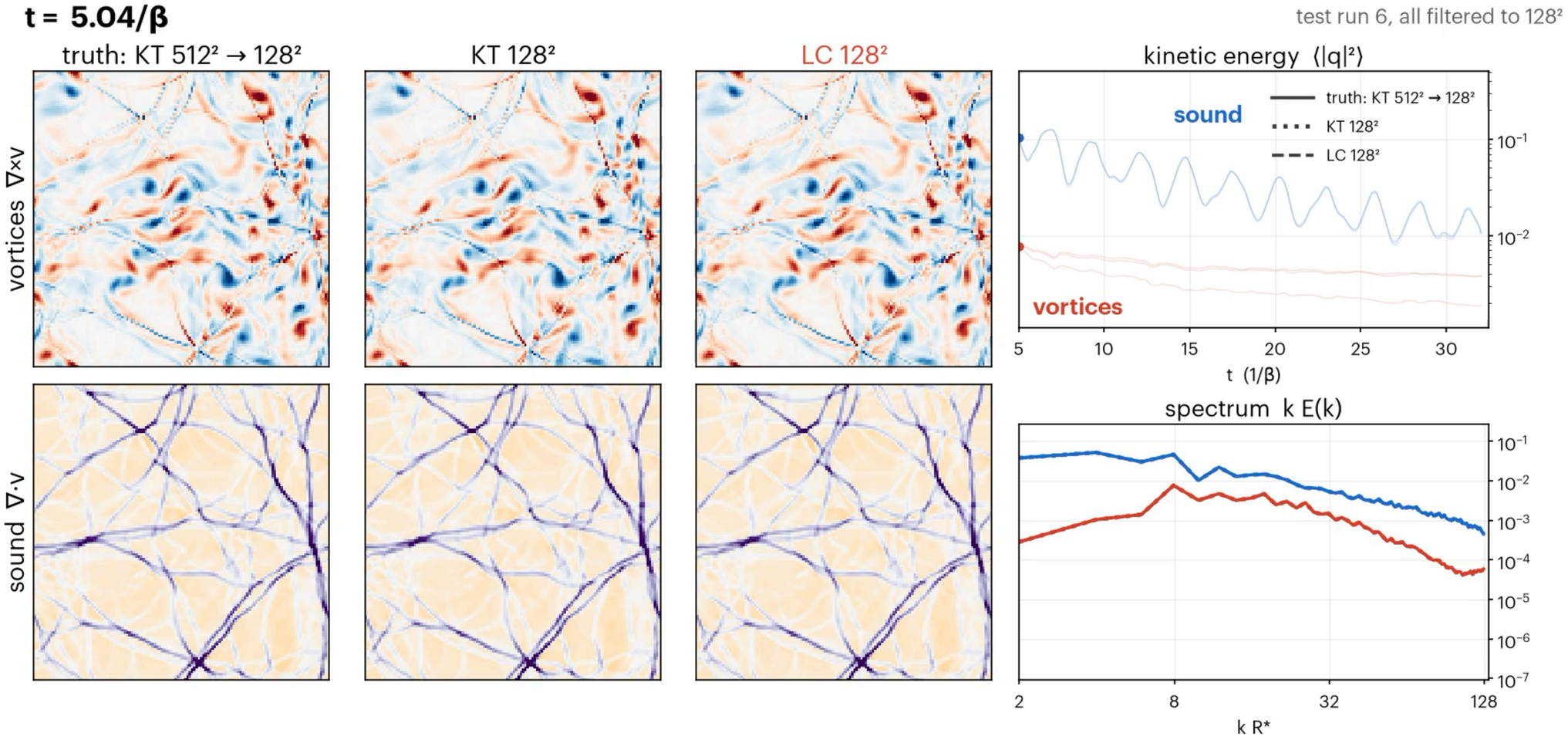


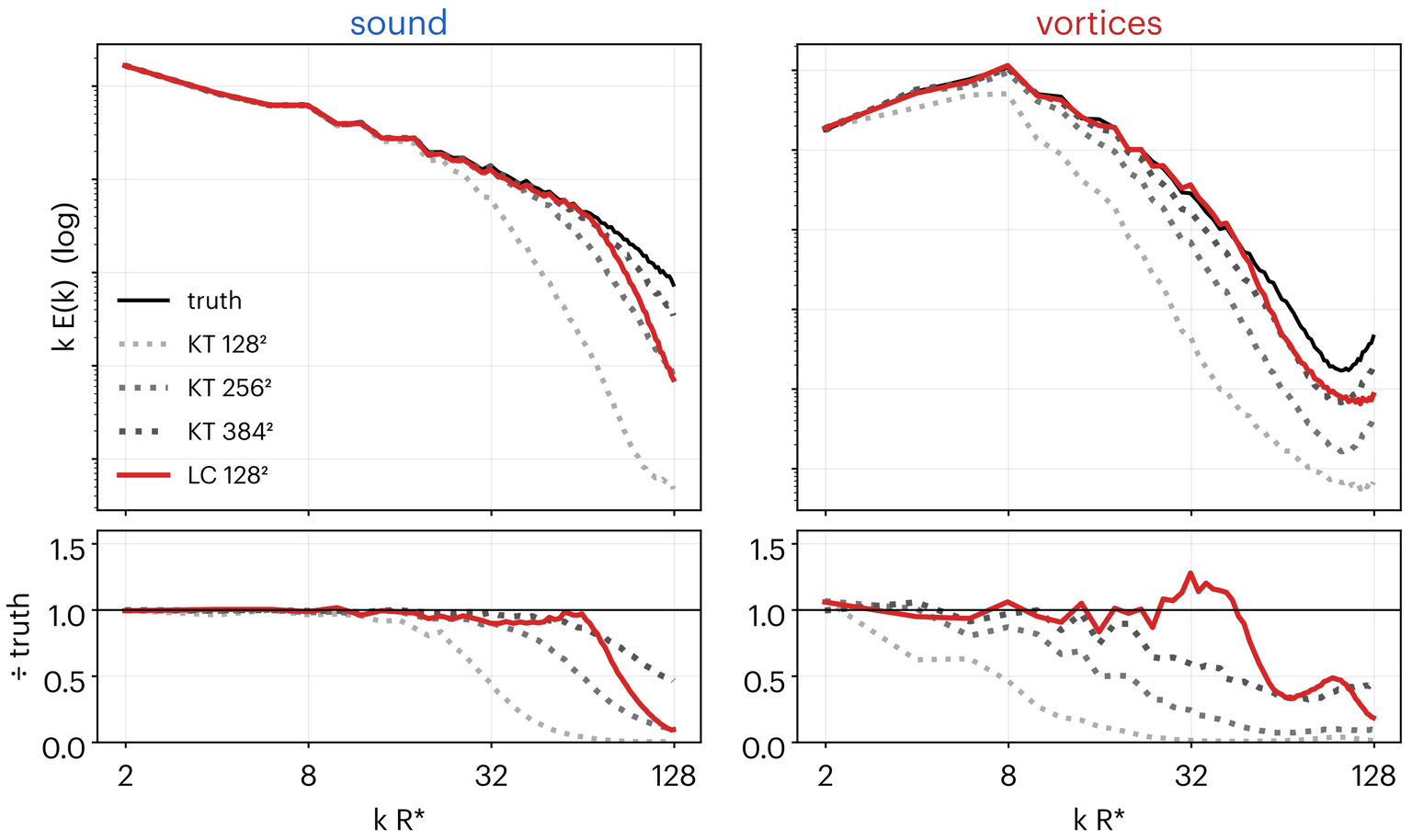
Results: truth, KT 128² and LC

Sound and vortices kept, in energy and spectrum



Results: truth, KT 128² and LC

At 32/β, the end of the training horizon

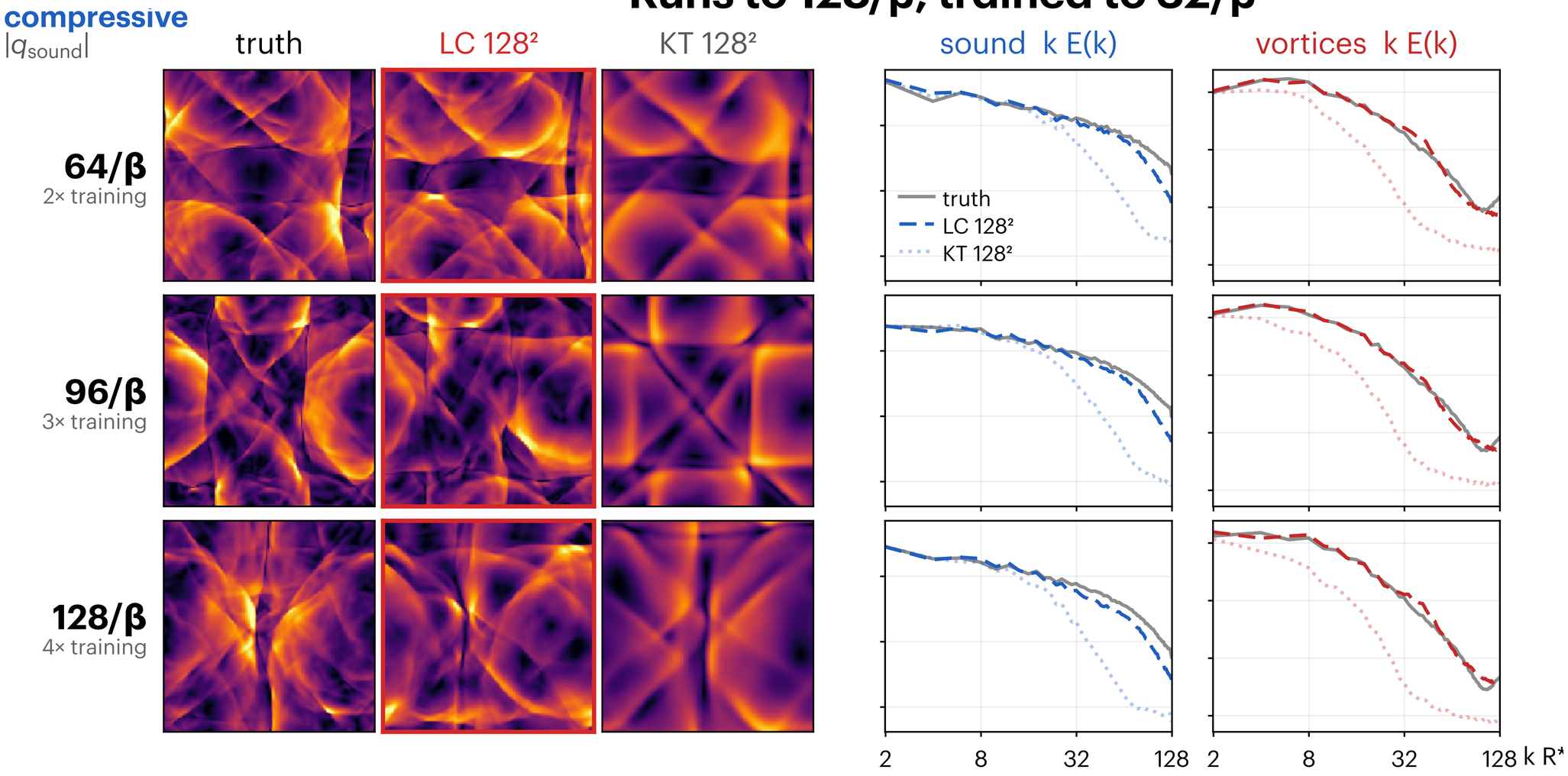


LC keeps the sound to k R* ≈ 70 (like KT 384²) and the vortices to ≈ 50 (≤ 30% above the truth); KT 384² loses the vortices from 16, bare KT 128² from 4 · mean of 4 validation runs

How well does it generalize to unseen physics?

Four times longer

Runs to 128/β, trained to 32/β



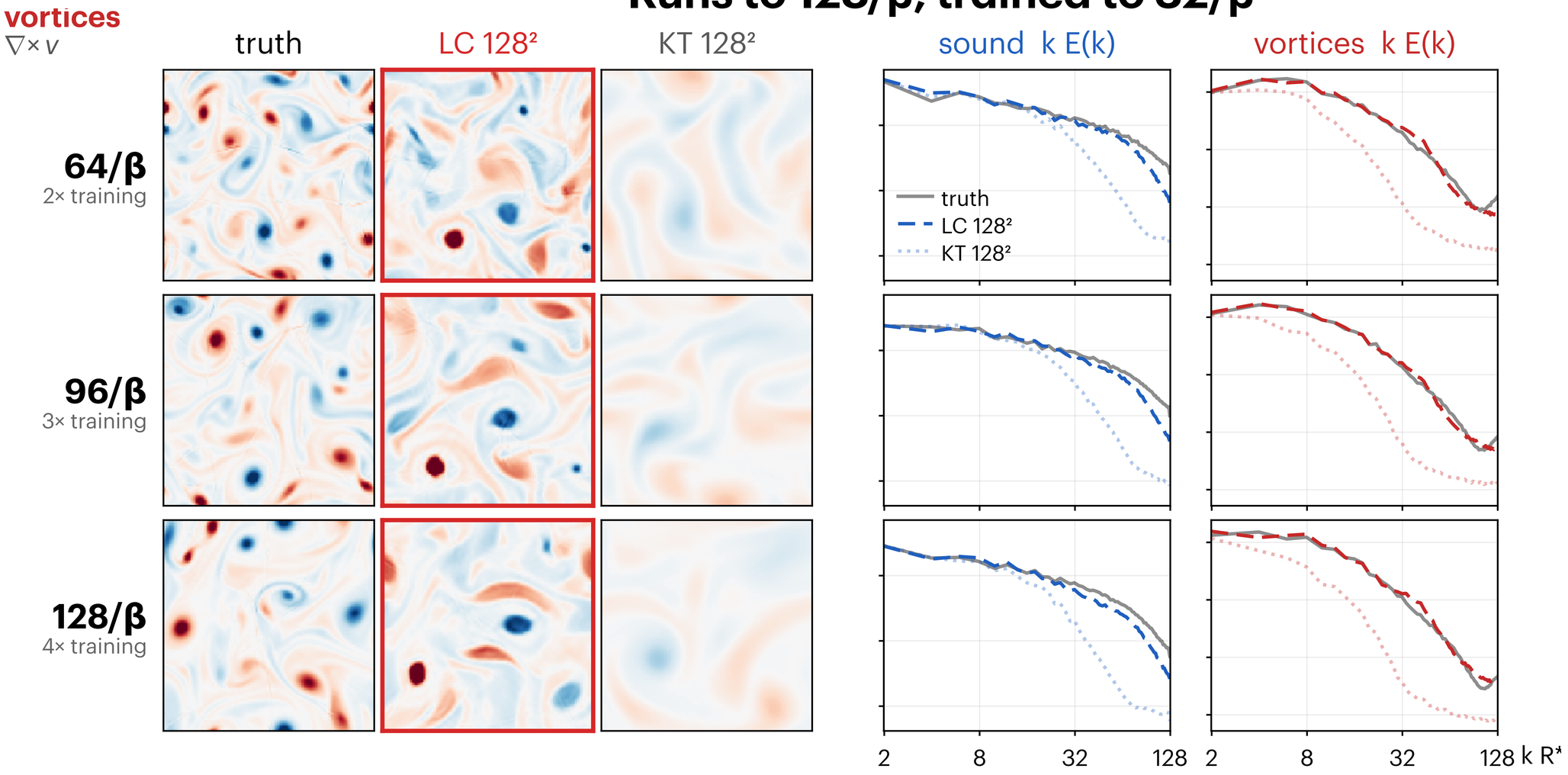
Stable
and **generalizes**:

four times its training horizon,

into **damped flow conditions**
it never saw.

Four times longer

Runs to 128/β, trained to 32/β



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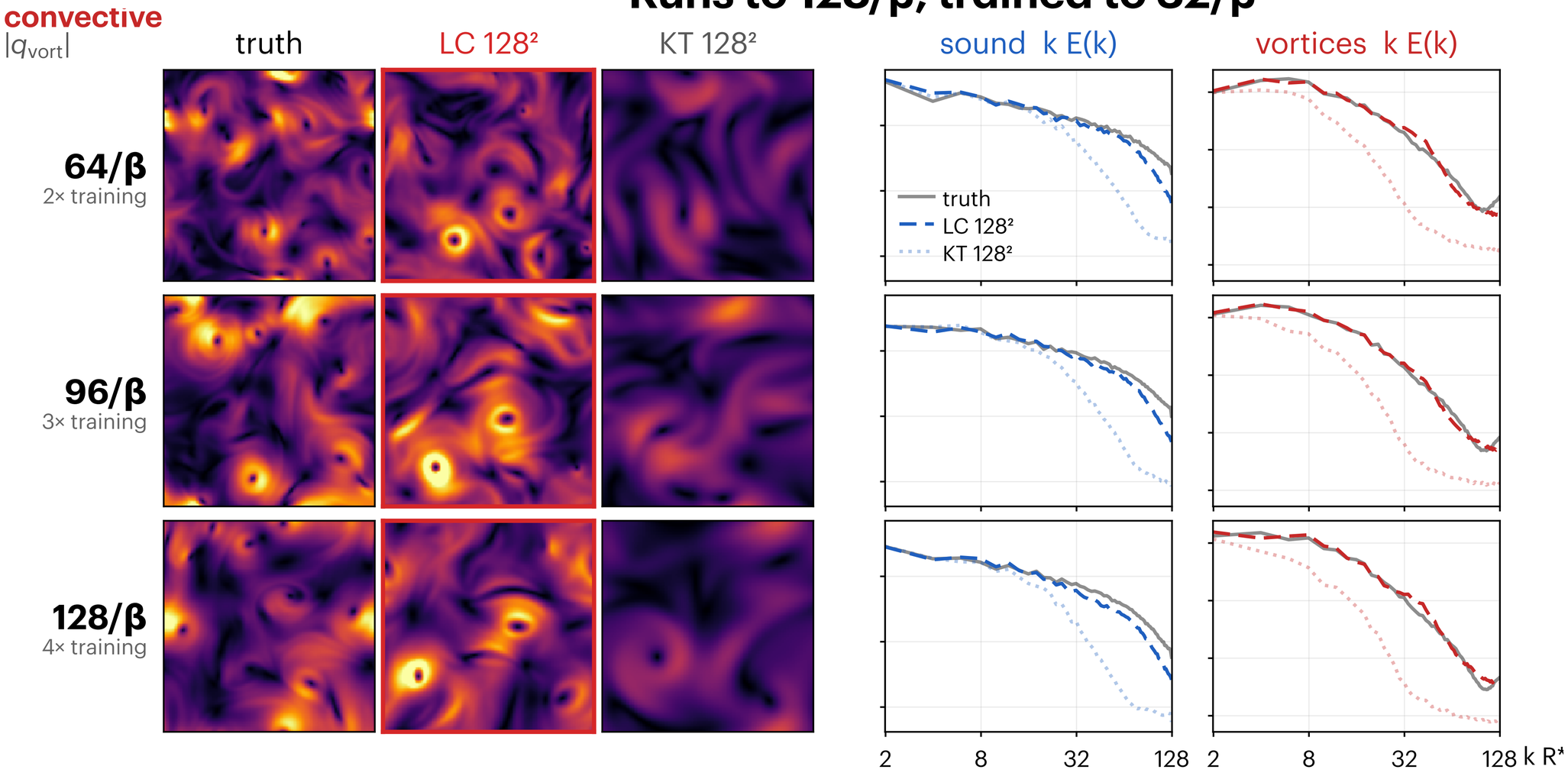

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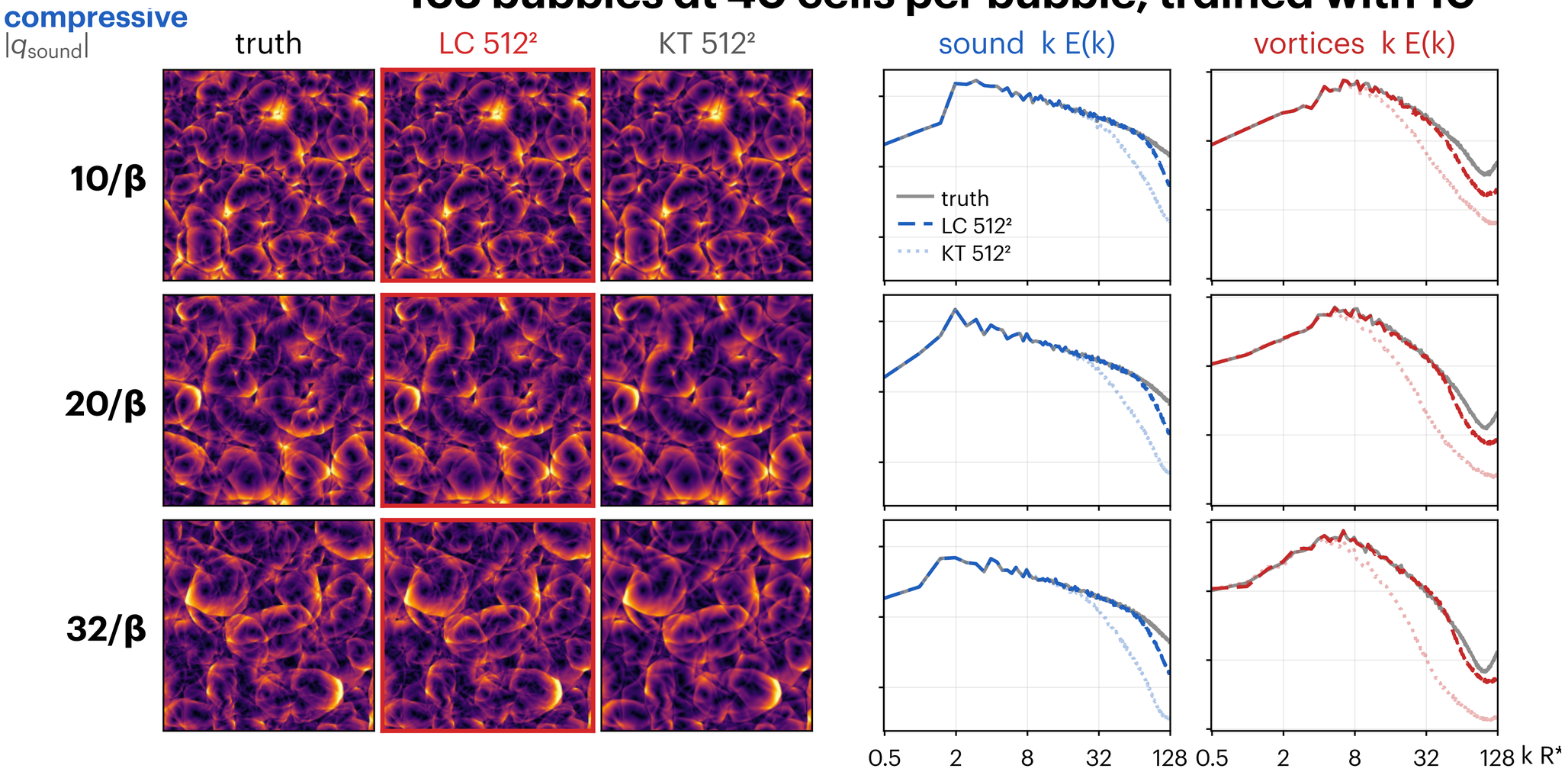
Sixteen times the bubbles

163 bubbles at 40 cells per bubble, trained with 10



Generalizes to **16× the bubbles**

at the same cells per bubble,

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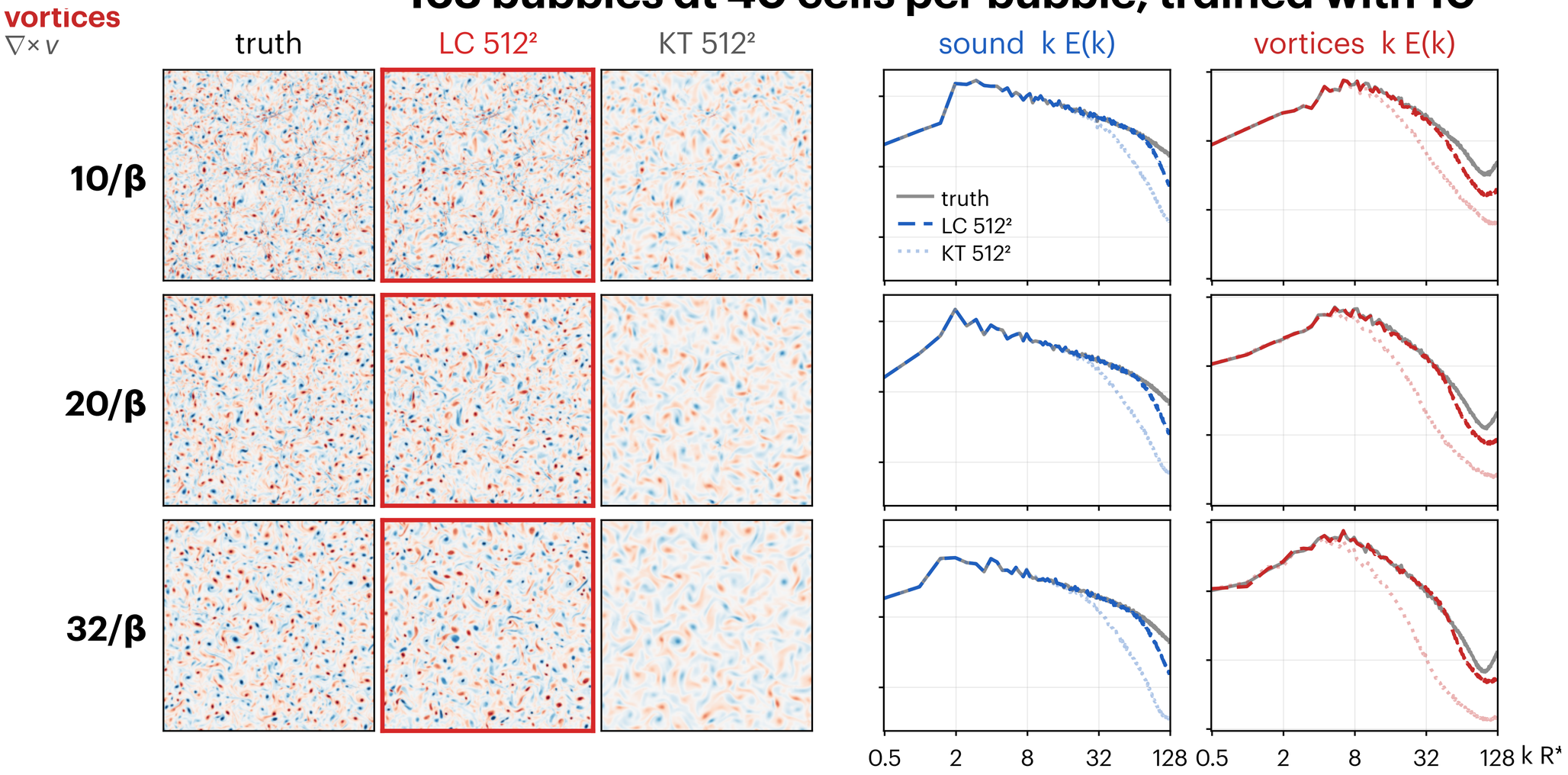

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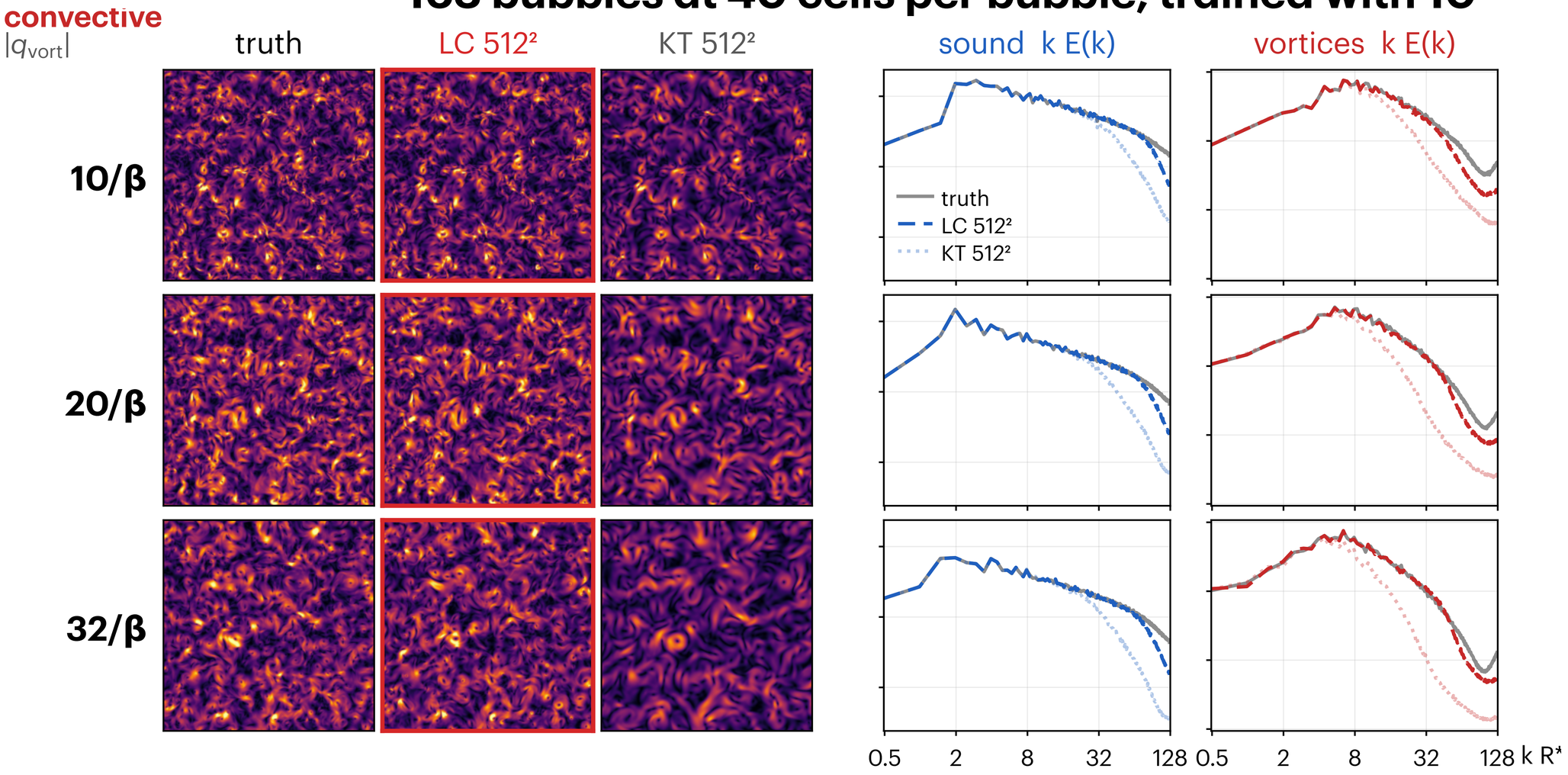

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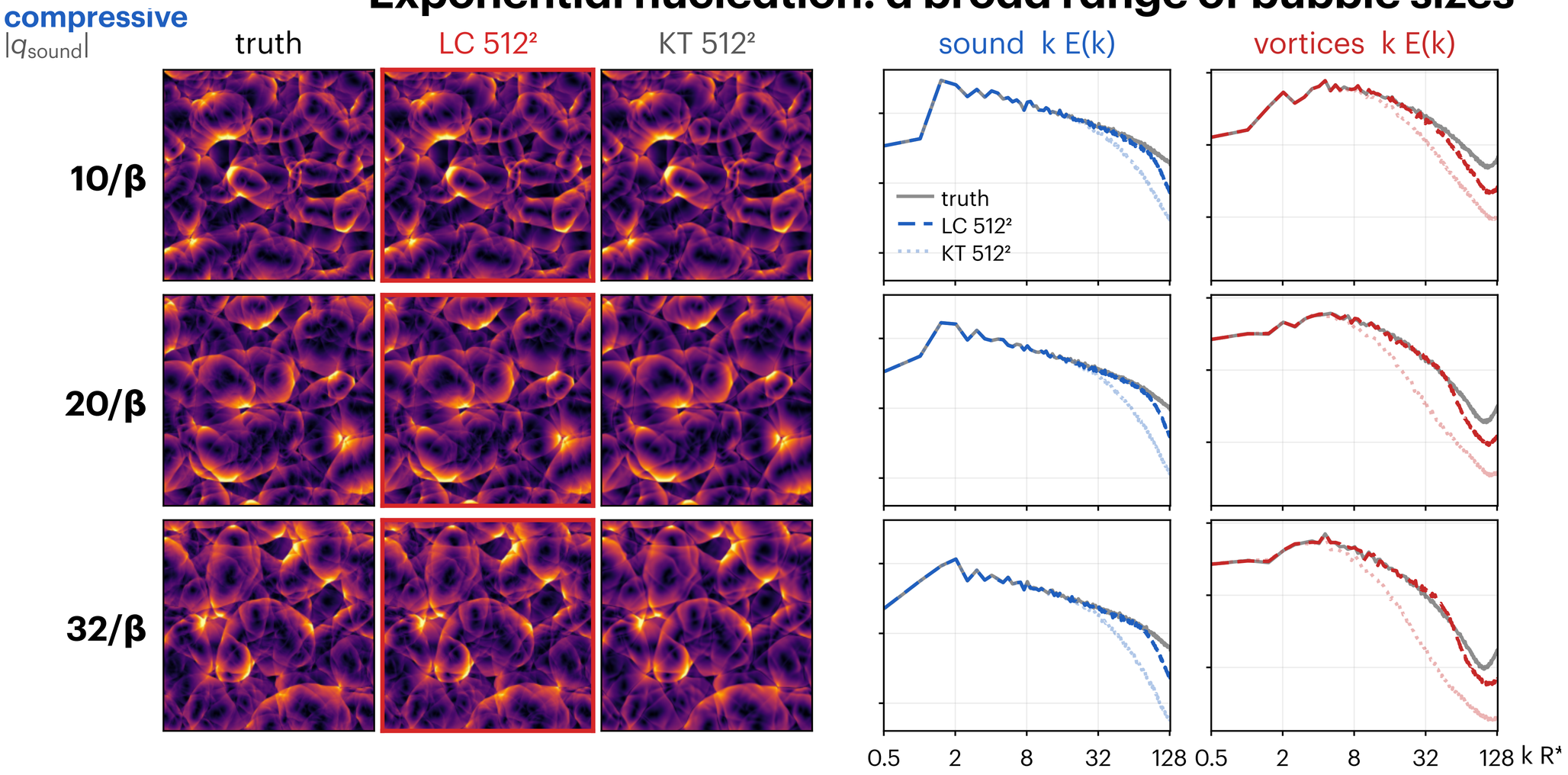
Another nucleation history

Exponential nucleation: a broad range of bubble sizes



Generalizes to a **nucleation history** it never saw:

bubbles of **all sizes**.

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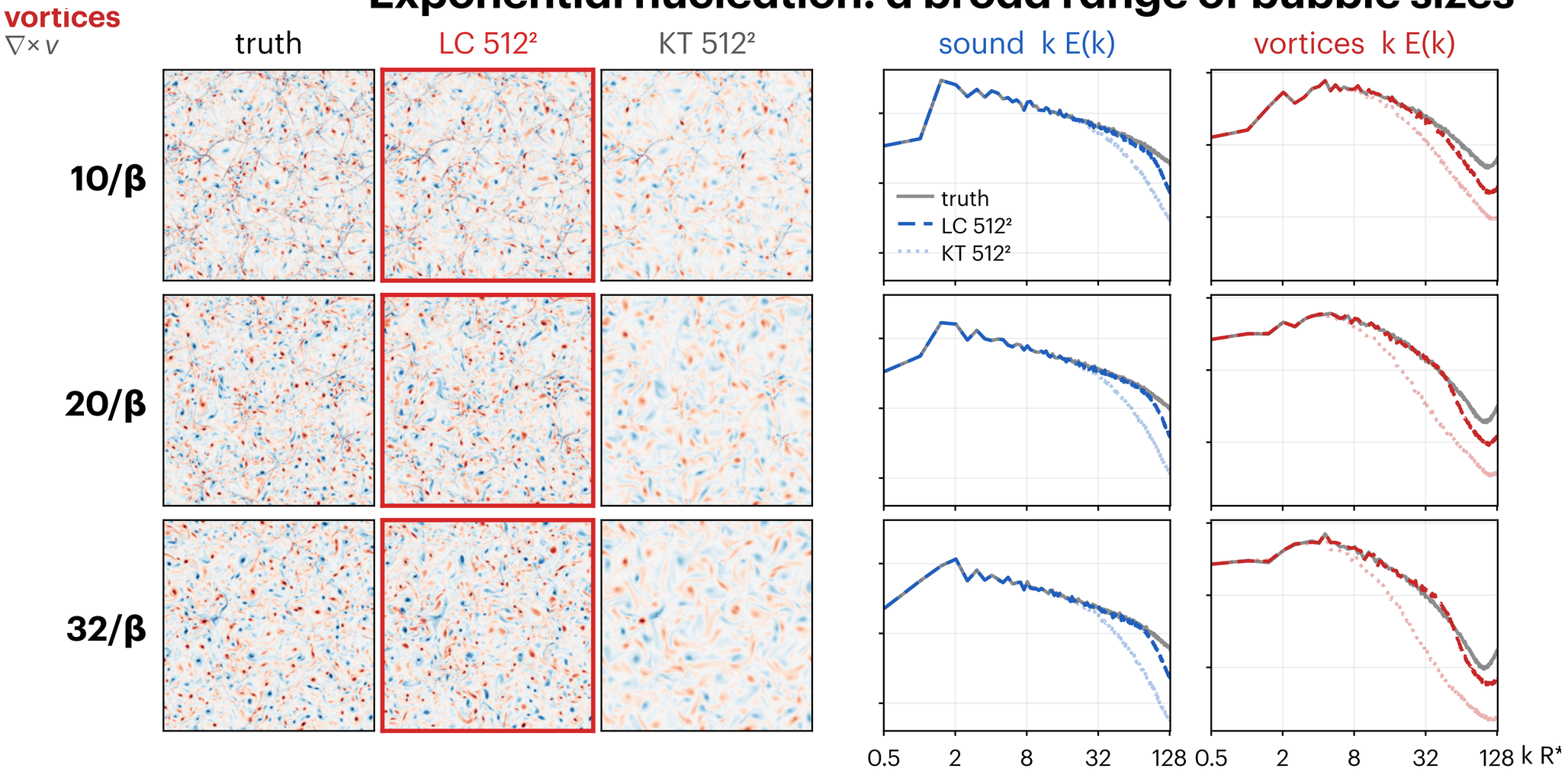
Exponential nucleation: a broad range of bubble sizes



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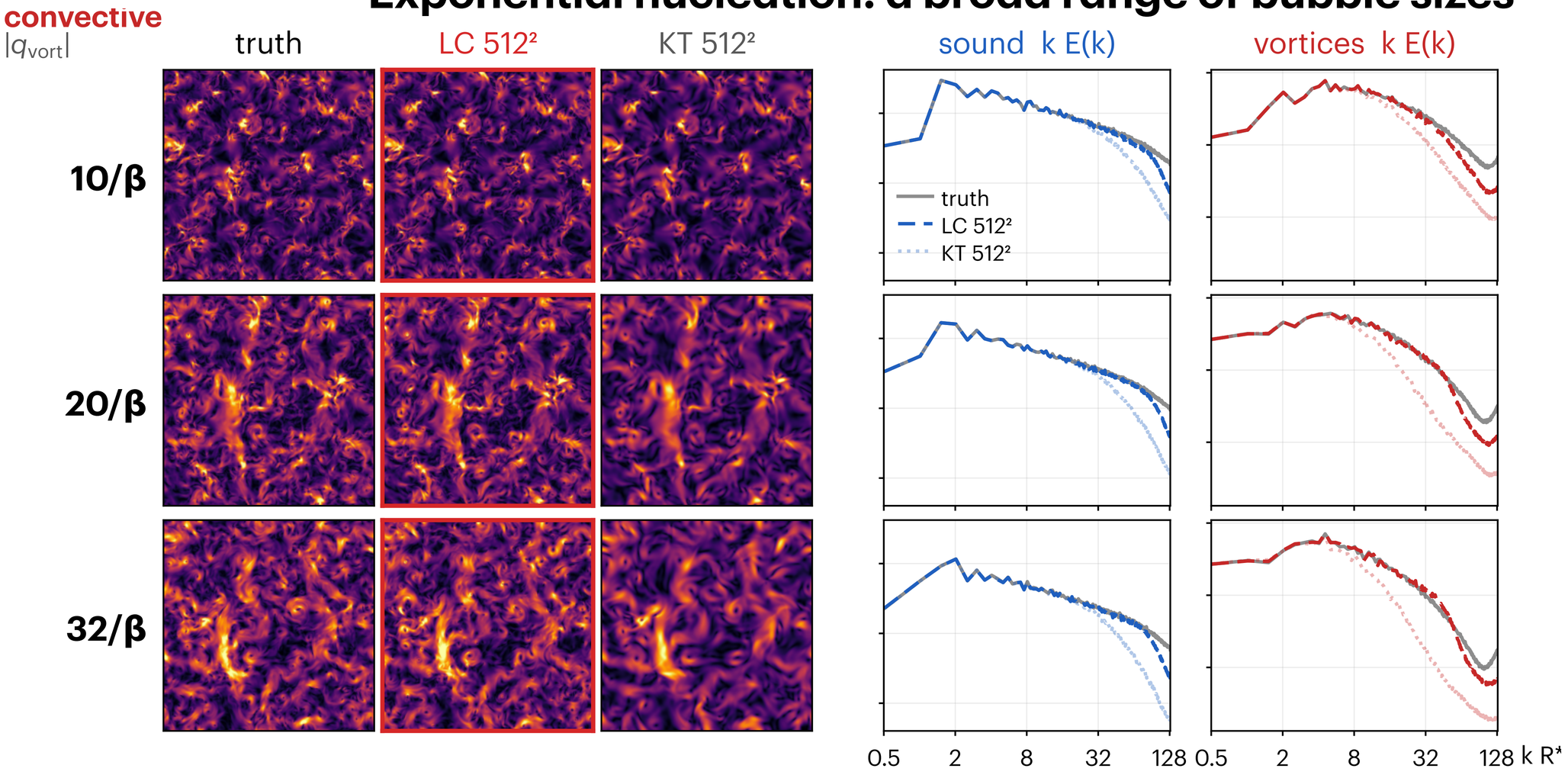
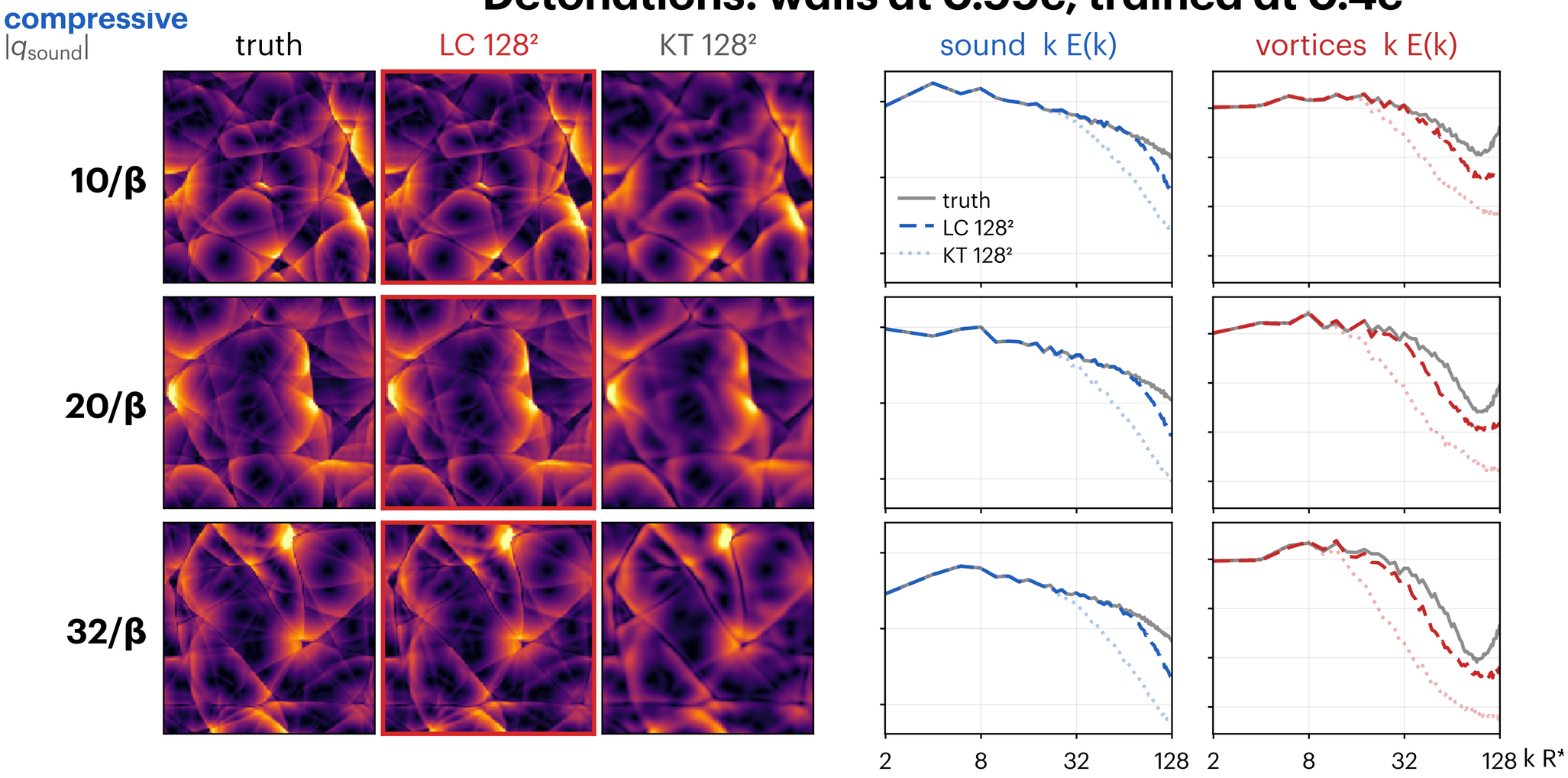


Generalizes to a **nucleation history** it never saw:

bubbles of **all sizes**.

Faster walls

Detonations: walls at 0.99c, trained at 0.4c

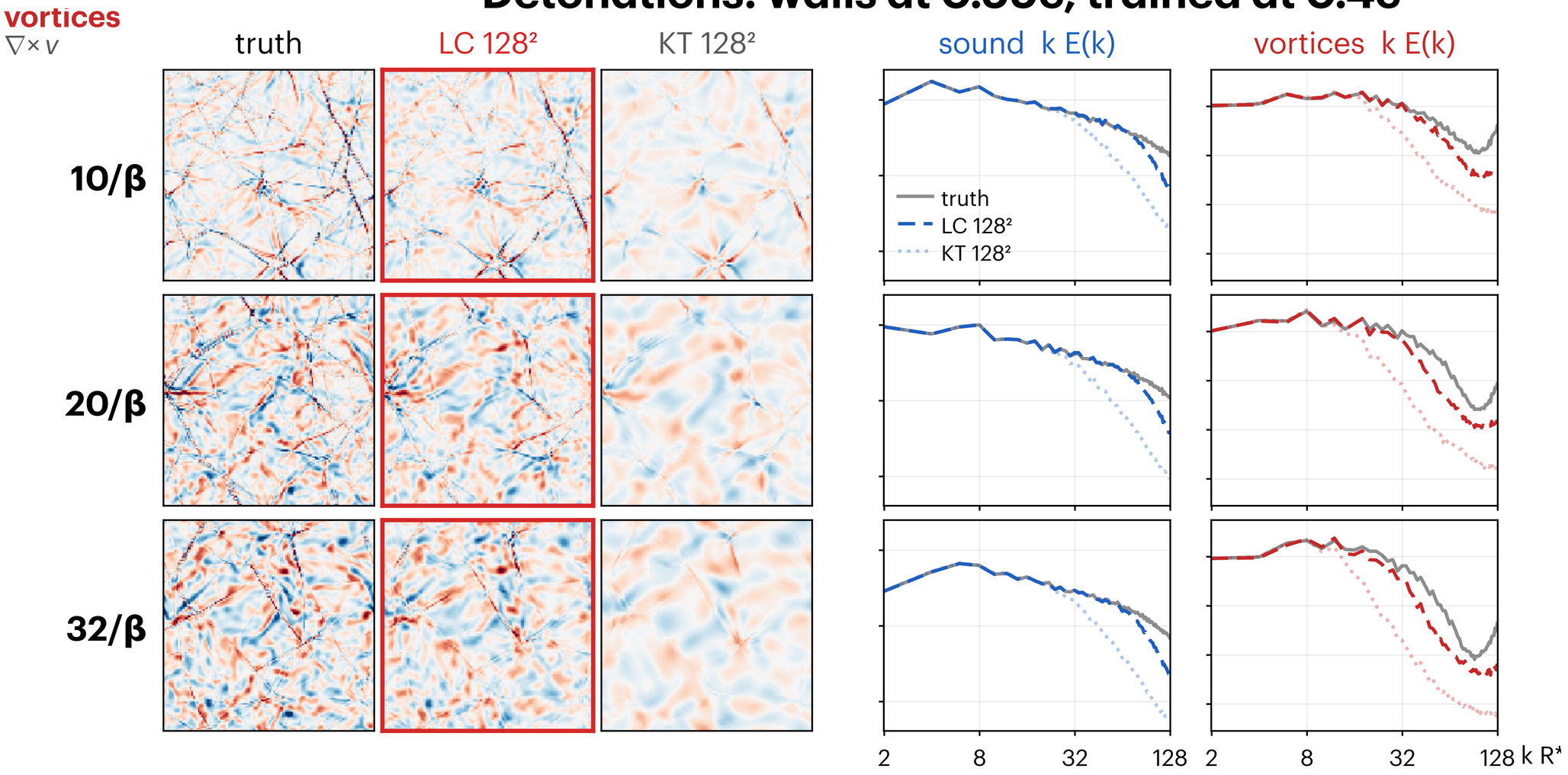


Generalizes to **walls at 0.99c**:

different shocks, **weaker vorticity** than anything in training.

Faster walls

Detonations: walls at 0.99c, trained at 0.4c

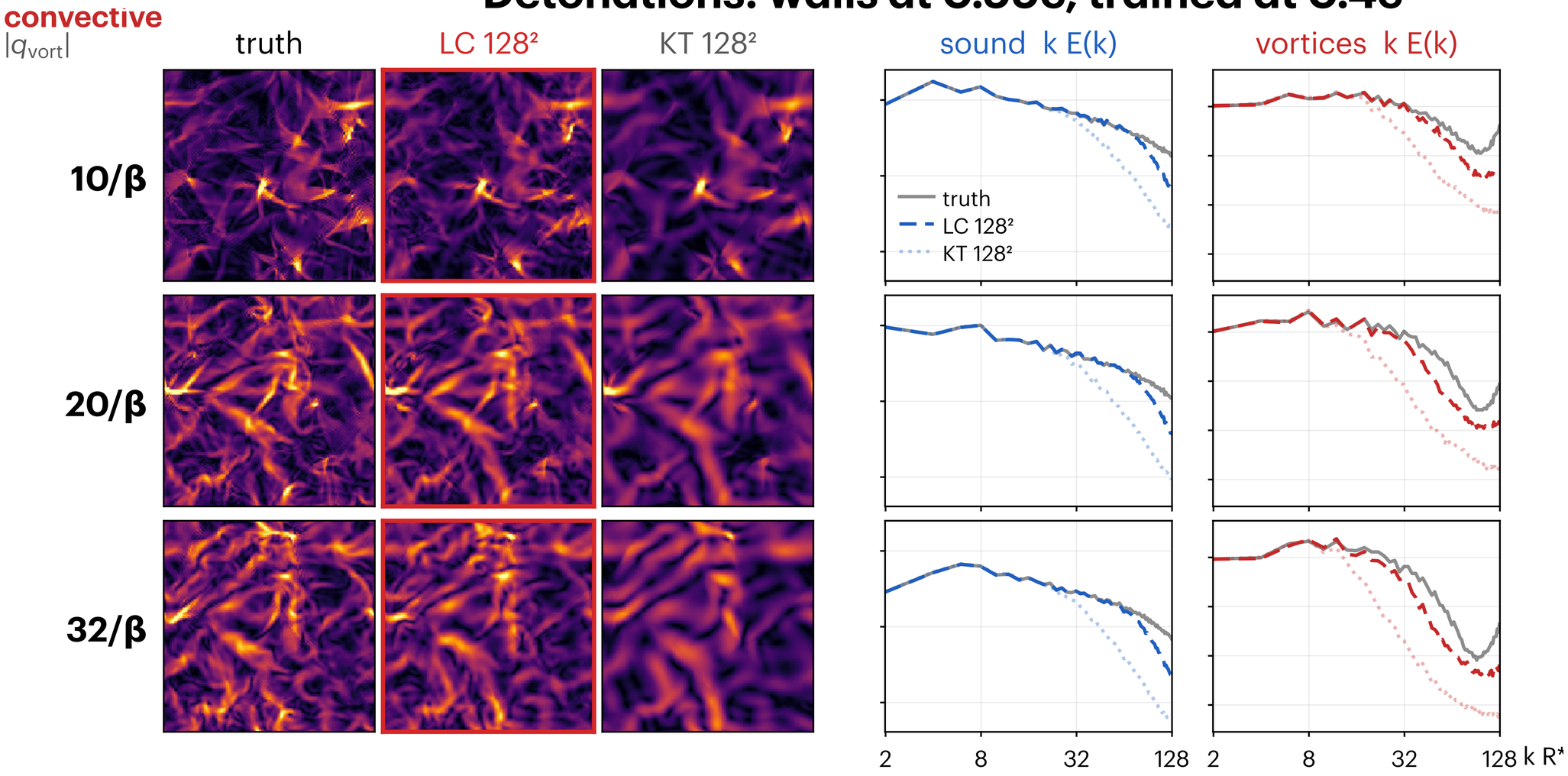


Generalizes to **walls at 0.99c**:

different shocks, **weaker vorticity** than anything in training.

Faster walls

Detonations: walls at 0.99c, trained at 0.4c

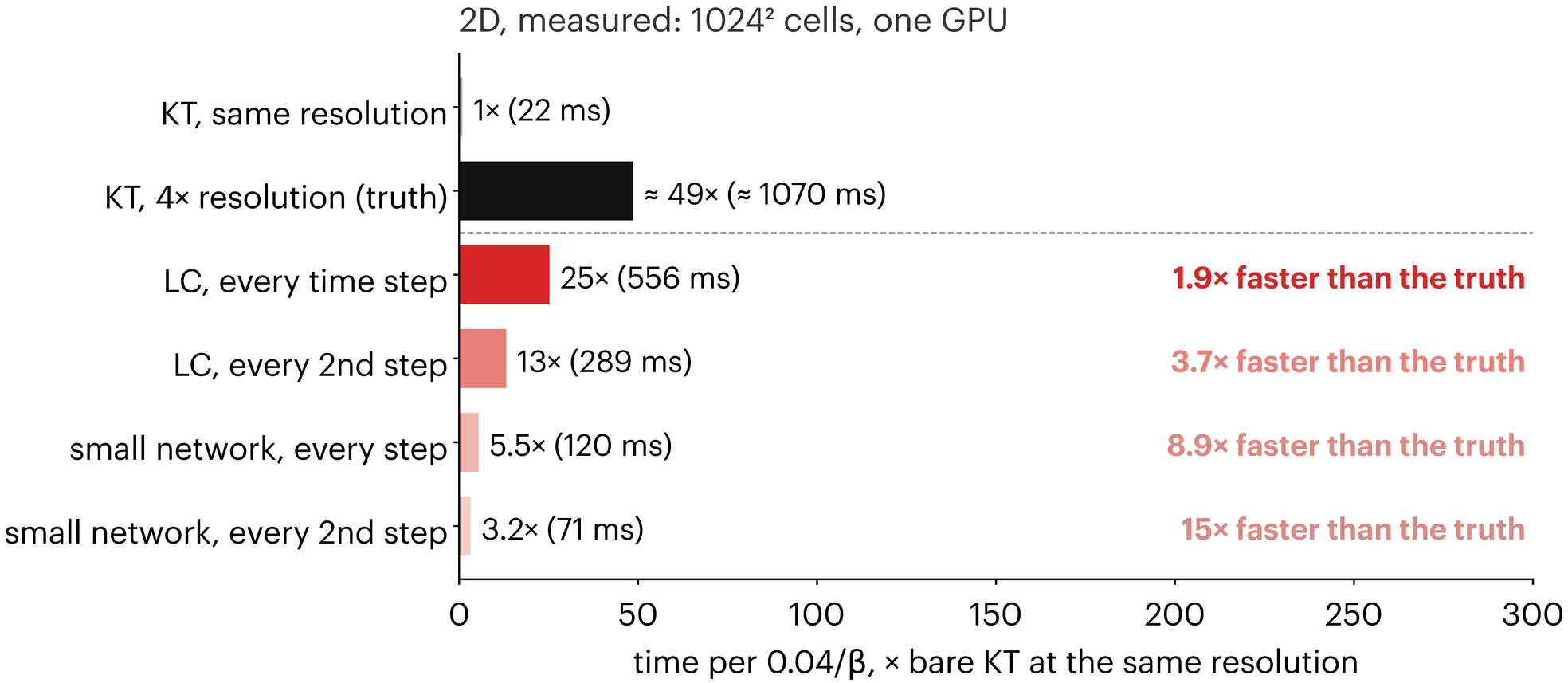


Generalizes to **walls at 0.99c**:

different shocks, **weaker vorticity** than anything in training.

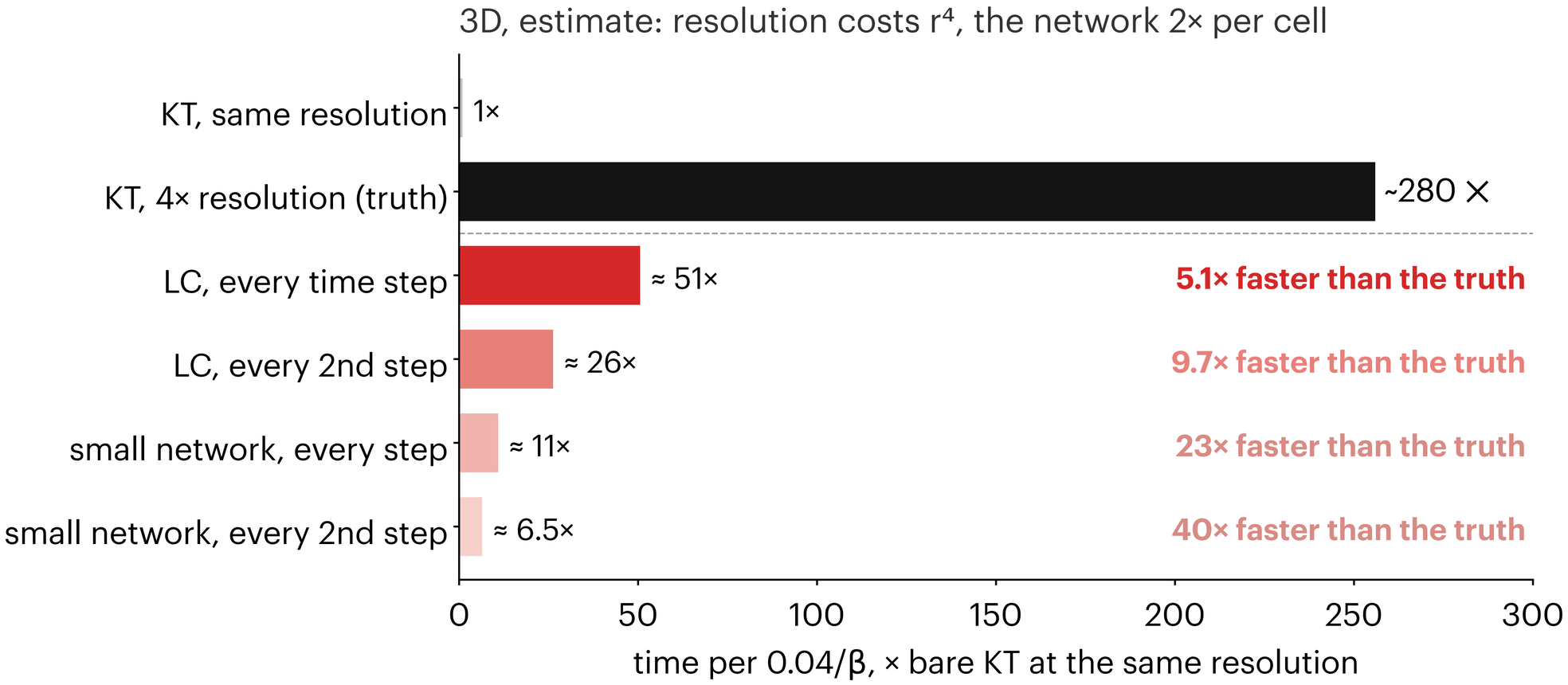
The elephant in the room: Cost

In 2D, LC costs about half of the truth's resolution



The elephant in the room: Cost

In 3D, about a fifth (estimate)



Conclusions

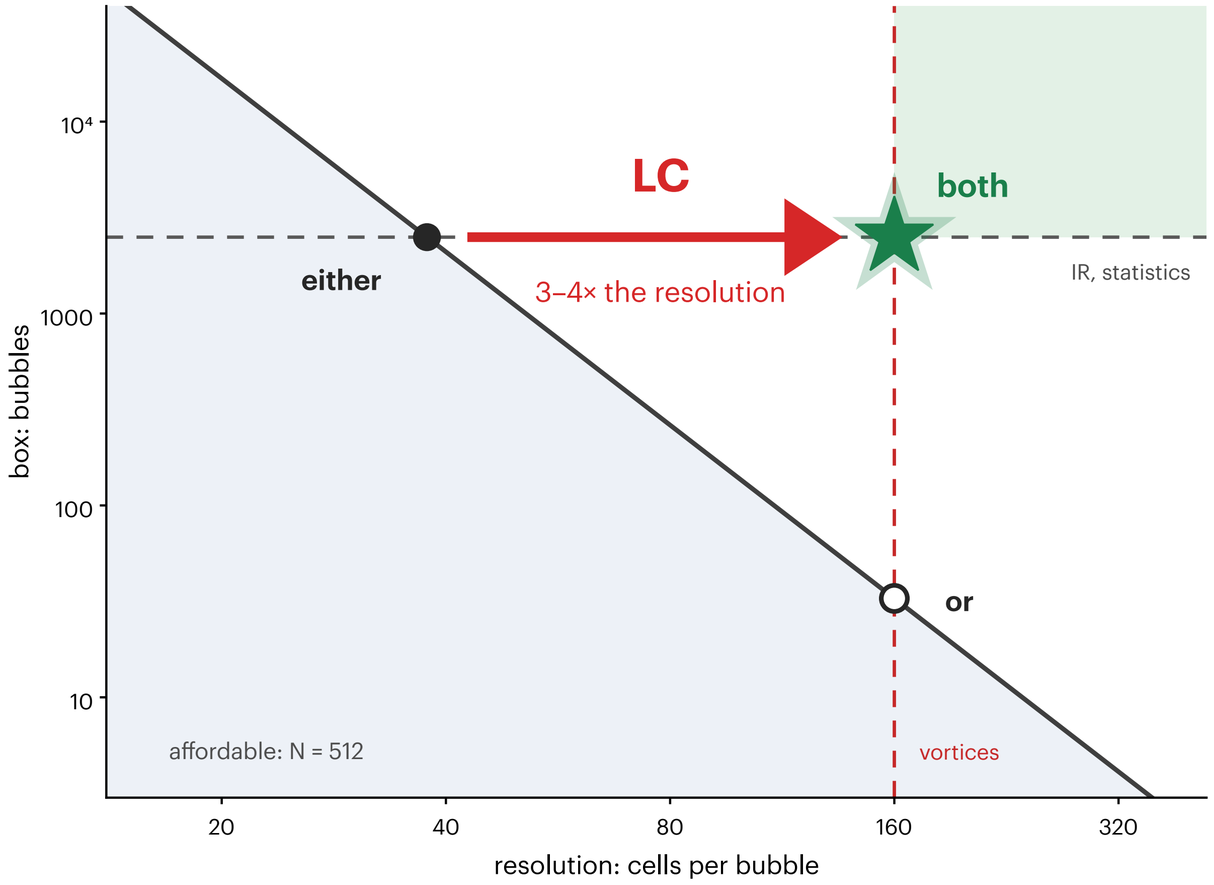
What we asked of a closure, and what LC achieved

What we asked of a closure	What LC achieved (2D, 128^2)
Keep the vortices the vortical energy and spectrum of high resolution	**Vortical energy within 2%** statistics like 3–4× the resolution; phases 1.5–2×
Keep the source alive conserve exactly, stable to saturation	**Exact and stable to 128/β** round-off drift; GW source Π^{TT} within 2–8% at the peak
Restore the small scales the UV statistics of high resolution	**Sound to k R* ≈ 70, vortices to ≈ 50** spectra like 3–4× the resolution
Generalize beyond training big boxes, long runs, other transitions	**Generalizes without retraining** 4× the training time, 16× the bubbles, exponential nucleation, 0.99c walls

Many bubbles at low resolution, with the GW source of high resolution, beyond its training: shown in 2D, next in 3D.

Learned closure closes the gap

Many bubbles and resolved vortices, on the grid strategy 2 can afford



LC takes us to the green star

3–4× the resolution
on N = 512

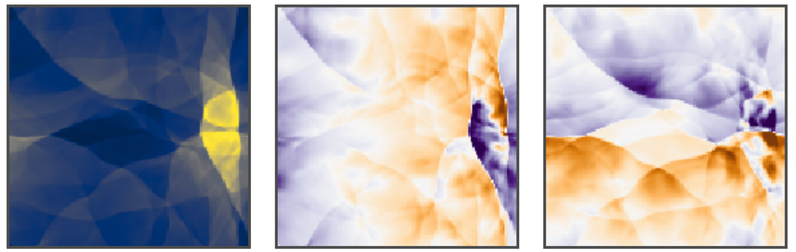
Thank you!

Backup

The loss

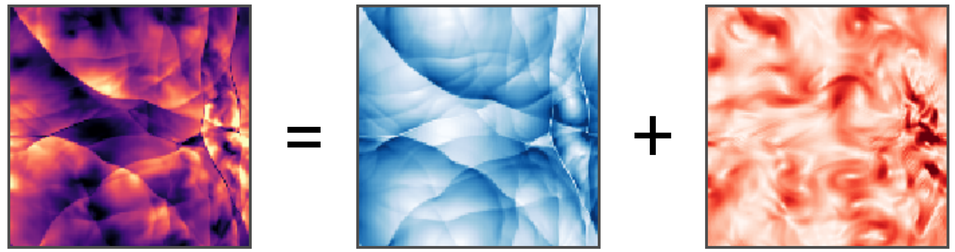
State, sound and vortices weighted equally, no spectral term

energy K^0 momentum K^x momentum K^y



$$\mathscr{L}^U = \frac{1}{3} \sum_{v \in \{K^0, K^x, K^y\}} \frac{\langle (U_v - U_v^{\text{true}})^2 \rangle_{\text{cells}}}{\sigma_v^2}$$

$q = \sqrt{w}\,\gamma v$ q^{sound} q^{vort}



89% of the energy 11% of the energy

$$\mathscr{L}^p = \frac{\langle |\, q^p - q^{p,\text{true}} \,|^2 \rangle_{\text{cells}}}{\sigma_p^2}, \quad p \in \{\text{sound}, \text{vort}\}$$

$$\mathscr{L} = \frac{1}{n} \sum_{\text{snapshots}} \left[\mathscr{L}^U + \mathscr{L}^{\text{sound}} + \mathscr{L}^{\text{vort}} \right]$$

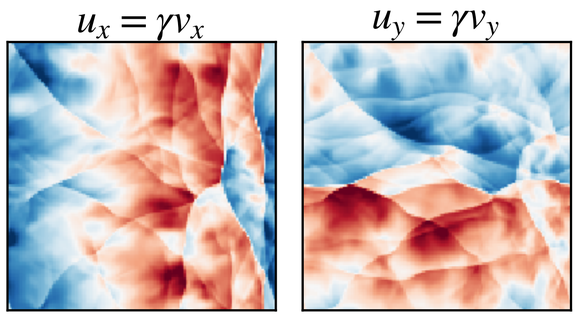
a truth snapshot every 0.04/β (8 coarse steps)
n: the snapshots in a training window
(80 for LC: 3.2/β)

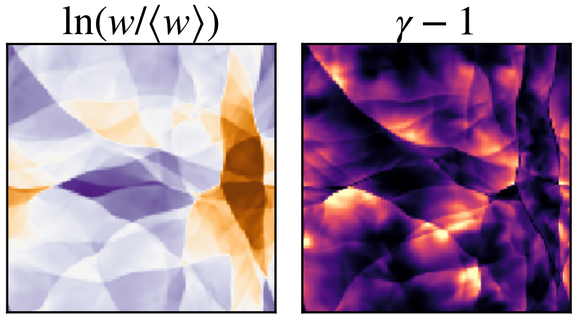
Three terms count equally: energy and momenta, sound, vortices, each over its spread in the truth.

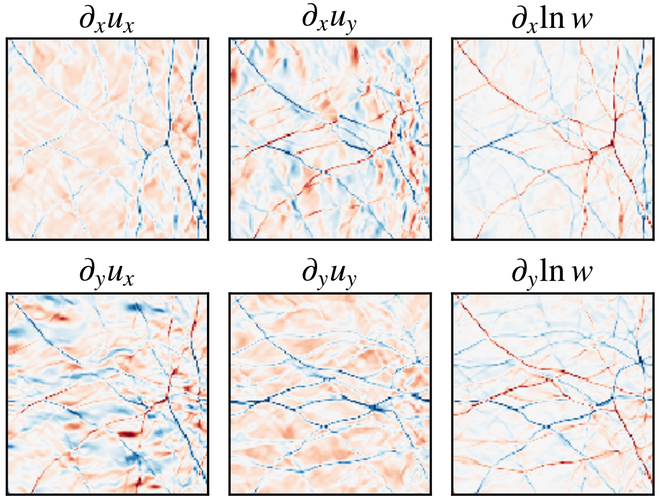
⟨·⟩: mean over cells; σ: the truth's spread over the training window (for the parts of q their RMS, their means being negligible); sound, vort: the compressive and solenoidal (Helmholtz) parts of q = √w γv

What the network sees

Ten inputs per cell: no energy scale, bounded, with gradients







Velocity, not momentum

γv is the relativistic velocity. The momentum $K^i = w\gamma^2 v^i$ would carry the energy scale, which differs between runs, boxes and times.

The state relative to the box

$\ln(w/\langle w \rangle)$ removes the energy scale and treats rarefied interiors and compressed shocks alike; $\gamma - 1$ is the speed, as it enters the fluxes (γ capped at 4).

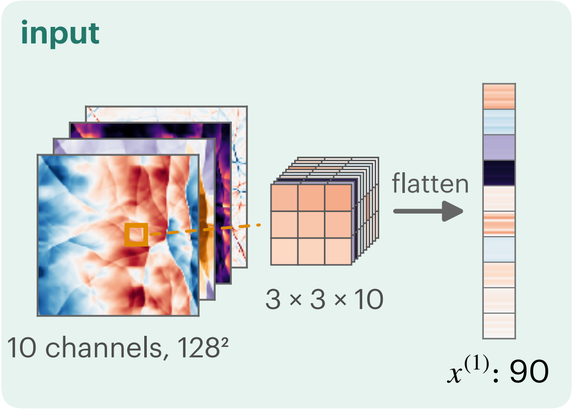
Gradients, limited

A subgrid stress depends to leading order on the resolved gradients (Smagorinsky, Clark): given, not learned. Minmod-limited one-cell slopes, $\div c_s$ and clipped, like KT's own: central differences blew up at shocks.

Inputs and outputs carry no scale: the same weights apply to every run, box, nucleation history and wall speed.

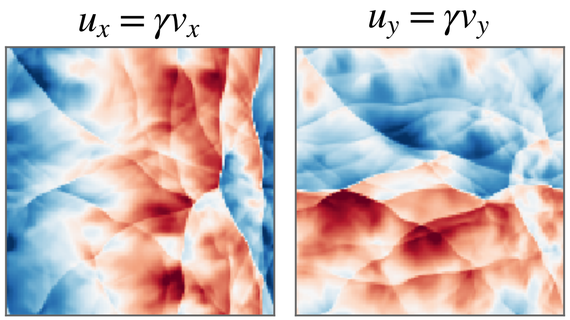
Inside the network

From the 128² grid to six face corrections per cell



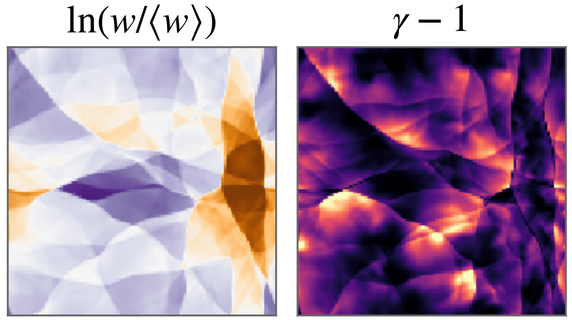
The 10 input channels

Velocity, not momentum



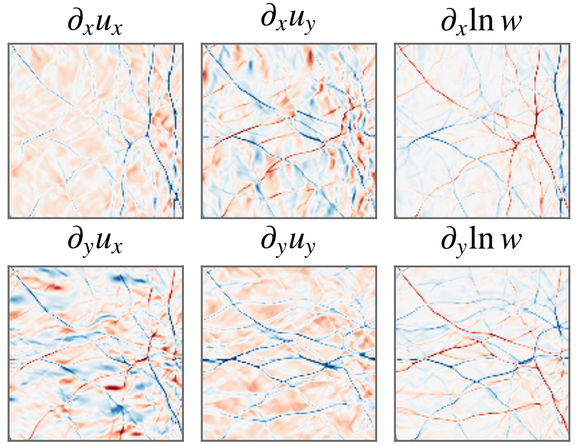
γv: bounded, no energy scale
(the momentum wγ²v carries one)

Enthalpy and speed



the enthalpy against its mean,
the speed as it enters the fluxes

Gradients



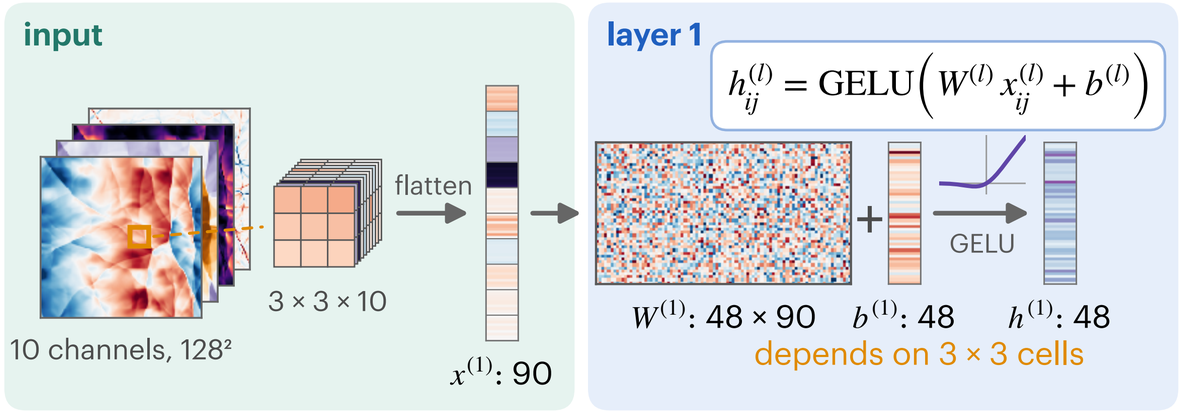
given to the network, not learned

No energy scale in or out: the same weights for every run, box, nucleation history and wall speed.

Each cell's **3 × 3 window** of the **10 input channels** becomes one vector **x** of 90 numbers.

Inside the network

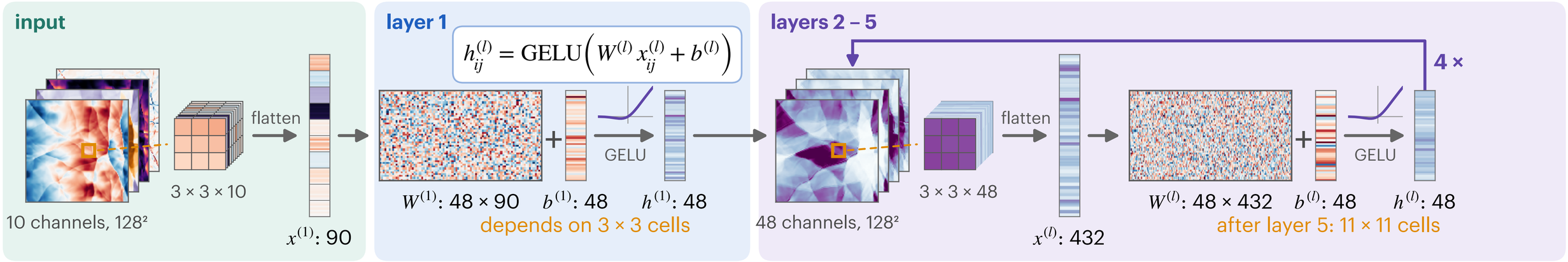
From the 128² grid to six face corrections per cell



Layer 1 is **W x + b**, then GELU: **one matrix product per cell**, the **same W and b at every cell**.

Inside the network

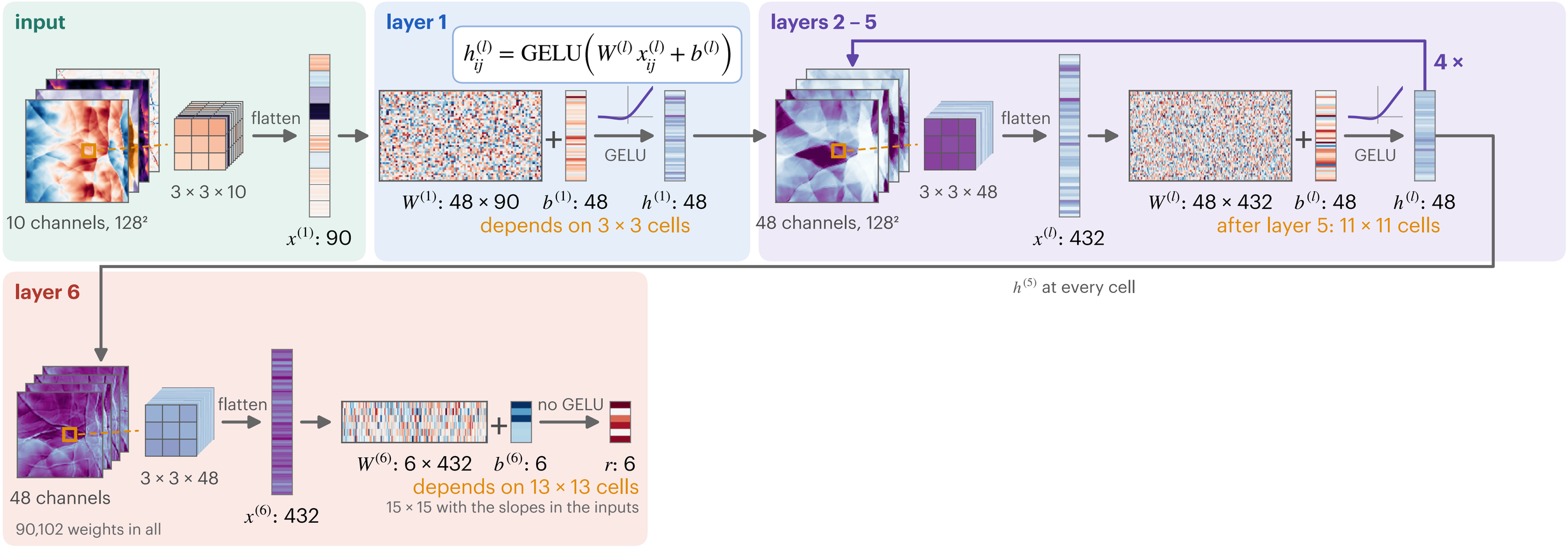
From the 128² grid to six face corrections per cell



Layers 2 – 5 flatten a **3 × 3 window of all 48 channels**: x now has **432 numbers**.

Inside the network

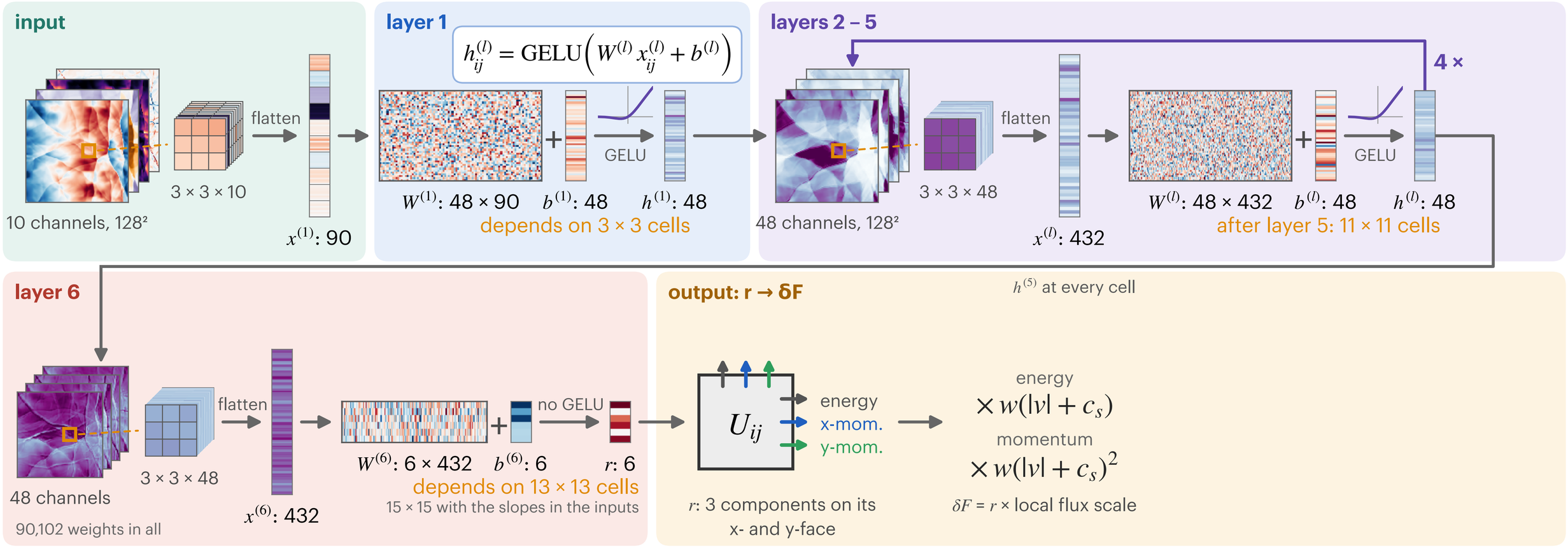
From the 128² grid to six face corrections per cell



Layer 6 returns r, **6 numbers per cell**. It **starts at zero**: untrained, the closure is the **bare scheme**.

Inside the network

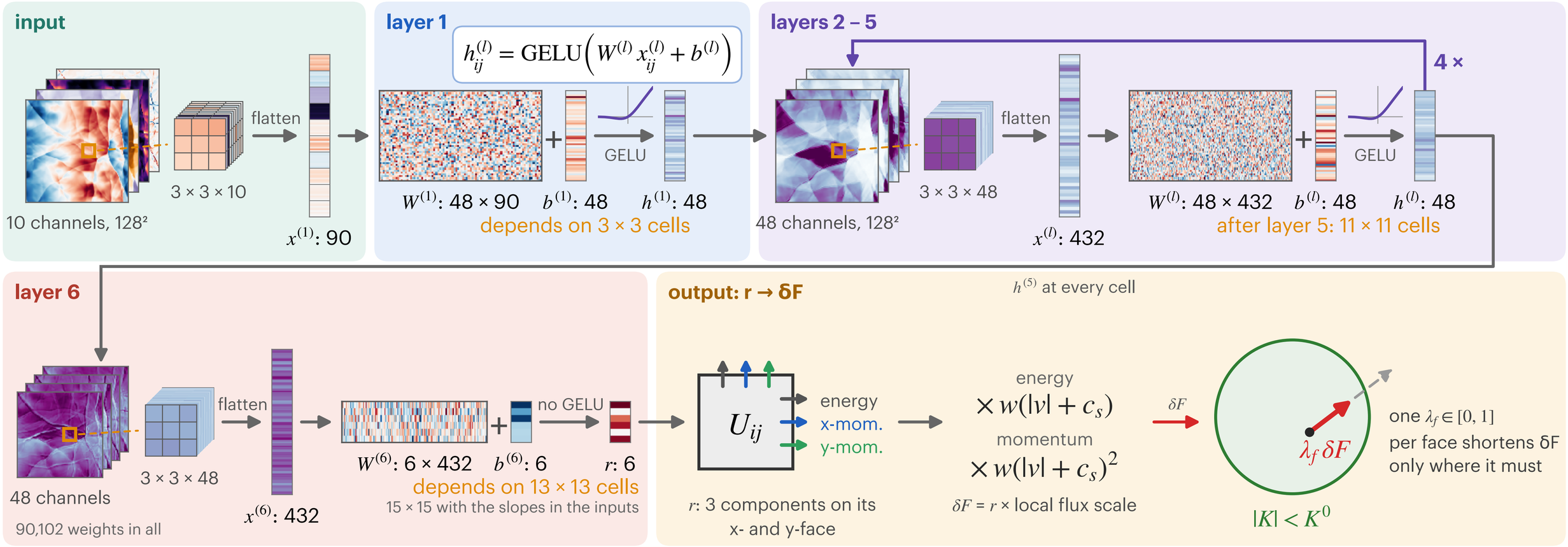
From the 128² grid to six face corrections per cell



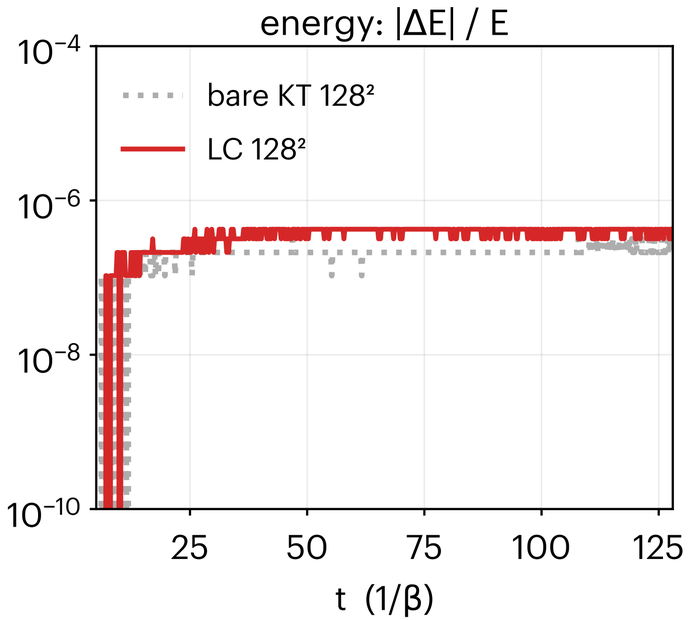
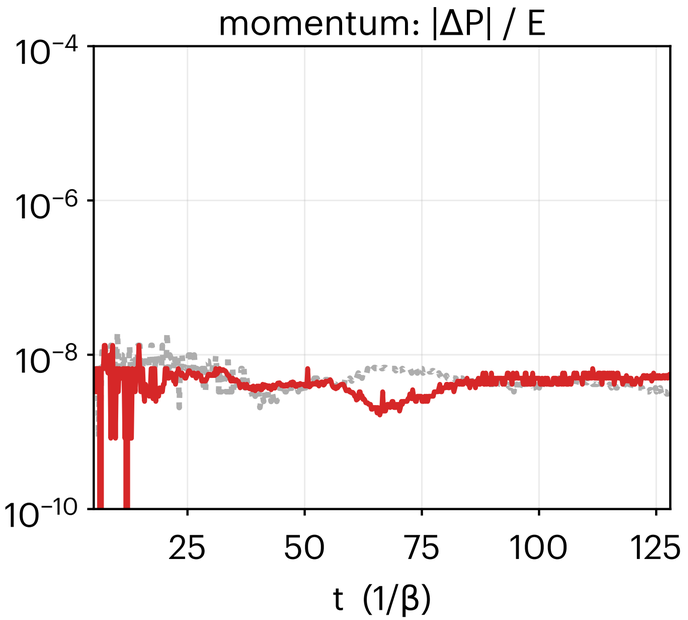
Each r is **dimensionless**: times a **local flux scale** it becomes **δF**, in quiet flow and shocks alike.

Inside the network

From the 128² grid to six face corrections per cell



A **factor per face**, at most 1, keeps every cell in the **light cone** and caps the energy moved per step.

Conservation and stability

Exact to round-off, admissible throughout, 128/β